Colorado State University - Department of Mechanical Engineering - MECH 486A

Preliminary Design Review Report

CSU IREC 2027 Competition Rocket (10,000 ft AGL, COTS Motor, Airbrakes, Payload)

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Table of Contents

Executive Summary 5
1. Problem Definition and Stakeholder Analysis 6
1.1 Problem Statement, Scope, and Assumptions 6
1.2 Concept of Operations 6
1.3 System Scope and Boundary 7
1.5 Stakeholder Needs 9
1.6 Safety/Ethical Considerations & Stakeholder Conflicts 11
1.7 Influence vs. Importance 12
2. Engineering Requirements & Standards 14
2.1 Requirement Levels and Types 14
2.2 Stakeholder Needs to “System Shall” Requirements 14
2.3 Standards to Requirements 18
2.4 Verification Methods 20
2.5 Measures of Effectiveness 21
3. Functional Decomposition 22
3.1 Function Tree 22
3.2 Function-to-Requirement Mapping 24
3.3 Derived Requirements 28
3.4 Airbrakes as a Derived Requirement 29
3.4 Traceability from Need to Verification 31
4. Concept Generation & Trade Studies 33
4.1 Concepts and Criteria 33
4.2 System Architecture Trade 34
4.3 Concepts and Criteria 35
4.4 Trade Studies 40
4.5 Selected Concept and Justification 43
4.6 Trade Decision Records 45
4.7 Tradeoffs and Assumptions 47
5. Preliminary Design 49
5.1 System Layout 49
5.2 Subsystem Interfaces 50
5.3 How the Architecture Meets Key Requirements 51
5.4 Safety-Critical Features 56
5.5 Preliminary Component and Material Selections 58
5.6 Engineering Budgets and Margins 60

5.7 Risk Management..... 61
5.8 Verification, Validation and Schedule..... 62
References..... 64
Appendix..... 65
A. Full Requirements List..... 65
B. Subsystem Decision Matrices..... 75
C. Black Powder Charge Calculations..... 78
D. Nose Cone, Fin and Motor Studies..... 79
E. Budget and Timeline..... 81
F. Risk Matrix / FMEA..... 83
G. Systems Engineering Calculations..... 86
H. Resource Allocation..... 88
I. Traceability Matrix..... 91
J. PDR Content Map..... 96

List of Figures

Figure 1. Mission sequence (CONOPS): 1 pre-launch, 2 powered ascent, 3 coast, 4 apogee and drogue deployment, 5 main deployment, 6 landing and recovery. 6

Figure 2. Three Systems: SOI, Enabling System, and Context System. Green marks the Context System, peach the Enabling System, and charcoal the SOI. 8

Figure 3. Enabling System: the people, rules, facilities and funding that support the SOI. 8

1.4 Operating Context and Assumptions 9

Figure 4. Stakeholder importance vs. influence. Points with identical scores are offset for legibility. 12

Figure 5. Functional decomposition to Level 2. Orange rows are safety-critical functions; † marks functions needed only if the option is flown (airbrakes or the SDL experiment). 23

Figure 6. Preliminary OpenRocket layout of the full-scale vehicle with an AeroTech M2500T [9]. 45

Figure 7. System architecture with function allocation and interfaces (ICD numbers match Table 36). ICD-010 and ICD-011 are proposed. 49

Table 35. Allocation of functions to subsystems. 50

Figure 8. OpenRocket results for the full-scale vehicle: (a) altitude at Midland conditions, (b) static margin with no wind. 54

Figure 9. Avionics wiring layout: each flight computer has its own 9 V battery and screw switch and fires its own main and drogue charges. 59

List of Tables

Table 1. Mission phases, functions, requirements and hazards. 7
Table 2. Project scope: owned, interfaced, and excluded. 9
Table 3. Operating context and assumptions. 9
Table 4. Stakeholder analysis summary. 10
Table 5. Safety considerations, conflicts, and tradeoffs. 11
Table 6. Stakeholder importance and influence scores, strategy and engagement approach. 13
Table 7. Requirement types used in this report. 14
Table 8. Stakeholder needs converted to “system shall” requirements. 15
Table 9. Requirements added or revised relative to the Master SRR (rev. 10/2/2026). 17
Table 10. Standards and documents converted to requirements. 19
Table 11. Test requirements and how they will be run. 20
Table 12. Measures of effectiveness. 21
Table 13. Function dictionary and function-to-requirement mapping. Shaded olive rows are Level 1; orange rows are safety-critical; † marks optional functions. 24
Table 14. Functions with no requirement in the Master SRR. 27
Table 15. Derived requirements revealed by functional decomposition. 29
Table 16. Derivation chain for the airbrake requirements. 29
Table 17. Gates for keeping the airbrakes in the design. 30
Table 18. Traceability summary by Level 1 function (a requirement counts under every Level 1 function it maps to). † Optional. 31
Table 19. Coverage check of the trace. 32
Table 20. Basis labels for scores. 33
Table 21. Scale anchors for the 1 to 10 scores. 33
Table 22. System architecture options. 34
Table 23. Facts that differ between the architecture options. 34
Table 24. System architecture decision matrix (weights 1 to 5, scores 1 to 10). Draft scores, proposed for team review. 35
Table 25. Trade studies, concepts compared, and criteria with weights († added by design decision). 36
Table 26. Motor candidates and past use in IREC entries. 37
Table 27. Avionics layout concepts. 38
Table 28. Airbrake concepts. 39
Table 29. SDL payload experiment concepts. 40
Table 30. Trade study roll-up (weights 1–5, scores 1–10). 43
Table 31. Trade decision register. 45
Table 32. Decisions made without a trade study. 47
Table 33. Key tradeoffs. 47

Table 34. Design assumptions. 48
Table 36. Subsystem interfaces. 50
Table 37. Key requirements against the current design (OpenRocket, Midland conditions). 52
Table 38. Safety-critical features. 57
Table 39. Engineering budgets and margins. 60
Table 40. Highest-scoring risks and next actions. 61
Table 41. Validation milestones. 62
Table 42. Master SRR requirements († changed in this report). 65
Table 43. Decision matrix: airframe material. 75
Table 44. Decision matrix: airbrake mechanism. 75
Table 45. Decision matrix: recovery parachute. 75
Table 46. Decision matrix: avionics layout. 76
Table 47. Decision matrix: SDL payload experiment. 76
Table 48. Evidence needed to replace judgment scores. 76
Table 49. Black powder charge sizing for the main and drogue bays. 78
Table 50. Decision matrix: nose cone shape. 79
Table 51. Decision matrix: fin planform. 79
Table 52. Decision matrix: fin cross-section. 79
Table 53. Mandrel material choices. 80
Table 54. Fixturing choices. 80
Table 55. Budget by subsystem. 81
Table 56. Funding sources. 81
Table 57. Flight timeline. 82
Table 58. Impact and probability scales and risk levels. 83
Table 59. Hazard list and risk matrix (I = impact, P = probability). 83
Table 60. Mass limit from the rail-exit requirement. 86
Table 61. Descent speed for three vehicle mass assumptions. 87
Table 62. Proposed mass growth allowance by stage. 88
Table 63. Mass allocation from the preliminary model. 88
Table 64. Geometry and volume allocation. 89
Table 65. Electrical energy budget (to be completed). 89
Table 66. Budget shares by subsystem. 90
Table 67. Requirements traceability matrix. * proposed. Verif.: T test, A analysis, I inspection, D demonstration. 91
Table 68. Mapping of generic PDR content to this report. 96

Executive Summary

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1. Problem Definition and Stakeholder Analysis

1.1 Problem Statement, Scope, and Assumptions

Problem statement. Design, manufacture, and test an SRAD (student researched and developed) high-powered rocket for the International Rocket Engineering Competition that reaches a nominal apogee of 10,000 ft above ground level (AGL) with accurate predictability of its trajectory, and carries a payload of at least 2.0 kg (4.4 lb).

Accuracy matters because apogee is scored by distance from the target: under the 2027 rules (R&R-6.6.1) the apogee score falls linearly to zero at ±30% of the target (7,000 ft and 13,000 ft) [1]. Safety is a separate condition that applies to every entry. Before range officers allow a launch, the team must show that the vehicle will fly stably, descend under control, and use hardware that meets the rules [1], [2]. A vehicle that passes those checks and reaches the window can fly.

1.2 Concept of Operations

The mission sequence in Figure 1, also called the concept of operations (CONOPS), follows one flight from start to finish in six phases: (1) pre-launch on the rail, (2) powered ascent, (3) coast, (4) apogee and drogue deployment, (5) main deployment and descent, and (6) landing and recovery. Each requirement applies in at least one phase. For example, the 25 m/s rail-exit speed applies in phase 2 and the drogue timing in phase 4. Listing the phases lets us check that no part of the flight is left without a requirement.

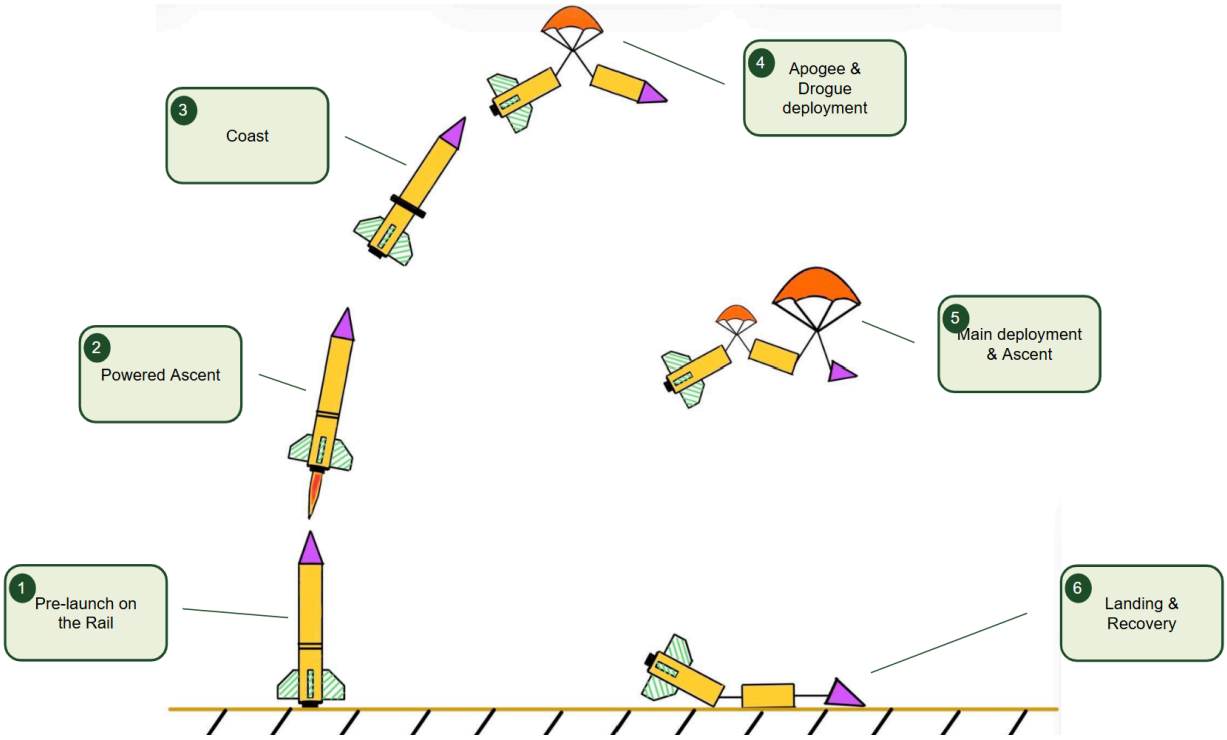


Figure 1. Mission sequence (CONOPS): 1 pre-launch, 2 powered ascent, 3 coast, 4 apogee and drogue deployment, 5 main deployment, 6 landing and recovery.

Table 1 breaks each phase into what the vehicle must do, the functions that act in it, the requirements that govern it and the hazard rows that apply. It is a first check of completeness: every phase has requirements, and every hazard row in Appendix F falls in at least one phase. Payload functions (6.1 to 6.3) apply in every phase after loading. The airbrake rows apply only if the option is flown. The format follows the concept-of-operations outlined in the NASA Systems Engineering Handbook, Appendix S [7].

Table 1. Mission phases, functions, requirements and hazards.

Phase	What the vehicle must do	Functions	Key requirements	Hazard rows
1. Pre-launch on the rail	Sit on the rail, stay inert until armed, acquire a GPS fix and show that it is ready.	2.1, 8.4, 9.4, 10.1, 10.5	AV-001, AV-008, TST-002, TST-003, PT-006, SYS-009	HZ-05, HZ-15, HZ-16
2. Powered ascent	Start the motor, leave the rail at 25 m/s or more, stay stable, carry the thrust loads.	1.1–1.4, 2.2, 3.1–3.3, 7.1	SYS-005, SYS-006, SYS-007, SYS-008, STR-001, PT-004, AERO-003	HZ-07, HZ-09, HZ-10, HZ-11, HZ-12
3. Coast	Stay stable and record flight data. If airbrakes are flown: estimate state, predict apogee, trim drag after burnout.	3.2, 9.2; option: 4.1–4.5	SYS-001, AV-003; option: SYS-013 (proposed), AB-001–007, AV-013, AV-014	HZ-19; option: HZ-A1, HZ-A2, HZ-A3
4. Apogee and drogue deployment	Detect apogee, fire the drogue charge within 100 ft, separate the sections, allow no early firing.	5.1, 5.2, 5.3, 10.2	REC-001, REC-005–008, AV-007	HZ-01, HZ-02, HZ-03, HZ-05, HZ-06, HZ-13
5. Main deployment and descent	Fire the main at or below 1,500 ft AGL and slow to a landing rate under 11 m/s.	5.2, 5.4, 5.5	REC-002, REC-003, REC-004, REC-009, REC-010, STR-004	HZ-04, HZ-08, HZ-14
6. Landing and recovery	Land intact, keep the flight record, report position, and make the vehicle safe to approach.	5.6, 5.7, 9.2, 9.3, 10.4	AV-002, AV-004, REC-014, DOC-007	HZ-14, HZ-15, HZ-16

1.3 System Scope and Boundary

The system of interest (SOI) is the competition vehicle. Figure 2 places it in three layers. The **SOI** is what the team designs and controls; the figure shows only the baseline vehicle that IREC requires. The **Context System** is what acts on the SOI directly: the launch rail, the airspace and landing area (where the general public is), and the atmosphere. The **Enabling System** is what makes the whole project cycle, and the vehicle's performance, possible: the people, rules, facilities and funding detailed in Figure 3. Every stakeholder in Table 4 sits in exactly one layer, and the "Layer and role" column of that table gives the layer by name. Table 2 lists what the team owns, what it designs to, and what is outside the project. The baseline payload need is only mass (at least 2.0 kg [1]), volume and a safe interface: the payload must not reduce stability or interfere with recovery. Airbrakes and the SDL Payload Challenge experiment are not in the baseline SOI. They are optional design decisions, compared in the architecture trade (Section 4.2) and shown in the architecture in Section 5.1.

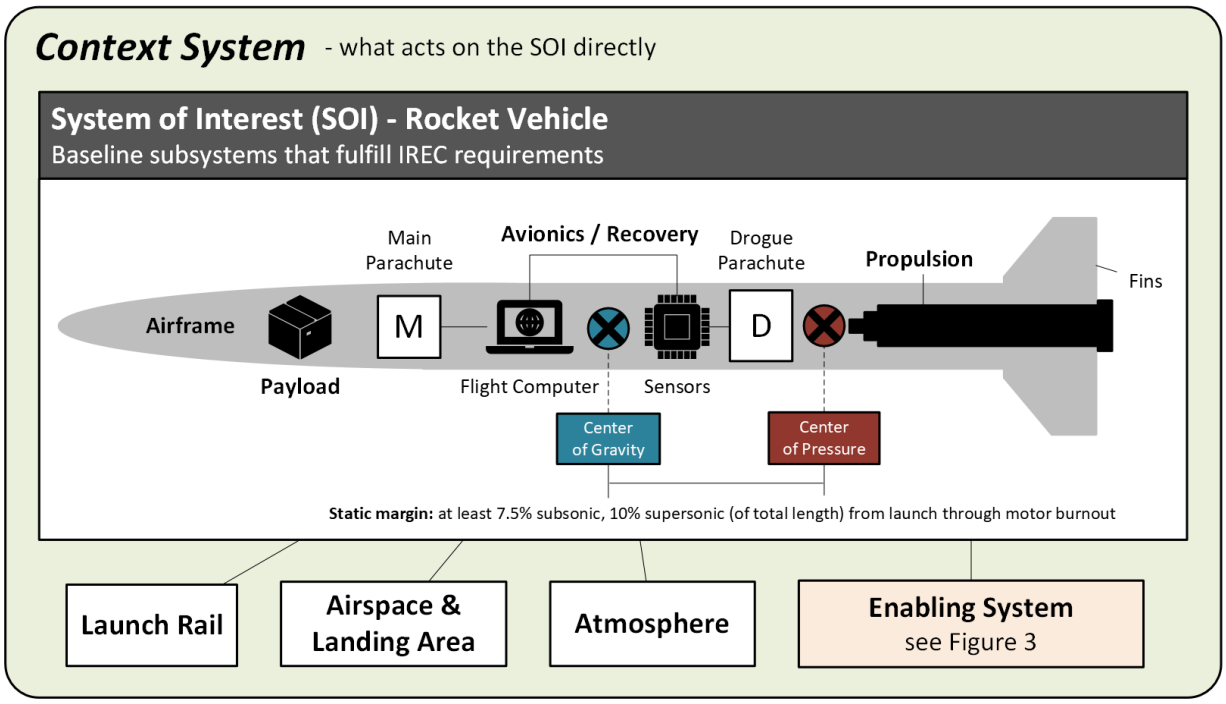


Figure 2. Three Systems: SOI, Enabling System, and Context System. Green marks the Context System, peach the Enabling System, and charcoal the SOI.

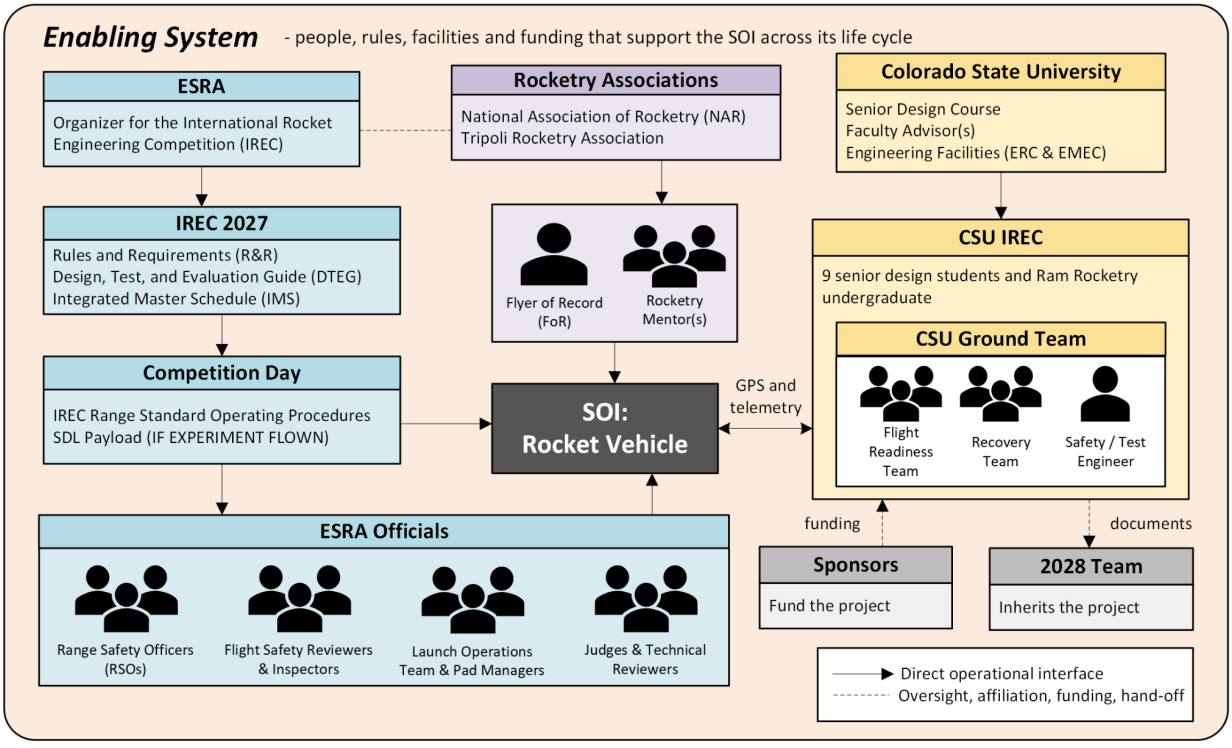


Figure 3. Enabling System: the people, rules, facilities and funding that support the SOI.

Table 2. Project scope: owned, interfaced, and excluded.

Category	Items
Baseline SOI (needed to fly)	Airframe and fins; integration of one COTS solid motor; dual-deployment recovery; avionics (flight computer, altimeters, GPS); a payload of at least 2.0 kg.
Added by design decision (Section 4)	Active airbrakes to trim apogee; a non-deployable thermodynamic-cycle experiment as the payload.
Interfaces (designed to, not owned)	Launch rail and pad hardware; ground support and range procedures; IREC Rules & Requirements and Design, Test & Evaluation Guide; ground tracking station.
Out of scope	Motor design and manufacture; launch rail design; range operations and launch authority; liquid or hybrid propulsion (2028 team); production versions of the vehicle

1.4 Operating Context and Assumptions

Operating context and assumptions. Table 3 lists the assumptions about the competition and launch site that bound the problem: the rules edition, the dates and location, the launch site elevation and rail, the supply of motors, the $20,000 budget and the flight-test schedule. If one of these changes, the requirements that depend on it must be re-checked. The design assumptions A1 to A5, which cover the team's own design choices, are in Section 4.7.

Table 3. Operating context and assumptions.

ID	Assumption	Basis
C1	Competition is IREC 2027, June 12–19, 2027 [12], Midland, TX.	ESRA calendar. The IMS lists June 12–19, 2027; June 20 is only the Media Challenge photo deadline [12].
C2	The 2027 REV P1 R&R, DTEG and IMS (effective October 15, 2026 and September 21, 2026) are the rules baseline [1], [2], [12]. The frequency plan is the 2026 Rev D plan, which ESRA lists as a work in progress for 2027 [13]. Section numbers use the 2027 numbering (R&R-x.y.z and D&T-x.y.z).	[1], [2]
C3	Launch site elevation is about 890 m (2,920 ft) MSL; apogee is measured AGL; the organizer supplies a rail of at least 17 ft.	DTEG descent calculations
C4	One COTS solid motor is used. The AeroTech M2500T is preferred; supply is limited, with M1939W, M2400T and M3400-P as alternates.	Motor survey and supply review
C5	The total budget is $20,000. Funding is predicted, not secured: senior design $7,500, Colorado Space Grant $5,000, RamFunder $4,000, travel grant $1,200, sponsorships $1,200, fundraisers $1,100.	Team budget plan
C6	Flight tests: subscale November 28, 2026; full-scale March 13 and May 15, 2027.	Team launch timeline

1.5 Stakeholder Needs

Table 4 lists the twelve stakeholders, how each relates to the system, and the outcome each needs. The "Layer and role" column places each stakeholder in the Context System or the Enabling System shown in Figure 2 and Figure 3. Needs are written as outcomes, not solutions, so that the requirements in Section 2

do not fix a design before the trade studies in Section 4. Priority is High where failing the need stops the flight, the project or the team's shop access (or sets the course deliverables), Medium where it shapes score, funding or procedures, and Low where it affects only future work.

Table 4. Stakeholder analysis summary.

1	Senior Design Course	Enabling. Sets deliverables and grading.	The Senior Design Course needs the project to show a complete, traceable systems engineering record across the MECH 486A deliverables to judge the team's readiness for detailed design.	High	Honest reporting of test and analysis results.
2	CSU IREC Team	Enabling. Designs, builds and flies the vehicle; owns the SOI.	The CSU IREC Team needs the vehicle to reach its target apogee and recover safely at the 2027 competition within the senior design schedule and budget to earn a competitive result	High	Team safety during build, test and launch; sustainable workload.
3	2028 Team (Liquids)	Enabling. Inherits the project after 2027.	The 2028 team needs the design, models, and documentation to be maintainable so the project can be inherited and improved.	Low	Lessons on hazards and safe practice carried forward.
4	CSU Engineering Faculty	Enabling. Oversees the program.	CSU Engineering Faculty need student projects to comply with department safety policy during design, fabrication and test to protect people and the program's reputation with sponsors and accreditation bodies.	Medium	University liability; compliance with safety policy.
5	CSU ERC & EMEC	Enabling. Provides fabrication facilities.	The CSU ERC and EMEC need all fabrication work to follow facility safety protocols and equipment procedures to avoid injury, equipment damage, and loss of shop access for the team and future users.	High	Resin and composite dust exposure, machine safety, PPE.
6	ESRA (IREC organizer)	Enabling. Sets rules and runs the event.	ESRA needs the team and vehicle to comply with IREC rules and safety standards throughout the event to keep the competition sanctioned.	High	Fair competition; rule compliance.
7	Range Safety Officers (RSOs)	Enabling. Grant the launch waiver.	RSOs need verifiable evidence of stability, recovery and safety compliance before launch to decide whether to grant a launch waiver.	High	Evidence must be complete and truthful.
8	Flight Safety Reviewers & Inspectors	Enabling. Certify flight readiness.	Flight Safety Reviewers need complete documentation and a physically inspectable vehicle at flight-readiness review to certify the vehicle for flight.	High	Accurate documentation of the as-built vehicle.
9	Judges & Technical Reviewers	Enabling. Score the entry.	Judges need clear design rationale and demonstrated flight performance at competition to score the vehicle fairly.	Medium	Honest representation of results.
10	Launch Operations & Pad Managers	Enabling. Run the pad and launch procedure.	Launch Operations need the vehicle to integrate with standard rail and pad hardware and a repeatable launch procedure to launch safely and on schedule.	Medium	Motor and pyrotechnic handling on the pad.
11	General Public / Local Community	Context. In the landing area near the range.	The general public needs launch and recovery operations to pose no risk to bystanders or property near the range to remain safe.	High	Controlled descent; no uncommanded deployment; hazard

					areas.
12	Sponsors	Enabling. Fund the project.	Sponsors need visibility of responsible use of funds and evidence of progress toward a successful flight to justify continued support and see a return on sponsorship.	Medium	Honest use and reporting of funds.

1.6 Safety/Ethical Considerations & Stakeholder Conflicts

The safety hazards in this project fall on four groups of stakeholders from Table 4, and each is tied to a need there. The General Public / Local Community could be exposed to an uncontrolled descent if the vehicle drifts too far, so the vehicle must separate and decelerate only on command and land at a limited speed (Section 2, recovery and safety requirements). The Range Safety Officers (RSOs) and the Flight Safety Reviewers and Inspectors will grant a launch waiver only on complete and truthful evidence of stability, recovery and compliance, so every flight-critical requirement has an assigned verification method and record (Section 2, stability, verification and process requirements). The CSU ERC and EMEC need fabrication to follow the facility protocols laid out by Justin Hollis, because composite layup and machining involve resin, dust and machine hazards that put the operator and building occupants at risk. The ERC will be used for all composite work, and the EMEC is still needed for preparation such as mandrel machining. The CSU IREC Team and the Launch Operations and Pad Managers handle the motor and the black powder ejection charges at the pad, so the design keeps pyrotechnics inert until the vehicle is armed and leaves handling steps to the operations procedures. This is controlled by safety checks during rocket preparation and by black powder calculations for the pressure needed. Table 5 lists the conflicts this creates between safety and performance requirements. In each one, flight stability and safe recovery take priority over apogee and mass, and the rest are accepted trade-offs managed through a mass budget and simulation.

Table 5. Safety considerations, conflicts, and tradeoffs.

#	Requirements A	Requirements B	Nature of Conflict	Resolution
1	SYS-001 (apogee of 7,000–13,000 ft AGL, nominal 10,000 ft)	SYS-007/008 (stability margin of at least 7.5% of vehicle length, and no more than 18% at launch and 25% in flight)	Tuning apogee by moving mass or ballast shifts the CG, which shifts the stability margin.	Stability takes priority for flight safety. Apogee is tuned only within the limits stability allows.
2	SYS-001	SYS-010 (total liftoff mass cap)	Added mass lowers apogee, and removing mass risks overshooting the 13,000 ft ceiling.	Set the mass cap from motor selection and structural margin, then manage mass as a budget with a reserve.
3	SYS-010	SYS-007/008	Where the mass sits affects stability, not just how much there is.	Allocate mass per function, with stability as the highest priority.
4	SYS-001	AB-001/002 (airbrakes control drag for apogee, within a required response time)	Airbrakes can only add drag, so they help only if the rocket would overshoot. Whether it overshoots depends on vehicle mass.	Run airbrake simulations iteratively against weather and trajectory variation. The un-braked trajectory must overshoot the target.

5	SYS-001	Motor selection	The motor's impulse largely sets the apogee.	State the impulse limit for the desired apogee and justify it with OpenRocket simulation.
6	SYS-001	Flight Readiness Team	Flight safety and resulting apogee depend on how well the team sets up the rocket.	Run several rehearsals and scaled flights to practice flight readiness.
7	AB-005 (airbrakes shall not reduce stability margin below the requirement)	AERO-005 (stability maintained in every airbrake state)	Airbrakes need enough drag authority for apogee precision, but deploying them must never push stability below the SYS-007 floor.	Cap airbrake authority at the stability margin floor and verify with CFD and wind-tunnel testing.

1.7 Influence vs. Importance

Figure 4 plots each stakeholder by importance (how much the project outcome matters to them) against influence (how much power they have to change the project's direction or success), using the scores in Table 6. Scores run 1 to 5, and a score of 3 or higher counts as high. The quadrants follow the standard grid: high importance and high influence is Manage Closely; high influence and low importance is Keep Satisfied; high importance and low influence is Keep Informed; low on both is Monitor [8]. The quadrant sets how much effort the team spends on that stakeholder and how it engages them.

We manage four stakeholders closely: the CSU IREC Team, the CSU fabrication facilities (the ERC and EMEC), Launch Operations and Sponsors. The ERC and EMEC can revoke shop access, and the project cannot build without it. Launch Operations decide whether our rail and pad interfaces work on launch day. Sponsors decide whether the budget exists. The Experimental Sounding Rocket Association (ESRA, the IREC organizer), the Range Safety Officers (RSOs), the Flight Safety Reviewers and the Judges sit in Keep Satisfied. They have little personal stake in our result, but they hold gate authority: the RSOs can deny the launch waiver, the reviewers certify flight readiness, and ESRA can disqualify. We do not negotiate with them. We satisfy them by meeting their rules in full and delivering the evidence they ask for, which is why their needs became hard requirements in Section 2. The General Public is the most affected and the least able to influence the design, so it is protected by hard safety requirements (Section 2) instead of consultation.

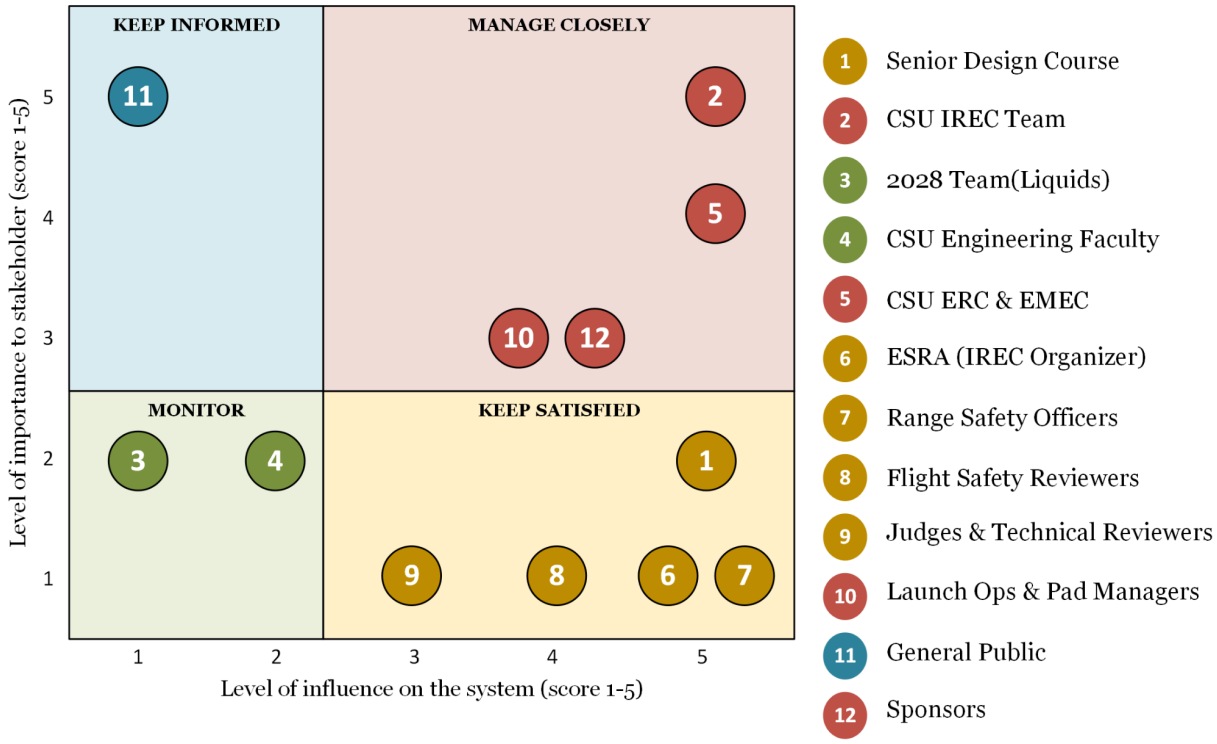


Figure 4. Stakeholder importance vs. influence. Points with identical scores are offset for legibility.

Table 6. Stakeholder importance and influence scores, strategy and engagement approach.

#	Stakeholder	Importance	Influence	Strategy	Engagement approach
1	Senior Design Course	2	5	Keep Satisfied	Meet every deliverable and rubric item; keep the traceability record current.
2	CSU IREC Team	5	5	Manage Closely	Weekly systems review; one owner for requirement IDs.
3	2028 Team (Liquids)	2	1	Monitor	Keep documents and models in a shared, maintained location.
4	CSU Engineering Faculty	2	2	Monitor	Report safety status at each design review.
5	CSU ERC & EMEC	4	5	Manage Closely	Review each layup and machining plan with shop staff before work starts.
6	ESRA	1	5	Keep Satisfied	Track the 2027 rules; submit every document on time.
7	Range Safety Officers	1	5	Keep Satisfied	Prepare stability, recovery and safety evidence before flight-readiness review.

#	Stakeholder	Importance	Influence	Strategy	Engagement approach
8	Flight Safety Reviewers	1	4	Keep Satisfied	Keep the vehicle inspectable and the configuration traceable.
9	Judges & Technical Reviewers	1	3	Keep Satisfied	Present design rationale and traceable evidence in the technical report.
10	Launch Ops & Pad Managers	3	4	Manage Closely	Confirm rail, pad and procedure compatibility early.
11	General Public	5	1	Keep Informed	Protect through requirements; follow range hazard areas.
12	Sponsors	3	4	Manage Closely	Regular progress and budget updates showing where funds go.

[Scores come from the team's stakeholder analysis in the system-level requirements work [8], rescaled from its 1–9 range to 1–5 as in the team's Figure 4. The earlier table and the PDR slide label the two off-diagonal quadrants the opposite way from the legend. This report uses the standard grid; update the slide to match.]

2. Engineering Requirements & Standards

2.1 Requirement Levels and Types

Requirements in this report sit at three levels. Stakeholder needs (Table 4) are outcomes. System-level requirements (prefix SYS) state what the vehicle as a whole shall do. Subsystem and process requirements (STR, AERO, AV, REC, PT, AB, PAY, INT, ENG, DOC, TST) state what one part of the system or the project shall do, and each traces up to a system requirement, a rule or a need. Each requirement is also one of four types, which tells a reviewer why it exists and what happens to it if the reason changes (Table 7).

Table 7. Requirement types used in this report.

Type	Why the requirement exists	Examples	If the reason changes
Rule-driven	A competition, FAA or university rule sets it. The team has no choice about the limit.	SYS-004, SYS-006, REC-005, STR-008	It changes only when the rule changes (check the 2027 edition).
Stakeholder-driven	A stakeholder need sets it, and the team chose the limit.	SYS-005, ENG-012, DOC-008	Revisit with the stakeholder.
Derived by analysis	Calculation on a higher-level requirement produces it.	SYS-010 (mass cap from SYS-006), AV-013 (from SYS-001 and REC-001)	Recompute when the inputs change.
Derived from a design decision	It exists only because the team chose a design. A different design would not need it.	SYS-013 (proposed), AB-001–007, AV-014	Withdraw it if the decision is reversed (Section 3.4).

The Source column of Table 8 and Appendix A gives the origin of each requirement, and so its type. Section 3.3 lists the derived requirements the decomposition exposed, and Section 3.4 follows one design-decision chain, the airbrakes, from need to verification.

2.2 Stakeholder Needs to “System Shall” Requirements

Each need in Table 4 was converted by asking what measurable condition would show that the need is met, then writing it as one “shall” statement with a limit, a source and a verification method. Statements follow the form “The system shall [action or condition] within [measurable limit]” so that two engineers read them the same way. Each requirement ID starts with a prefix for its subject: SYS (system level), STR (structures), AERO (aerodynamics and stability), AV (avionics), REC (recovery), PT (propulsion), AB (airbrakes), PAY (payload), INT (interfaces), ENG (engineering process), DOC (documentation and compliance) and TST (test provisions). The Systems Engineer owns the IDs so that each one stays unique and traceable. Table 8 shows the requirements that answer the needs, grouped by topic. The first column gives the need code (N1 to N12, matching the # column of Table 4) that each requirement answers. The complete list of 112 requirements (105 carried from the System-Level Requirements [8] and seven

proposed) is in Appendix A. The first groups in the table are the baseline: what any IREC entry must meet. The last group is derived from the apogee-control decision (Section 3.4).

Section numbers cite the 2027 REV P1 R&R and DTEG [1], [2]. A dagger (†) marks wording or sources changed from the System-Level Requirements [8], which this report revises; "proposed" marks a requirement not in that baseline (Table 9).

Table 8. Stakeholder needs converted to "system shall" requirements.

Need	The system shall…	ID	Source	Verification
Performance and propulsion				
N2, N9	The vehicle shall achieve an apogee between 7,000 ft and 13,000 ft AGL, with a nominal target of 10,000 ft AGL.	SYS-001†	R&R-6.6.1	Test
N2	The vehicle shall achieve a liftoff thrust-to-weight ratio of at least 5:1.	SYS-005	Team, launch safety	Analysis
N7, N10	The vehicle shall achieve a minimum launch-rail exit velocity of 25 m/s.	SYS-006	D&T-5.3.1	Analysis
N6, N8	The vehicle shall use a COTS-certified solid rocket motor permitted by the applicable IREC competition category.	SYS-004†	D&T-5.1.1	Inspection
N6, N7	The propulsion system shall have a total impulse of no more than 40,960 N·s.	PT-008 (proposed)	FAA JO 7400.2R §31-1-5	Inspection
N2	The propulsion system shall provide sufficient total impulse to enable the vehicle to satisfy the required apogee performance.	PT-001	SYS-001	Analysis
Stability				
N6, N7, N8	The vehicle shall maintain a static stability margin of at least 7.5% of vehicle length while subsonic and at least 10% while supersonic, from launch through drogue deployment.	SYS-007†	D&T-10.2.1	Analysis
N7	The vehicle shall maintain a static stability margin no greater than 18% at launch and 25% during flight.	SYS-008†	D&T-10.3.1	Analysis
Recovery				
N7, N11	The recovery system shall initiate the drogue recovery event at or within 100 ft of apogee.	REC-001†	D&T-6.3.1; ±100 ft is team-derived	Test
N7, N11	The recovery system shall produce a drogue descent rate between 20 m/s	REC-002	D&T-6.3.3	Test

Need	The system shall…	ID	Source	Verification
	and 40 m/s at main-parachute deployment.			
N7, N11	The recovery system shall initiate main-parachute deployment at or below 1,500 ft AGL.	REC-003	D&T-6.4.1	Test
N2, N11	The recovery system shall produce a main-parachute landing descent rate below 11 m/s (36 ft/s).	REC-004	D&T-6.4.3	Test
N7, N11	The recovery system shall provide two functionally independent deployment channels.	REC-005	D&T-6.9.1	Inspection
N11	The recovery system shall prevent unintended deployment before the intended recovery event.	REC-008	Recovery safety	Test
Payload and structures				
N6, N9	The vehicle shall carry a payload with a mass of at least 2.0 kg (4.4 lb).	PAY-006 (proposed)	R&R-3.4.1.1	Inspection
N2, N7	The primary structure shall withstand all predicted flight loads with a structural factor of safety of 3.	STR-001	D&T-8.2.1; NASA-STD-5001B	Analysis
N7	The vehicle shall have a predicted fin flutter velocity of at least 150% of the maximum expected vehicle velocity.	AERO-003	D&T-8.2.2, 8.2.3	Analysis
N7	The vehicle shall have a fin semi-span of at least 0.8 times the largest airframe diameter.	AERO-002	D&T-10.2.4	Inspection
Safety, process and programme				
N11	The vehicle shall complete a controlled flight from launch through recovery without loss of the vehicle or required flight functions.	SYS-002	Stakeholder safety need	Test
N4, N6, N11	The vehicle shall remain within all applicable IREC, launch-range, and university safety constraints during launch and flight operations.	SYS-009	R&R; DTEG; range	Inspection
N7, N8	The avionics system shall provide an unambiguous indication of flight-critical system readiness before launch.	AV-008	Flight Readiness Team	Demonstration
N4, N5	All fabrication shall follow the facility safety protocols and equipment procedures of the CSU ERC and EMEC.	ENG-011 (proposed)	CSU ERC/EMEC rules	Inspection

Need	The system shall…	ID	Source	Verification
N12	Total project expenditure shall not exceed the $20,000 baseline budget.	ENG-012 (proposed)	Team budget	Inspection
N1, N6, N8, N9	The team shall submit all required IREC competition documentation by the applicable competition deadlines.	DOC-001	R&R-4.1.1 to 4.1.3	Inspection
N1	Requirements, design elements, interfaces, and verification evidence shall maintain traceability throughout the project.	ENG-010	MBSE / INCOSE practice	Inspection
N3	The team shall maintain documentation sufficient to support transfer of the vehicle design and engineering knowledge to future project teams.	DOC-008	Future-team need	Inspection
Derived from the apogee-control decision (Section 3.4)				
N2	The vehicle shall provide a means of controlling apogee after motor burnout so that apogee error stays within ±500 ft of 10,000 ft AGL (TBR).	SYS-013 (proposed)	Derived from SYS-001 and measure M1 by the apogee-control decision	Analysis
N2	The airbrake system shall control aerodynamic drag during the designated portion of flight to support the vehicle's apogee objective.	AB-001†	SYS-013 (proposed)	Analysis
N7, N11	The airbrake system shall prevent unintended aerodynamic deployment before the defined active-control flight state.	AB-003†	D&T-7.4.1	Test
N7, N11	The airbrake system shall transition to a defined safe state following loss of valid control data or control power.	AB-004†	D&T-7.3.1	Test
N6, N7	The airbrake system shall not reduce the vehicle's stability margin below the applicable requirement during operation.	AB-005†	D&T-7.2.1, D&T-10.2.1	Analysis

Table 9 lists every change this report proposes to the System-Level Requirements [8]. Values set for the previously open requirements (SYS-012, AV-005, AV-006, AV-012, AB-002, STR-009, PAY-002 and INT-003) are listed with their sources in Appendix K (Table K-1). The mass cap in SYS-010 is derived in Section 3.3 and Appendix G.

Table 9. Requirements added or revised relative to the System-Level Requirements [8].

ID	Change	Reason
SYS-001†	Source changed from the 2026 R&R §2.1 to R&R-6.6.1.	The apogee window and scoring are in R&R-6.6.1 and R&R-3.1.2 (Table 3-1) of the 2027 REV P1 R&R.
SYS-003†	Source changed from R&R §2.1 to D&T-5.1.1.	IREC permits multistage vehicles (two-stage categories in R&R Table 3-1); single-stage is our choice.
SYS-004†	Source changed from R&R §2.1 to DTEG §5.1.1.1.	D&T-10.2.1 sets 7.5% subsonic and 10% supersonic, from launch through motor burnout.
SYS-007†	Add 10% minimum while supersonic; measure “from launch”.	DTEG §10.2.1 sets 7.5% subsonic and 10% supersonic.
SYS-010	Keep the cap open until CDR: the mass budget is outside the scope of a PDR. About 28 kg (62 lb) is a screening value used in Appendix D.3, not a requirement.	The 25 m/s rail-exit requirement sets the mass limit. See Section 3.3.
REC-001†, AB-003†, AB-004†, AB-005†	Sources made specific; stale “existing SYS-0xx” pointers removed from Source columns.	Cited IDs pointed to unrelated requirements.
AB-001†	Source changed from SYS-001 to SYS-013 (proposed).	AB-001 exists because of the airbrake decision; SYS-013 states the need it serves.
PT-008 (new)	The propulsion system shall have a total impulse of no more than 40,960 N·s.	FAA Class 2 limit [3]; team survey table uses the same cap.
PAY-006 (new)	The vehicle shall carry a payload with a mass of at least 2.0 kg (4.4 lb).	R&R-3.4.1.1 floor (4.4 lb; 95% accepted without penalty, R&R-3.4.1.5); PAY-002 now has capacity values (Appendix K).
ENG-011 (new)	All fabrication shall follow CSU ERC and EMEC facility safety protocols.	Stakeholder 5 had no requirement.
ENG-012 (new)	Total project expenditure shall not exceed the $20,000 baseline budget.	Stakeholder 12 and MoE M4 had no requirement.
AV-013 (new)	The avionics system shall measure altitude with an uncertainty of no more than ±50 ft up to 13,000 ft AGL (initial allocation, TBR).	Derived from SYS-001 and REC-001. See Section 3.3.
AV-014 (new)	The avionics system shall detect motor burnout and shall not enable airbrake deployment before burnout is detected.	Derived from AB-003. See Section 3.3.
SYS-013 (new)	The vehicle shall provide a means of controlling apogee after motor burnout so that apogee error stays within ±500 ft of 10,000 ft AGL (TBR).	Derived from SYS-001 and measure M1 by the apogee-control decision (Section 3.4); it is the parent of AB-001 to AB-007.

2.3 Standards to Requirements

Table 10 lists the standards and governing documents behind the requirements. IREC's Rules & Requirements and DTEG, the FAA rule through ESRA's launch waiver, and CSU facility rules are mandatory. NASA-STD-5001B, SAE ARP4754B, INCOSE and NASA systems engineering guidance are adopted as good practice. They are written for spacecraft, civil aircraft and general systems work, so we apply them by analogy and say so.

Table 10. Standards and documents converted to requirements.

Standard or document	What it covers and how we use it	Requirement IDs
IREC Rules & Requirements (2027 REV P1, effective Oct. 15, 2026) [1]	Apogee window and scoring (R&R-6.6.1); payload minimum mass (R&R-3.4.1.1); category and motor rules (R&R-3.1.2, D&T-5.1.1); documentation deadlines (R&R-4.1.1 to 4.1.3; IMS [12]).	SYS-001, SYS-004, PAY-006, DOC-001; AV-001–004 (as cited in the System-Level Requirements [8])
IREC DTEG (2027 REV P1, effective Oct. 15, 2026) [2]	Motor and rail (§5), recovery and avionics (§6), airbrakes (§7), structures (§8), stability (§10).	SYS-004, SYS-006–008, REC-001–007, REC-010–014, AV-009–011, AB-003–005, STR-001, STR-004, STR-008, AERO-001–003
FAA JO 7400.2R, Ch. 31 [3]	Amateur rocket classes: Class 2 (high power) is limited to 40,960 N·s total impulse (§31-1-5). Waiver and launch conditions: cloud cover at most five-tenths, visibility at least five miles, no launch into cloud, and separation from uninvolved people of at least the greater of one-quarter of the maximum altitude or 1,500 ft (§31-2-2). ESRA holds the waiver; the vehicle must fit inside it.	PT-008 (proposed). Wind limit SYS-012 and the range recovery area are still open.
NASA-STD-5001B with Change 3 [4]	Minimum structural design and test factors of safety for spaceflight hardware. For composite structures the minimum ultimate factor is 1.4 for uniform material and 2.0 at discontinuities (prototype approach). It excludes solid propellant and engines. We use it to set and defend the project factor of safety; the project factor of 3 exceeds these minima.	STR-001, STR-004; closes the TBR "structural factor of safety" (set at 3)
SAE ARP4754B (2023; replaces ARP4754A) [5]	Process guidance for developing civil aircraft and systems: capturing and validating requirements, verifying the implementation, and flowing derived requirements down. It excludes structures and detailed software and hardware development. We use its validate-then-verify structure.	ENG-001, ENG-002, ENG-009, ENG-010, DOC-005, TST-007
INCOSE Systems Engineering Handbook [6]; NASA SE Handbook, App. C [7]	Requirement quality rules and the definition of a derived requirement (Section 3.3).	ENG-010, DOC-002; AV-013, AV-014
CSU ERC / EMEC facility safety rules	Training, PPE, layup and machining procedures for the team's in-house composite fabrication.	ENG-011 (proposed)
CSU department and university safety policy	Conduct of student projects, including testing and launch activity	SYS-009
NASA Systems Engineering Handbook [7]	Generic aerospace PDR content and systems engineering process: requirements traceability, interfaces, budgets and	ENG-001, ENG-002, DOC-003

and NPR 7123.1C [11]	margins, risk, verification planning. We use their PDR content as a checklist for this report (Appendix J), by analogy to a spacecraft programme.	

2.4 Verification Methods

Every requirement has one verification method. **Test** operates the item under controlled or representative conditions and measures the result. **Analysis** uses calculation, simulation or a model (OpenRocket, Ansys, CFD) to show compliance where flight conditions cannot be reproduced on the ground. **Inspection** examines hardware, documents or records against the requirement. **Demonstration** operates the item and observes that it performs the function, with a pass or fail result and no measurement.

We assign Analysis to apogee-related and stability requirements because they cannot be measured directly on the ground. An analysis is only as good as its inputs, so each one must be fed with measured values, not estimates. For example, SYS-007 (stability margin) is verified by analysis, but the analysis needs the as-built mass and center of gravity (CG), which come from the weighing and balance requirements ENG-003, ENG-004 and TST-004/005. Of the 112 requirements, 39 use Inspection, 36 Analysis, 34 Test and 3 Demonstration (Appendix A and Appendix I).

Table 11 shows how the Test requirements will be run. For each group it gives the facility, what is measured, the acceptance criterion, who runs the test and when. Values marked TBR or [confirm] are still to be set by the owners. Rows marked proposed were added so that every one of the 34 Test requirements belongs to a group; their facilities and criteria are drafts for the owners to confirm.

Table 11. Test requirements and how they will be run.

Group	Requirements	Facility	What is measured	Acceptance criterion	Run by	When
Structural	STR-001 to 004, 006, 007, 010	ERC/EMEC	Failure load of layup coupons and an airframe section; pull load on recovery attachments; laminate voids	Failure load at or above predicted maximum load × 3; attachments hold the peak recovery load × 3 (load calculated at CDR); no voids beyond the limit (TBR)	Structures subteam; Safety/Test Engineer witnesses	Coupons before layup; section test before FS1
Recovery	REC-001 to 004, 007, 008, 010; TST-001, 006	Bench; vacuum chamber; supervised ejection-test site	Each channel fires at its event; separation at the calculated charge; deployment timing, descent rates, landing speed	Each channel fires alone; no output before arming; drogue within 100 ft of apogee; main at or below 1,500 ft AGL; drogue 20–40 m/s; landing under 11 m/s	Recovery team; Safety/Test Engineer	Bench and vacuum before SS1; ejection test before each flight; descent data from SS1, FS1, FS2
Avionics	AV-001 to 007, 009, 010, 013	Electronics bench; vacuum	GPS fix time; altitude against a reference; circuit	GPS fix at least 10 min before launch; altitude record	Avionics subteam	Bench before SS1; integrated

Group	Requirements	Facility	What is measured	Acceptance criterion	Run by	When
	(proposed), 014 (proposed); INT-002; TST-003, 008	chamber; vibration and thermal GPS station	continuity; connector retention; event timing; battery runtime; altitude uncertainty against a reference; burnout detection time	survives landing; no disconnects under vibration; battery covers the mission with margin (TBR); altitude within ±50 ft (TBR); burnout detected before any airbrake enable		test before each flight
Propulsion and mass properties	PT-004; SYS-010; ENG-003, 004; TST-004, 005	ERC scale and CG fixture; retainer load fixture; launch rail	Motor retention load; as-built mass; CG; margin from measured CG; rail-exit speed; thrust-to-weight	Retainer holds maximum thrust × 3; mass at or below the cap (TBD); margin at least 7.5%, within 18% at launch and 25% in flight; T/W at least 5:1; rail exit at least 25 m/s	Propulsion subteam; Flight Readiness Team	Mass and CG before every flight; retainer test before FS1
Airbrakes (proposed)	AB-003, AB-004, AB-006	ERC bench; wind tunnel	Airbrake stays retracted until the active-control state; moves to the safe state on loss of data or power; mechanism stays attached under load	No motion before burnout is detected; safe state reached within the TBR time; no detachment at the TBR load	Airbrake subteam; Safety/Test Engineer	Bench before FS1; wind tunnel before CDR
Payload (proposed)	PAY-003	ERC bench; vibration	Payload mount under flight and recovery loads	No payload movement beyond the TBR limit	Payload subteam	Before FS1
Flight (proposed)	SYS-001, SYS-002	Subscale and full-scale flights	Apogee against the window; completion of a controlled flight with both recovery events	Apogee inside 7,000 to 13,000 ft; both events occur; vehicle recovered intact	Flight Readiness Team	SS1, FS1, FS2, competition flight

2.5 Measures of Effectiveness

Measures of effectiveness (MoEs) describe project success from the stakeholders' point of view, not component performance. Table 12 gives six: apogee accuracy (team and judges), stability (RSOs and ESRA), mass (launch operations and the rail-exit limit), cost (sponsors), schedule (the course and faculty), and safe recovery (the public and RSOs). Each has a success target and the risk that threatens it.

Table 12. Measures of effectiveness.

ID	Measure	Success target	Risk to success	Linked requirements
M1	Apogee error vs. SYS-001	Apogee within ±500 ft of 10,000 ft AGL.	Variability in flight prediction puts apogee outside the target.	SYS-001, SYS-013 (proposed), AB-001
M2	Stability margin vs. SYS-007/008	Margin stays at least 1.0 percentage point inside every limit: 8.5% or more subsonic, 11% or more supersonic, 17% or less at launch, 24% or less in flight.	CG and CP shift pushes stability outside the margin and breaks pre-flight rules.	SYS-007, SYS-008, AERO-005
M3	Liftoff mass vs. SYS-010	Liftoff mass at or below the SYS-010 cap, set by the 25 m/s rail-exit limit (preliminary 28 kg, TBR), with thrust-to-weight of at least 5:1.	Excess mass lowers rail-exit speed below 25 m/s and breaks pre-flight rules.	SYS-005, SYS-006, SYS-010
M4	Cost vs. allocated budget	Total cost within the $20,000 budget.	Travel and part costs run over and cut project scope.	ENG-012 (proposed)
M5	Schedule risk vs. CDR and flight test	All design work complete 2 weeks before CDR and the following flight test.	Missed deadlines lead to a failed CDR and rushed flight tests.	DOC-003, DOC-004
M6 (proposed)	Safe recovery	Both recovery events occur, landing speed is below 11 m/s, and the vehicle is recovered intact on the competition flight.	Missed or early deployment causes a ballistic descent and vehicle loss.	REC-001–005, REC-008

M1 (apogee accuracy) has the clearest effect on score: under the 2027 scoring formula, a 500 ft miss costs about 58 of the 350 apogee points [1] (Appendix G), so even small errors matter. M6 (safe recovery) is proposed because recovery is part of the flight score and is the main safety control for the public, yet it was not among the five measures in the System-Level Requirements [8].

3. Functional Decomposition

3.1 Function Tree

Figure 5 shows the functional decomposition [17] the team developed from the top function, **Perform Mission Flight.** It has ten Level 1 functions and 41 Level 2 functions; the figure stops at Level 2. The 90 Level 3 functions are in the team's functional decomposition deck and are not repeated here. A function says what the system must do, not how it is built. Each name is a verb and a noun, with no part or technology in it (for example, Modulate Drag rather than Deploy Airbrake Flaps), so the same tree holds for any concept the trade studies in Section 4 might select.

The decomposition stops at a function when four things are true: it can be written as a requirement, it can be verified, one owner can be assigned to it, and more than one concept could still perform it. Assembly, transport, inspection, range clearance and retrieval are left out on purpose. They are things people do around the vehicle, so they belong in the concept of operations (Figure 1) and not in the vehicle's functions.

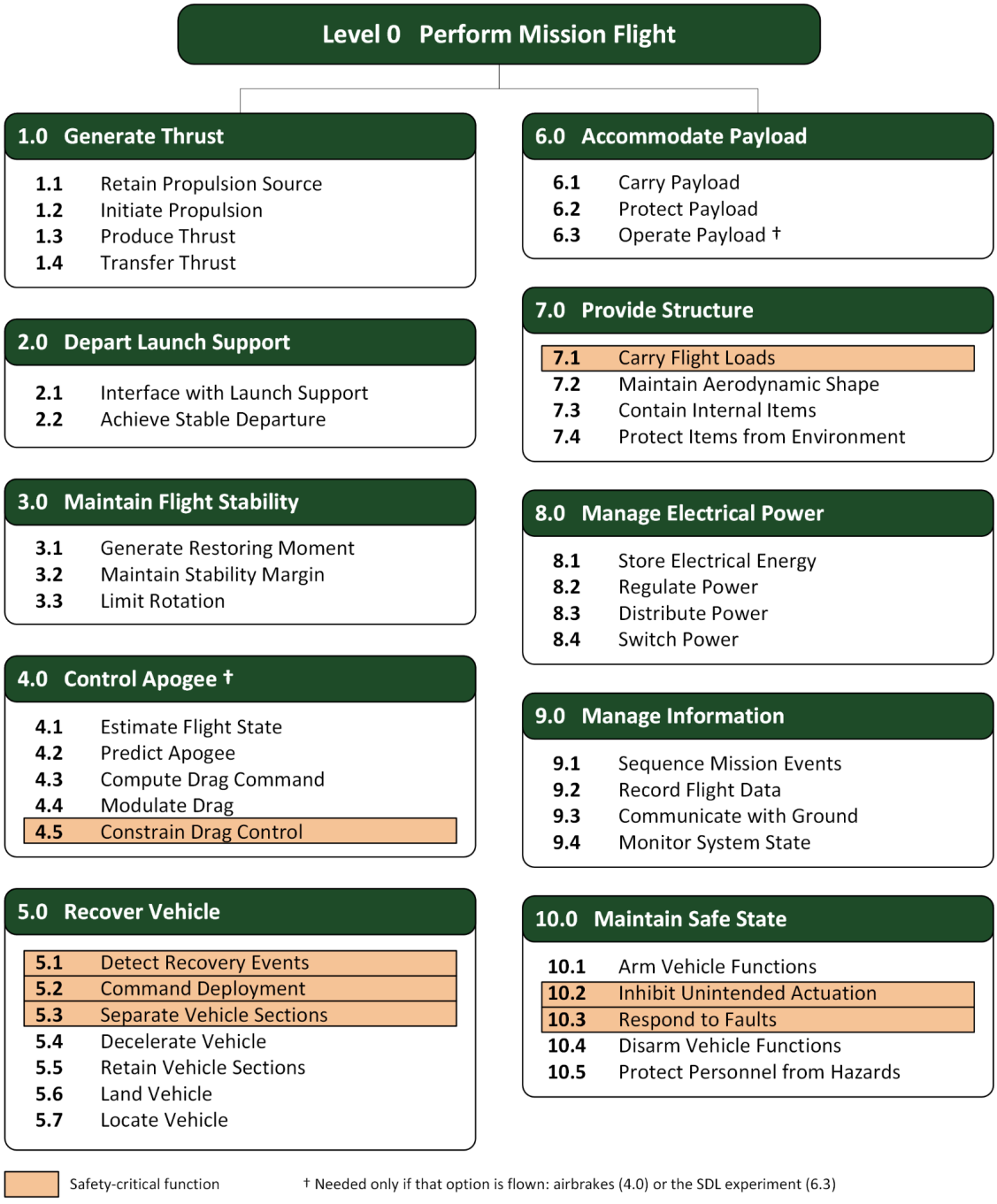


Figure 5. Functional decomposition to Level 2. Orange rows are safety-critical functions; † marks functions needed only if the option is flown (airbrakes or the SDL experiment).

Seven Level 2 functions are safety-critical because a failure in them can lose the vehicle or endanger people: **Constrain Drag Control** (4.5), **Detect Recovery Events** (5.1), **Command Deployment** (5.2), **Separate Vehicle Sections** (5.3), **Carry Flight Loads** (7.1), **Inhibit Unintended Actuation** (10.2) and **Respond to Faults** (10.3). Section 5.4 describes how the design protects each of them. Their verification gets the earliest and most careful planning, and their hazards are rows in Appendix F.

Only two parts of the tree are optional. IREC does not require Control Apogee (4.0): a vehicle with no way to trim apogee can still fly. It is on the tree because apogee is scored by its distance from the target, so the system benefits from controlling it. Airbrakes are the solution we carry for it (Section 4.2), and the airbrake requirements follow from that choice (Section 3.4). Operate Payload (6.3) is optional in the same way: the baseline payload only has to be carried and protected (6.1 and 6.2), and the SDL experiment is a design decision. Every other function is needed by any IREC entry.

3.2 Function-to-Requirement Mapping

Table 13 is the function dictionary and the requirement mapping in one table. It gives each function a one-line statement of what must happen, marks the safety-critical ones, and lists the requirements from Appendix A that govern it. Every one of the 112 requirements (105 carried from the System-Level Requirements [8] and seven proposed) maps to at least one function, to the system as a whole, or to the programme and process row. The mapping also works in the other direction: a function with no requirement is something the system must do that nobody has yet specified. 9 of the 41 Level 2 functions have no requirement in the System-Level Requirements [8]. Two of them (4.1 and 4.2) are covered only by the proposed AV-013 and AV-014. Table 14 lists the gaps and the requirement each one needs.

Table 13. Function dictionary and function-to-requirement mapping. Shaded olive rows are Level 1; orange rows are safety-critical; † marks optional functions.

ID	Function	What must happen	Crit.	Requirements
1.0	**Generate Thrust**	Produce the thrust that accelerates the vehicle to the speed and altitude the mission needs.		See Level 2
1.1	Retain Propulsion Source	Hold the propulsion energy source in the vehicle and limit its effects on the surrounding structure.		STR-010, PT-004, PT-005
1.2	Initiate Propulsion	Accept ignition energy and start the release of propulsion energy.		None yet
1.3	Produce Thrust	Convert released energy into directed thrust for the required duration.		SYS-001, SYS-003–005, PT-001–003, PT-008 (proposed)
1.4	Transfer Thrust	Pass thrust into the vehicle so that the whole vehicle accelerates.		PT-007
2.0	**Depart Launch Support**	Leave the launch support in a controlled and stable way.		See Level 2
2.1	Interface with Launch Support	Mate with the launch support, hold the launch orientation, and release cleanly.		PT-006

ID	Function	What must happen	Crit.	Requirements
2.2	Achieve Stable Departure	Reach a speed and attitude at departure that allow stable flight.		SYS-005, SYS-006, SYS-012
3.0	**Maintain Flight Stability**	Keep the vehicle's attitude predictable throughout flight.		See Level 2
3.1	Generate Restoring Moment	Create a moment that opposes disturbances in angle of attack.		AERO-002, AERO-004
3.2	Maintain Stability Margin	Keep the balance between center of mass and center of pressure within safe limits.		SYS-007, SYS-008, AERO-001, AERO-005, AB-005, PAY-004, INT-005, ENG-003–005, TST-004, TST-005
3.3	Limit Rotation	Limit unwanted rotation about every axis.		AERO-006
4.0	**Control Apogee †**	Adjust the vehicle's energy so that apogee lands on the target.		See Level 2
4.1	Estimate Flight State	Measure motion and attitude and produce a trusted estimate of vehicle state.		AV-013 (proposed)
4.2	Predict Apogee	Determine when coasting begins and project where the trajectory will peak.		AV-014 (proposed)
4.3	Compute Drag Command	Decide how much drag change is needed to move predicted apogee toward the target.		SYS-001, SYS-013 (proposed)
4.4	Modulate Drag	Change the vehicle's drag on command and confirm that the change occurred.		AERO-005, AB-001, AB-002, AB-006, SYS-013 (proposed)
4.5	Constrain Drag Control	Limit when and how drag control may act so that it cannot harm the flight.	Y	AB-003, AB-004, AB-007
5.0	**Recover Vehicle**	Slow the vehicle after apogee, bring it to the ground intact, and help the team find it.		See Level 2
5.1	Detect Recovery Events	Recognize the flight conditions that call for each descent event.	Y	AV-007, REC-001, AV-013 (proposed)
5.2	Command Deployment	Issue each descent command at the right time, with redundancy.	Y	REC-001, REC-003, REC-005–007, TST-001, TST-006
5.3	Separate Vehicle Sections	Divide the vehicle into the sections needed for descent and confirm it happened.	Y	STR-007
5.4	Decelerate Vehicle	Reduce speed in stages to the initial and landing descent rates while limiting drift.		REC-002, REC-004
5.5	Retain Vehicle Sections	Keep separated sections together and limit deployment shock.		STR-004, STR-005, REC-009, REC-010

ID	Function	What must happen	Crit.	Requirements
5.6	Land Vehicle	Bring the vehicle to rest while limiting landing loads and post-landing drag.		REC-004
5.7	Locate Vehicle	Determine where the vehicle landed and make that position available outside the vehicle.		AV-001, AV-002, REC-014
6.0	**Accommodate Payload**	Carry, protect, and operate the payload throughout the mission.		See Level 2
6.1	Carry Payload	Mount the payload, keep it attached through landing, and allow its removal.		PAY-001–003, PAY-005, PAY-006 (proposed)
6.2	Protect Payload	Limit shock, vibration, and environmental exposure at the payload.		None yet
6.3	Operate Payload †	Power, command, and collect data from the payload.		None yet
7.0	**Provide Structure**	Hold the vehicle together and give it the shape it needs in every load condition.		See Level 2
7.1	Carry Flight Loads	Resist the loads of every flight event without failure.	Y	STR-001–003, STR-006, STR-008, AERO-003
7.2	Maintain Aerodynamic Shape	Keep the outer profile that the flight depends on, including under load.		SYS-011
7.3	Contain Internal Items	Locate and hold everything carried inside the vehicle.		STR-009, REC-012, INT-001, INT-003
7.4	Protect Items from Environment	Shield contents from heat, vibration, and contamination.		STR-010, AV-006, REC-011
8.0	**Manage Electrical Power**	Supply electrical energy to every onboard function.		See Level 2
8.1	Store Electrical Energy	Hold enough electrical energy for the whole mission.		AV-011, AV-012
8.2	Regulate Power	Keep supplied power within limits that every user can accept.		None yet
8.3	Distribute Power	Deliver power to each onboard user.		AV-009, AV-010, INT-002
8.4	Switch Power	Turn power to each user on or off on command.		None yet
9.0	**Manage Information**	Sense, decide, record, and communicate what the mission needs.		See Level 2
9.1	Sequence Mission Events	Know where the mission stands and trigger the actions that go with it.		None yet
9.2	Record Flight Data	Capture flight data and keep it available after landing.		AV-003–005, DOC-007, TST-008

ID	Function	What must happen	Crit.	Requirements
9.3	Communicate with Ground	Exchange data and commands with people and systems outside the vehicle.		AV-002
9.4	Monitor System State	Check that the vehicle is healthy and make its status known.		AV-008, TST-002, TST-003
10.0	**Maintain Safe State**	Keep the vehicle and the people around it safe in every state, including during faults.		See Level 2
10.1	Arm Vehicle Functions	Enable the hazardous or safety-critical functions only when they are needed.		AV-008
10.2	Inhibit Unintended Actuation	Prevent any hazardous function from acting at the wrong time.	Y	STR-007, REC-008, AB-003
10.3	Respond to Faults	Move the vehicle to a safe condition when something goes wrong.	Y	REC-013, AB-004, INT-004, ENG-008
10.4	Disarm Vehicle Functions	Remove the hazard from hazardous functions once the mission ends.		None yet
10.5	Protect Personnel from Hazards	Limit the hazards the vehicle presents to people near it.		SYS-009
—	System level	Properties of the whole vehicle that no single function owns.		SYS-002, SYS-009, SYS-010, INT-006
—	Programme and process	Requirements on the project, not the vehicle (engineering, documentation and test provisions).		ENG-001, ENG-002, ENG-006, ENG-007, ENG-009, ENG-010, DOC-001–006, DOC-008, TST-007, ENG-011, ENG-012 (proposed)

The gaps in Table 14 are mostly in functions that the baseline vehicle needs and the rules touch only lightly: starting the motor, powering and switching the electronics, deciding when each event fires, and making the vehicle safe after landing. Closing these gaps is the next work item for Section 2 and Appendix A. The requirement text below is a topic, not final wording; each owner writes the measurable limit and chooses the verification method.

Table 14. Functions with no requirement in the Master SRR.

Function	Why the system needs a requirement	Requirement to write	Owner
1.2 Initiate Propulsion	Ignition starts the only source of thrust, and the launch rules govern ignition and arming.	Ignition circuit stays inert until the vehicle is armed; igniter continuity is confirmed before launch (ICD-007).	Propulsion and avionics
6.2 Protect Payload	The payload must survive boost, deployment shock and landing; the mount loads depend on this.	Shock and vibration limits at the payload, and its environment, with a limit and a test.	Payload

Function	Why the system needs a requirement	Requirement to write	Owner
6.3 Operate Payload †	The SDL experiment has no requirements of its own; PAY-001 to PAY-005 cover only the generic interface.	Experiment power, data recording, ground test and success criteria, written when the experiment is confirmed.	Payload
8.2 Regulate Power	AV-012 covers stored energy but not the voltage each flight computer and the servo accept.	Supply voltage at each flight-critical user stays inside its limit for the whole mission.	Avionics
8.4 Switch Power	Arming from outside the vehicle is how the pad crew keeps charges and the airbrake inert until needed.	Each flight-critical circuit can be switched on and off from outside the airframe (check the DTEG arming rules).	Avionics
9.1 Sequence Mission Events	REC-001 and REC-003 set when events must occur, but no requirement says what logic issues the commands.	Each event command is issued only in its defined flight state, including the burnout lockout for the airbrakes.	Avionics
10.4 Disarm Vehicle Functions	After landing, unfired charges and powered circuits still present a hazard to people.	Hazardous functions are disabled or confirmed inert by a defined method after landing. Procedures live in operations documents.	Recovery and safety

3.3 Derived Requirements

Following INCOSE practice, a derived requirement is a requirement developed through analysis, design decisions, or decomposition of higher-level requirements that is not explicitly stated in the original stakeholder requirements [6]. Table 15 lists the ones the decomposition exposed. Two come from analysis and are not stated in any rule or stakeholder need. The first is the mass cap: the 25 m/s rail-exit requirement (SYS-006) limits liftoff mass, while the 5:1 thrust-to-weight requirement (SYS-005) would allow far more, so SYS-005 is not the binding limit. The preliminary 28 kg cap in SYS-010 comes from a constant-average-thrust estimate (Appendix G). The OpenRocket thrust curve gives about 29 m/s at 28 kg (Section 5.3.4), so the cap is conservative, and the apogee prediction is the tighter limit on mass for this motor. The second is altitude accuracy: the drogue must fire within 100 ft of apogee (REC-001), so the altitude measurement must be accurate to well under 100 ft, or the drogue could fire outside its window even when the sensor is working.

The INCOSE definition also covers requirements that come from design decisions, and the airbrakes are the clearest case. IREC does not require airbrakes, so the seven airbrake requirements (AB-001 to AB-007), the burnout-detection requirement (AV-014) and the system-level requirement SYS-013 that states what they are for exist only because of the apogee-control decision. Section 3.4 follows that chain from need to verification and sets the conditions under which the decision stands. Once the decision is made, DTEG §7 adds its own rules for active control. If the team dropped the airbrakes, these requirements would go with them and the baseline requirements would stay. The payload experiment has no requirements of its own in the Master SRR yet (function 6.3 in Table 14).

Table 15. Derived requirements revealed by functional decomposition.

#	Function	Derived requirement	ID	Derived from	Status
1	2.2 Achieve Stable Departure	Liftoff mass shall not exceed about 28 kg (62 lb) (TBR).	SYS-010	SYS-006 by $v^2 = 2aL$, with M2500T average thrust and a 13 ft effective rail (Appendix G).	Value proposed
2	5.1 Detect Recovery Events (and 4.1)	Altitude shall be measured to within ±50 ft up to 13,000 ft AGL (TBR).	AV-013	SYS-001, REC-001. Event triggers depend on this reading.	Proposed
3	4.2 Predict Apogee	The avionics shall detect motor burnout and not enable airbrakes before it.	AV-014	AB-003 and DTEG §7.4.1.	Proposed
4	5.2 Command Deployment	Two independent channels, each with its own power and trigger; no early actuation.	REC-005–008	DTEG §6.9; hazard analysis.	In SRR
5	10.3 Respond to Faults	A single fault or loss of data or power shall lead to a defined state.	AB-004, REC-013, ENG-008	Single-fault analysis (assumption A2).	In SRR
6	10.2 Inhibit Unintended Actuation	No airbrake or recovery actuation before its defined flight state.	AB-003, REC-008	Flight-event logic.	In SRR
7	3.2 Maintain Stability Margin	Stability shall hold in every commanded airbrake state.	AERO-005, AB-005	SYS-007 with airbrake deployment.	In SRR
8	4.0 Control Apogee †	The vehicle shall control apogee after burnout so that apogee error stays within ±500 ft (TBR).	SYS-013	SYS-001 and measure M1, by the apogee-control decision (Section 3.4).	Proposed
9	4.0 Control Apogee †	All other airbrake requirements follow from SYS-013.	AB-001–007	SYS-013 and DTEG §7 (Section 3.4).	In SRR

3.4 Airbrakes as a Derived Requirement

IREC does not require airbrakes. This section shows how the airbrake requirements follow from a stakeholder need and the scoring rule, and which conditions must hold for the airbrakes to stay in the design. Each link in the chain can be checked without the OpenRocket numbers, which are preliminary (Section 5.3).

Table 16. Derivation chain for the airbrake requirements.

Step	Statement	Source	Type
1. Need	N2: the CSU IREC Team needs the vehicle to reach its target apogee and recover safely within the schedule and budget (Table 4).	Stakeholder analysis	Need

Step	Statement	Source	Type
2. Rule	Apogee is scored by its distance from 10,000 ft, falling linearly to zero at ±30% [1]. SYS-001 sets the 7,000 to 13,000 ft window.	R&R §2.8.1.4	Rule-driven
3. Measure	M1: apogee within ±500 ft of 10,000 ft AGL. Each 100 ft of error costs 11.7 of the 350 apogee points (Appendix G).	Team measure	Stakeholder-drive n
4. Analysis	A motor-only vehicle cannot change its energy after burnout, so its apogee error is the full prediction error of motor impulse, mass, drag and wind. Meeting M1 against that dispersion needs something that acts after burnout. A fixed offset can be removed before flight (ballast or a lower-impulse motor); only an in-flight means trims the dispersion.	Analysis of SYS-001 and M1	Derived by analysis
5. System requirement	SYS-013 (proposed): the vehicle shall provide a means of controlling apogee after motor burnout so that apogee error stays within ±500 ft of 10,000 ft AGL (TBR).	Steps 2 to 4; architecture A1 (Section 4.2)	Derived from a design decision
6. Subsystem requirements	AB-001 to AB-007 for the airbrake mechanism chosen in Section 4.4.5, with DTEG §7 rules for active control (AB-003, AB-004, AB-005).	SYS-013; DTEG §7	Derived from a design decision
7. Supporting requirements	AV-013 (altitude within ±50 ft), AV-014 (burnout detected before airbrake enable), AERO-005 (stability in every airbrake state); interfaces ICD-004 and ICD-005; hazards HZ-A1 to HZ-A3.	AB-003, AB-005, SYS-001	Derived by analysis and by design decision
8. Verification	SYS-013, AB-001, AB-002: Monte Carlo analysis (RocketPy), CFD and wind tunnel, then flight at FS1 and FS2. AB-003, AB-004, AB-006: test (Table 11).	Table 11; Appendix I	Analysis, test

The chain explains why the airbrake requirements exist. It does not show that airbrakes are worth their cost. That is the question the gates in Table 17 answer. SYS-013 and the airbrake requirements remain in the baseline of requirements only while all five gates hold. If a gate fails, the team withdraws SYS-013, AB-001 to AB-007 and AV-014 and flies the baseline (architecture A0); the baseline requirements do not change.

Table 17. Gates for keeping the airbrakes in the design.

Gate	Condition to keep the airbrakes	Evidence	Owner (proposed)	Due (proposed)
G1 Authority	Monte Carlo analysis shows that full deployment removes at least the baseline overshoot plus the high side of the dispersion in impulse, mass, drag and wind, and holds apogee within ±500 ft in at least a team-set share of runs (TBR).	RocketPy Monte Carlo with the updated mass model	Systems Engineer with airbrake lead	Before CDR
G2 Net benefit	Airbrakes give a better expected apogee score than ballast or a lower-impulse motor once their mass, cost and risk are counted.	Re-run of decision DR-01 with measured data; Appendix G	Systems Engineer	Before CDR

Gate	Condition to keep the airbrakes	Evidence	Owner (proposed)	Due (proposed)
G3 Stability	Static margin stays at or above the SYS-007 floor with the 1.7 kg assembly (the model carries 0.97 kg) in every airbrake state (AB-005, AERO-005).	Updated OpenRocket model, CFD, wind tunnel	Aerodynamics lead	Before CDR
G4 Safety	Hazards HZ-A1 to HZ-A3 are reduced to Medium or accepted in writing by the Safety/Test Engineer with the mitigations in Appendix F.	Updated FMEA and test plan	Safety/Test Engineer	Before CDR
G5 Mass and schedule	Liftoff mass stays under the SYS-010 cap with the airbrakes, and hardware is ready for test before FS1 (March 13, 2027).	Mass budget (Appendix H); schedule (Section 5.8)	Systems Engineer	Design freeze

3.4 Traceability from Need to Verification

The trace runs in one direction: stakeholder need, requirement, function, subsystem, interface, hazard, verification method and test group. Appendix I holds the full 112-row matrix. Table 18 summarizes it by Level 1 function, and Table 19 states how complete the trace is. A coverage check like this is how the team finds requirements that have no parent, functions that have no requirement, and requirements that have no test.

Table 18. Traceability summary by Level 1 function (a requirement counts under every Level 1 function it maps to). † Optional.

Level 1 function	Level 2 functions	With a requirement	Requirements mapped	Safety-critical	Verification: Test / Analysis / Inspection / Demonstration
1.0 Generate Thrust	4	3	12	0	2 / 6 / 4 / 0
2.0 Depart Launch Support	2	2	4	0	0 / 3 / 1 / 0
3.0 Maintain Flight Stability	3	3	15	0	4 / 10 / 1 / 0
4.0 Control Apogee †	5	5	11	1	6 / 5 / 0 / 0
5.0 Recover Vehicle	7	7	19	3	11 / 4 / 3 / 1
6.0 Accommoda te Payload	3	1	5	0	1 / 1 / 3 / 0
7.0 Provide Structure	4	4	14	1	1 / 6 / 7 / 0

Level 1 function	Level 2 functions	With a requirement	Requirements mapped	Safety-critical	Verification: Test / Analysis / Inspection / Demonstration
8.0 Manage Electrical Power	4	2	5	0	3 / 1 / 1 / 0
9.0 Manage Information	4	3	9	0	6 / 0 / 1 / 2
10.0 Maintain Safe State	5	4	9	2	3 / 4 / 1 / 1
System level and programme	—	—	19	—	see Appendix I

Table 19. Coverage check of the trace.

Check	Result	Reading
Needs with at least one requirement	12 of 12	Every need N1 to N12 appears in Table 8.
Requirements listed with a need code in Table 8	31 of 112	The others trace to a rule, a higher-level requirement or a process source (Appendix A, Source column).
Requirements mapped to a function, the system level or the programme	112 of 112	Table 13 and Appendix I.
Level 2 functions with a requirement	32 of 41 in the Master SRR; 34 of 41 with proposed requirements	Seven functions still need new requirements (Table 14).
Level 2 functions allocated to a subsystem	39 of 41	Functions 2.2 and 10.5 belong to the whole vehicle (Table 35).
Requirements with a verification method	112 of 112	Section 2.4 and Appendix A.
Test requirements assigned to a test group	34 of 34 (26 in the original table, 8 proposed rows)	Table 11.
Requirements linked to an interface	26 of 112	Requirements named in the “Traces to” column of Table 36.
Requirements linked to a hazard row	59 of 112	Table 59 in Appendix F.
Hazard rows with a linked requirement	24 of 25	HZ-21 (funding shortfall) has none.
Requirements on safety-critical functions with no hazard link	6 of 23: STR-008, AB-007, INT-004, ENG-008, TST-006, AV-013	Each needs a hazard row or an explanation (action for the Safety/Test Engineer).

4. Concept Generation & Trade Studies

4.1 Concepts and Criteria

Every trade in this report follows the same five steps.

1) **Define the decision** and the function it serves (Figure 5).
2) **List the options**, including the baseline of doing nothing where there is one.
3) **Apply pass/fail gates first**. Requirements every option must meet (stability, DTEG rules, structural factor of safety) are checked before scoring and are not traded for points.
4) **Score the remaining options** against weighted criteria (weights 1 to 5, scores 1 to 10) and label the basis of every score (Table 20).
5) **Test the result and record it**. Move each weight by one point and note whether the winner changes. Write a decision record (Section 4.6) that states the choice, the evidence, the confidence and what would reopen it.

Table 20. Basis labels for scores.

Score basis	Meaning	Evidence a reviewer should expect
Calculated	Taken from a calculation or a model.	Inputs, method and tool version (for example the OpenRocket file or a hand calculation).
Sourced	Taken from a data sheet, quote, survey or past project.	Citation or file; date.
Judgment	An estimate by a named person.	Name, short reason, and what evidence would replace it.

Table 21. Scale anchors for the 1 to 10 scores.

Score	Anchor (proposed for team use)
1–2	Fails the criterion or barely works; a serious workaround is needed.
3–4	Below what a typical option delivers.
5–6	Adequate; meets the need without margin.
7–8	Better than a typical option; some margin.
9–10	Best realistic option; clear margin.

The matrices in Appendix B and Appendix D were scored before these anchors and basis labels existed, so their scores have no recorded basis. Table 48 in Appendix B lists the evidence that would replace each judgment score. The architecture trade in Section 4.2 uses the labels from the start.

4.2 System Architecture Trade

The trade studies in Section 4.3 choose among concepts for each subsystem. Before them there is a higher decision: which capabilities the vehicle carries at all. IREC requires a COTS motor, a stable airframe, dual-deployment recovery and a payload of at least 2.0 kg. It does not require airbrakes or an experiment. We compared four architectures that differ only in those two options (Table 22), all built on the same baseline.

Table 22. System architecture options.

Option	Description
A0 Baseline	The baseline vehicle only: airframe, COTS motor, dual-deployment recovery, avionics, and an inert payload of at least 2.0 kg.
A1 Baseline + airbrakes	A0 plus linear-track airbrakes for apogee control (SYS-013, AB-001 to AB-007, AV-014).
A2 Baseline + SDL experiment	A0 with the thermodynamic-cycle experiment as the payload.
A3 Both options	A0 plus airbrakes plus the SDL experiment.

Table 23. Facts that differ between the architecture options.

Fact	A0	A1	A2	A3
Requirements added	None	9: SYS-013, AB-001 to AB-007, AV-014	Experiment requirements for function 6.3 (not yet written, Table 14)	Both sets
Hazard rows added (Table 59)	None	3 (HZ-A1 to HZ-A3)	1 (HZ-P1)	4
Interfaces added	None	ICD-004, ICD-005	ICD-003 if the payload needs power or data	ICD-003, ICD-004, ICD-005
Mass added (OpenRocket model)	None	0.97 kg in the model; 1.7 kg in Section 5.5.5	Experiment 2.3 to 2.7 kg replaces an inert payload of at least 2.0 kg	Both
Budget line (Table 55)	None	Airbrakes $800	Payload $800 (inert payload not costed separately)	$1,600

Table 24 scores the options. These scores are the author’s draft for team review, and most are judgment. They are shown so that the reasoning can be argued with, and the gates in Section 3.4 exist to replace the judgment with evidence.

Table 24. System architecture decision matrix (weights 1 to 5, scores 1 to 10). Draft scores, proposed for team review.

Criterion (weight)	A0	A1	A2	A3	Basis of the scores
Apogee accuracy (M1) (5)	3	8	3	8	Sourced and judgment: preliminary OpenRocket overshoot (+1,484 ft); RocketPy authority not yet run
SDL opportunity (2)	1	1	8	8	Judgment
Flight safety risk (4)	9	6	8	5	Judgment, from hazard rows added (Table 59)
Mass and stability margin (3)	9	6	7	5	Sourced: 3.88 kg reserve; airbrake mass 0.97 kg in model, 1.7 kg in Section 5.5.5
Schedule risk (4)	9	6	7	5	Judgment
Cost (2)	9	7	8	6	Sourced: budget lines (Table 55)
Weighted total	134	122	128	123	

The baseline A0 has the highest total (134), ahead of A2 (128), A3 (123) and A1 (122). It wins on the criteria where an added capability costs something: flight safety risk, mass and stability margin, schedule risk and cost. Moving one weight by one point changes the winner in 1 of 11 cases (raising the SDL opportunity weight from 2 to 3 makes A2 the winner by one point). The result is more sensitive to the apogee scores than to the weights. If apogee accuracy is scored 9 for the options with airbrakes and 1 for those without, A3 leads with 128 against 124 for A0. The airbrake decision therefore rests on whether the airbrakes can in fact deliver apogee accuracy at an acceptable cost in mass, stability and risk, which is what gates G1 to G5 test (Table 17).

Decision DR-01. Fly the baseline (A0) as the committed design, carry airbrakes and the SDL experiment as gated options, and keep designing toward A3. The airbrakes and the experiment enter the committed design only when their gates pass. This keeps a complete, flyable vehicle at every point in the schedule.

4.3 Concepts and Criteria

Each subsection below names the function the subsystem serves (Figure 5), lists the concepts compared, and states the criteria. Table 25 gives the criteria and weights for every trade. Weights run from 1 to 5 and scores from 1 to 10, and a weighted total is the sum of weight times score. The full matrices are in Appendix B (airframe material, airbrakes, recovery, avionics and payload) and Appendix D (nose cone, fins and motor), and Section 4.4 discusses the results. The four baseline subsystems come first, then the two capabilities added by design decision. Dual-deployment recovery is not traded, because the recovery requirements call for it.

Table 25. Trade studies, concepts compared, and criteria with weights († added by design decision).

Trade	Function	Concepts compared	Criteria (weight)	Matrix
System architecture	1.0–10.0	4: A0 baseline, A1 + airbrakes, A2 + SDL experiment, A3 both (Section 4.2)	Apogee accuracy (5), SDL opportunity (2), flight safety risk (4), mass and stability margin (3), schedule risk (4), cost (2)	Table 24
Airframe material	7.1–7.4	7: carbon fiber, fiberglass (each prepreg or wet layup), commercial carbon and fiberglass tubing, 6061 aluminum	Performance (5), bonus scoring (4), manufacturability (3), customizability (2), cost (1)	Table 43
Nose cone shape	3.1, 3.2, 7.2	6: conical, ogive, ellipsoid, power, parabolic and Haack series	Drag (5), manufacturability (3), stability effect (2)	Table 50
Fin planform	3.1, 3.2	3: clipped delta, trapezoidal, tapered swept	Apogee (5), flutter margin (4), stability (3)	Table 51
Fin cross-section	3.1, 3.2	3: rounded, double wedge, airfoil	Maximum velocity (4), manufacturability (4), stability (3)	Table 52
Motor	1.1–1.4	5 COTS motors (Section 4.3.2)	Being rebuilt (Section 4.4.2)	Appendix D
Parachute type	5.4	3: IRIS Ultralight, printed nylon, rip stop	Performance (4), packability (3), weight (2), cost (1)	Table 45
Avionics bay layout	7.3, 8.3	3: vertical, horizontal, hybrid panel assemblies	Ease of integration (4), compactness (4), manufacturability (3), cost (1)	Table 46
Airbrake mechanism †	4.4	4: linear track, linkage, rack and pinion, no airbrakes	Manufacturability (5), compactness (5), apogee control (5), ease of integration (4)	Table 44
Payload experiment †	6.3	5 SDL experiments (Section 4.3.6)	Scientific value (5), clarity of hypothesis (4), reliability (4), ground testability (2), package and mass difficulty (1)	Table 47

4.3.1 Airframe and Structures

The airframe serves the functions Provide Structure (7.0) and Maintain Flight Stability (3.0). It must carry the flight loads (7.1) and keep the outer shape (7.2) at the lowest practical mass. The team decided to manufacture the airframe in house instead of buying commercial tubing. IREC rewards student researched and developed (SRAD) components, and building the airframe ourselves gives control of the laminate: the number of plies, the fiber orientations and the wall thickness. That lets us fit the structure to the expected flight loads and avoid carrying unneeded weight. The alternative was commercial composite tubing, which saves build time but gives up the SRAD bonus and the control of the laminate.

The team compared prepreg carbon fiber, wet-layup carbon fiber, prepreg fiberglass, wet-layup fiberglass, commercial carbon and fiberglass tubing, and 6061 aluminum against five criteria: performance, bonus scoring, manufacturability, customizability and cost. Performance has the highest weight (5) because a structural failure in flight would end the mission. Bonus scoring has a weight of 4 because SRAD components raise the competition score. Manufacturability has a weight of 3, customizability 2 and cost 1. Cost matters, but structural performance and successful in-house fabrication come first for the airframe. The nose cone shape (six concepts), fin planform (three) and fin cross-section (three) were compared with the criteria in Table 25; their matrices are in Appendix D.

4.3.2 Propulsion

The propulsion subsystem serves Generate Thrust (1.0). IREC requires a COTS-certified solid motor (SYS-004), so the concepts are candidate motors and not motor designs. The shortlist is the AeroTech M2500T, M1939W, N2220DM and M2400T and the Cesaroni M3400-P. Table 26 shows how often each was flown by past IREC entries in the team's survey of 56 entries, with the average total impulse and the typical predicted apogee those teams reported. The survey vehicles are other teams' vehicles, so the apogees are context, not predictions for ours.

Table 26. Motor candidates and past use in IREC entries.

Motor	Entries (of 56)	Average total impulse (N·s)	Typical predicted apogee in the survey
AeroTech M2500T / M2500T-PS	26 (46%)	9,671	10,000–10,800 ft
AeroTech M1939W / M1939W-PS	5 (9%)	10,482	10,000–10,700 ft
AeroTech M2400 / M2400T	5 (9%)	7,680–7,717	9,900–10,300 ft
Cesaroni M3400-P	1 (2%)	9,995	11,100 ft
AeroTech N2220DM	Not in the survey	Not in the survey	Not in the survey

The criteria come from the requirements: predicted apogee against the 7,000 to 13,000 ft window (SYS-001) from OpenRocket, rail-exit speed against 25 m/s (SYS-006), total impulse against the FAA limit of 40,960 N·s (PT-008) [3], integration with the 98 mm motor mount (PT-004, ICD-006), and cost

and supply. Supply is limited (assumption C4 in Table 3), so the alternates matter as much as the first choice.

4.3.3 Recovery

The recovery subsystem serves Recover Vehicle (5.0). It brings the vehicle and everything inside it down unharmed. A small drogue parachute deploys at apogee to slow the first part of the descent, and a large main parachute deploys lower to bring the vehicle to a slow landing. Black powder charges separate the vehicle at set points to release each parachute. A redundant avionics system determines the altitude and sends the signals that fire the charges at the correct altitudes.

Parachute sizing starts from the descent ranges the competition sets for the drogue and the main and from the vehicle mass without propellant. The team rearranged the drag-force equation (Section 5.5.3) to find the diameters, and then selected several parachutes of those sizes. It compared them on performance, packability, weight and cost. Performance has the highest weight (4) because it sets how fast the parachute opens and how much drag it gives. Packability is next (3) because space in the vehicle is limited. Weight (2) is mainly a matter of material, and extra weight should not be added without need. The parachutes were all in a similar price range, so cost has the lowest weight (1).

4.3.4 Avionics

The avionics subsystem serves Manage Information (9.0), Manage Electrical Power (8.0) and two recovery functions, Detect Recovery Events (5.1) and Command Deployment (5.2). It measures altitude with altimeters and detects flight events such as apogee. It controls the drogue and main parachute deployments through two independent systems for redundancy. It includes GPS tracking to find the vehicle after landing, and it is powered by batteries secured in the avionics bay.

Three layouts for the electronics panel were compared (Table 27). They were scored on compactness, ease of integration, manufacturability and cost. Compactness matters because the bay has little room for the electronics, batteries and wiring. Ease of integration covers how easily the parts are installed and connected, and how easily they can be reached for maintenance, wiring changes and battery replacement before launch. Manufacturability matters because the panels will be 3D printed. Cost has the lowest weight (1).

Table 27. Avionics layout concepts.

Concept	Functional description
Vertical assembly	The electronics attach to a vertical panel. This gives more room for hands-on work.
Horizontal assembly	The electronics attach to a horizontal panel. This is more compact and leaves room for additional electronics.
Hybrid assembly	The electronics attach to both a horizontal and a vertical panel. This keeps the layout compact while leaving enough room to reach and maintain the components.

4.3.5 Airbrakes (derived requirement, Section 3.4)

The airbrake subsystem serves Control Apogee (4.0). IREC awards more points the closer the vehicle gets to 10,000 ft (measure M1), and the OpenRocket model overshoots that target (Section 5.3.1), so a way to remove altitude after burnout is worth considering. The subsystem is designed to meet AB-001 and AB-002 and to help the vehicle reach the IREC target apogee. Four concepts were compared (Table 28). “No airbrakes” is the baseline, so the decision is justified against flying without them.

Table 28. Airbrake concepts.

Concept	Functional description
No airbrakes	A fixed vehicle with no active apogee control. This has the least mass and mechanical complexity but allows no adjustment in flight.
Linear track assembly	A sliding track mechanism extends and retracts the airbrakes.
Linkage assembly	Connected mechanical links extend and retract the airbrakes.
Rack and pinion assembly	A rotating gear and a linear rack extend and retract the airbrakes.

The concepts were scored on ease of integration (4), manufacturability (5), compactness (5) and apogee control (5). Apogee control matters because the mechanism must provide enough drag to adjust the altitude. Stability in every airbrake position (AB-005, AERO-005), response time (AB-002) and the DTEG rules for active control (§7) are pass-or-fail requirements, not scored criteria: any concept must meet them, and they are checked in the CFD and wind-tunnel work.

4.3.6 Payload Experiment (added by design decision)

The payload serves Accommodate Payload (6.0). The baseline payload is mass only: at least 2.0 kg [1]. The SDL Payload Challenge is scored separately and allows a more complex experiment, so we considered adding one. A payload can be ejectable or non-ejectable. The team chose a non-ejectable payload. An ejectable one needs its own recovery system, with a minimum descent speed and a GPS unit, and it adds the risks of entanglement and failure to eject. Those risks and requirements outweighed the benefit. For a non-ejectable payload the requirements are a mass of at least 2.0 kg (PAY-006) and a form factor of 10 × 10 × 30 cm. PAY-003, PAY-004 and PAY-005 cover the safety of the vehicle: the payload must not affect stability, must not interfere with any other subsystem, and must stay mechanically secured in flight. The baseline concept is an inert payload of the required mass. Five experiments were compared (Table 29).

Table 29. SDL payload experiment concepts.

Experiment	Description
Thermodynamic cycle	A small vapor-compression refrigeration system stays attached to the vehicle and runs from before launch through flight. Temperature and pressure sensors at the compressor,

Experiment	Description
	the heat exchangers and the expansion device feed a computer that records them with altitude and acceleration. The data give the coefficient of performance (COP) of the cycle in flight, compared with ground tests.
RF attenuation	Two antennas, one inside the carbon-fiber airframe and one in a reference position, receive the same signal while the system switches between them. The difference in received signal strength, signal-to-noise ratio and packet loss estimates the attenuation caused by the carbon-fiber structure. Ground tests at several frequencies set a baseline.
Internal flight environment mapping	A central data logger and several small sensor modules at different positions inside the vehicle record temperature, pressure, acceleration and vibration at the main flight events. After recovery the data form a map of the conditions at each location.
Atmospheric particulate sampling	A particle sensor behind a small flush inlet measures airborne particle concentration. The computer records it with altitude, pressure, temperature and time, to plot a vertical profile after recovery.
Soil radiation sampling	The payload self-rights after landing, deploys a drill to collect up to 4 cubic inches of soil, and measures its radiation. It compares this with the ambient air radiation measured at apogee and at landing, and grades the landing site against “normal” levels.

The five criteria are scientific value, clarity of the hypothesis, ground testability, reliability, and package and mass difficulty. They map onto the main categories of the SDL rubric: scientific value and clarity of the hypothesis to the scientific or technical objective, package and mass difficulty and reliability to payload construction, and ground testability to the execution of the objectives.

4.4 Trade Studies

This section gives the result of each trade and how far it can be trusted. For each trade we state the winner and the lead over the runner-up, and whether the result holds when any one weight moves by one point. All scores are on the same scale: weights 1 to 5, scores 1 to 10, and the weighted total is the sum of weight times score. The full matrices are in Appendix B and Appendix D. Table 30 in Section 4.5 collects the results.

4.4.1 Airframe and Structures

A weighted decision matrix (Table 43) compared the airframe materials and manufacturing methods. Prepreg carbon fiber scored highest at 121, ahead of prepreg fiberglass at 113 and wet-layup carbon fiber at 111, a lead of 8 points (7%). It was selected mainly for its high strength-to-weight and stiffness-to-weight properties and because it can be manufactured in house. Compared with aluminum, composites also allow thin-walled structures with fiber orientations chosen for the expected loads. Prepreg carbon fiber may cost more and need more manufacturing care, but its structural performance and the SRAD bonus made it the preferred option. The final laminate will come from structural analysis and manufacturing trials, so that the airframe meets the required factor of safety.

The SRAD bonus weight of 4 has little effect on this result. With a bonus weight of 0, commercial carbon tubing (82) would edge out the prepreg airframe (81), and any weight of 1 or more restores the prepreg

lead. At the weight of 4 the lead over commercial carbon is 39 points. Moving any one weight by one point does not change the winner.

The other airframe trades follow the same method (Appendix D). The Haack series won the nose cone trade with 92 against 87 for the power series (lead 5, 6%), mainly on drag. The clipped delta won the fin planform with 97 against 87 for the tapered swept fin (lead 10, 11%), mainly on flutter margin. The fin cross-section is a near tie: the double wedge scored 80 and the rounded section 78, and the lead is inside scoring noise: moving one weight by one point changes the result in 2 cases (Stability from 3 to 2; Manufacturability from 4 to 5). The tie-breaker we propose is maximum velocity, because the vehicle reaches Mach 0.96 (Section 5.3.1) and the double wedge scores higher than the rounded section on that criterion (7 against 6).

4.4.2 Propulsion

The AeroTech M2500T is the selected motor. Its use in the survey supports the choice (26 of 56 past entries, Table 26). In the team's OpenRocket model with the M2500T and Midland conditions, the vehicle reaches 11,484 ft AGL, leaves the rail at 31.9 m/s, and has a thrust-to-weight ratio of 10.6:1. Total impulse is 9,671 N·s, well inside the 40,960 N·s limit (Section 5.3.1). The motor therefore meets every requirement it touches. Its weakness is that the predicted apogee is 1,484 ft above the nominal target. A motor with less total impulse, such as the M2400T, would reduce the overshoot, and that is why the alternates are kept on the list.

4.4.3 Recovery

A trade study (Table 45) compared three parachutes. The IRIS Ultralight (96 in main) scored highest with 92, ahead of the rip stop at 84 and the printed nylon at 83, a lead of 8 points (10%). It was chosen for its balance of performance, repeatable and efficient packing, and low weight. It gives a landing descent rate of about 6 m/s (Section 5.3.3), well inside the 11 m/s limit (REC-004, measure M6), and a rate that should not damage the structure or components. Moving any one weight by one point does not change the winner. Dual deployment itself was not traded, because it follows from the recovery requirements.

4.4.4 Avionics

The hybrid assembly received the highest weighted score (Table 46) with 98, followed by the horizontal stack with 97 and the vertical panel with 96. It was selected because it offered the best overall balance of cost, manufacturability, ease of integration and compactness: horizontal and vertical panels together use space efficiently and keep the electronics easy to reach.

This result is a near tie, and the lead is inside scoring noise: moving one weight by one point changes the result in 3 cases (Manufacturability from 3 to 4; Ease of integration from 4 to 3; Compactness from 4 to 5). The tie-breaker is integration fit: the hybrid has the highest ease-of-integration score (9 against 7 and 8). The answer would change if the printed panels prove hard to make (manufacturability) or if the bay length cannot hold the hybrid layout (compactness). Both can be checked with a trial print before the

Critical Design Review. The PDR deck also showed an avionics ranking on a success-probability scale (9.30, 9.25 and 9.15). The report uses only the 1 to 10 scale.

4.4.5 Airbrakes (derived requirement, Section 3.4)

Four concepts were scored (Table 44). The linear track assembly scored highest with 99, followed by the linkage assembly and the no-airbrake baseline at 89 each and the rack and pinion at 69, a lead of 10 points (11%). The linear track was selected for its balance of integration, manufacturability and compactness, and its ability to fit with the other subsystems.

The no-airbrake baseline matters here, because the decision to fly airbrakes rests on one criterion. The baseline scores 1 on apogee control and the linear track scores 7. The linear track keeps its lead only while the apogee-control weight is 4 or 5: at a weight of 4 it wins 92 to 88, and at a weight of 3 the baseline wins 87 to 85. Two things support the weight of 5. The first is the baseline overshoot: the OpenRocket model reaches 11,484 ft, so without airbrakes the apogee error is 1,484 ft, outside the ±500 ft target of measure M1, and the apogee score is 176.9 of 350. If the airbrakes hold the error to ±100 ft the score is at least 338.3, a gain of about 161 points (Appendix G). The second is the cost: the airbrake assembly adds 1.7 kg (Section 5.5.5) and three failure modes (HZ-A1 to HZ-A3 in Appendix F). The airbrake authority needed, at least 1,484 ft plus the dispersion, still has to be shown by the RocketPy Monte Carlo before the Critical Design Review. Section 3.4 sets five gates that test this weight with evidence.

4.4.6 Payload Experiment (added by design decision)

The decision matrix (Table 47) points to two answers: the thermodynamic cycle (134) and RF attenuation (132). The lead is 2 points (2%), so the result is a near tie and the lead is inside scoring noise: moving one weight by one point changes the result in 4 cases (Scientific value from 5 to 4; Clarity of hypothesis from 4 to 3; Reliability from 4 to 5; Package and mass difficulty from 1 to 2). What separates the two is the system-level requirements on mass and form factor. Both experiments fit the form factor, and the thermodynamic cycle uses more of the space. For mass, the team estimated the weight of each experiment's components. RF attenuation came to about 1.5 to 2.2 kg, which could miss the 2.0 kg minimum, and the thermodynamic cycle to 2.3 to 2.7 kg (Section 5.3.4). That makes the thermodynamic cycle the more robust choice for meeting PAY-006.

4.5 Selected Concept and Justification

Section 1 defined the baseline vehicle: a COTS solid motor, an airframe with fins, dual-deployment recovery with avionics, and a payload of at least 2.0 kg. The architecture trade (Section 4.2, record DR-01) selected that baseline (A0) as the committed design and carries two capabilities as gated options, with the target configuration being A3, the baseline plus airbrakes plus the SDL experiment. We carry airbrakes because IREC scores apogee by its distance from the target [1], and a motor-only vehicle cannot trim its apogee after burnout. The airbrake requirements are derived in Section 3.4, and the airbrakes stay in the design only while gates G1 to G5 hold. We carry a payload experiment to compete in the SDL

Payload Challenge, which is scored separately from IREC [1]. The vehicle must carry at least 2.0 kg of payload in any case, so the experiment takes the place of mass the baseline would carry anyway.

In full, the target configuration (A3) is an SRAD carbon-fiber prepreg airframe with a Haack-series nose cone and clipped-delta fins, flown on an AeroTech M2500T with linear-track airbrakes, dual-deployment recovery with IRIS Ultralight parachutes, a hybrid-stack avionics bay, and a non-deployable thermodynamic-cycle payload. Table 30 rolls up the subsystem trade studies. Every choice except the motor is the highest-scoring option in its own weighted matrix, and each matrix uses criteria drawn from the requirements and stakeholder priorities. Figure 6 shows the preliminary layout.

Table 30. Trade study roll-up (weights 1–5, scores 1–10).

Subsystem	Options	Highest-scoring concept	Score (weighted)	Runner-up	Lead	Holds if one weight moves ±1 point?
System level						
System architecture (Section 4.2)	4	A0 baseline (committed); A3 is the target	134	128 (A2)	+6 (5%)	Holds in 10 of 11 weight changes; the apogee scores matter more (Section 4.2)
Baseline subsystems						
Airframe material	7	Carbon fiber, prepreg layup	121	113 (fiberglass, prepreg)	+8 (7%)	Yes
Nose cone shape	6	Haack series	92	87 (power series)	+5 (6%)	Yes
Fin planform	3	Clipped delta	97	87 (tapered swept)	+10 (11%)	Yes
Fin cross-section	3	Double wedge	80	78 (rounded)	+2 (3%)	No (2 of the weight changes flip it)
Recovery parachute	3	IRIS Ultralight	92	84 (rip stop)	+8 (10%)	Yes
Avionics layout	3	Hybrid assembly	98	97 (horizontal stack)	+1 (1%)	No (3 of the weight changes flip it)
Motor	5	AeroTech M2500T (selected)	Matrix being rebuilt	—	—	—
Added by design decision (Section 4.5)						
Airbrake mechanism	4	Linear track	99	89 (linkage; no airbrakes 89)	+10 (11%)	Only if the apogee-control weight stays at 4 or 5
Payload experiment	5	Thermodynamic cycle	134	132 (RF attenuation)	+2 (2%)	No (4 of the weight changes flip it)

Scores from different matrices use different weights and are not summed across subsystems. The combination is justified by how the choices fit together. First, the airframe, payload and motor must stay within the mass cap from Section 3.3: the model's liftoff mass is 24.12 kg against a preliminary cap of 28 kg (Section 5.3.4). Second, the airbrake controller needs altitude and velocity from the avionics bay (ICD-005). Third, the carbon-fiber body must carry the 98 mm motor and the recovery loads with the project factor of safety, and a fiberglass nose cone keeps the GPS signal path open. Fourth, the combined cost must stay inside the $20,000 budget (Appendix E). Two weak points remain in the evidence. The first is close scores. The avionics, payload and fin cross-section trades are won by 1 to 3 points, so they rest on tie-breakers: integration fit for the hybrid avionics stack, scientific value and hypothesis clarity for the payload, and maximum velocity for the fin cross-section. The second is the motor matrix, which is being rebuilt. Until then the motor choice rests on the survey and the simulation results.

One decision needs more attention than the matrices give it: how the vehicle gets from its predicted apogee to the target. The baseline with the M2500T overshoots by 1,484 ft (Section 5.3.1). Airbrakes can only remove altitude, so they can correct dispersion in weather, thrust and drag, but a fixed offset of that size is better removed before flight. Ballast does that. The model already carries 0.57 kg of nose ballast, and the mass cap leaves 3.88 kg of reserve, although each kilogram of ballast also lowers the rail-exit speed (about 29 m/s at 28 kg) and shifts the stability margin. A lower-impulse motor does the same without added mass. The team should decide which of these sets the baseline apogee, and then size the airbrake authority for the remaining dispersion. Choosing the airbrakes without that decision leaves the system relying on them for an error that a simple change could remove. This is gate G2 in Section 3.4.

Several interfaces between these choices are not yet closed, so the following must be checked before the Critical Design Review (CDR):

- **Airframe diameter, payload envelope and recovery bay.** The vehicle diameter is 6.0 in in the OpenRocket model, the payload form factor is 100 × 100 × 300 mm (a 141 mm diagonal), and the recovery-bay calculations in Appendix C use a 99 mm inside diameter. SYS-011 (maximum diameter) is still TBD. One diameter must be fixed, and the payload fit and the black powder charge sizes recalculated from it.
- **Payload mass margin.** The thermodynamic-cycle experiment is 2.3 to 2.7 kg against a minimum of 2.0 kg (PAY-006). The lower end leaves only 0.3 kg of margin, and the rules limit non-functional mass in the payload to 0.52 kg, so reducing the condenser or refrigerant fill carries a compliance risk (Section 5.3.4).
- **Airbrake mass and stability.** The airbrake assembly is 1.7 kg in Section 5.5.5 but 0.97 kg in the OpenRocket model. The extra mass sits aft of the center of gravity, which moves it aft and lowers the stability margin, so the margin must be recomputed. AB-005 and AERO-005 also require stability in every airbrake state; CFD and wind-tunnel results are needed (conflict 7, Table 5).
- **Avionics to airbrake data.** The update rate and latency in ICD-005 are TBD and set the AB-002 response time.
- **Primary and backup altimeter.** Section 5.3.3 names the TeleMetrum as primary and the RRC3+ as backup, and the wiring diagram in Section 5.5.4 labels them the other way round. Section 5.3.1 also uses the TeleMetrum as the official apogee record.

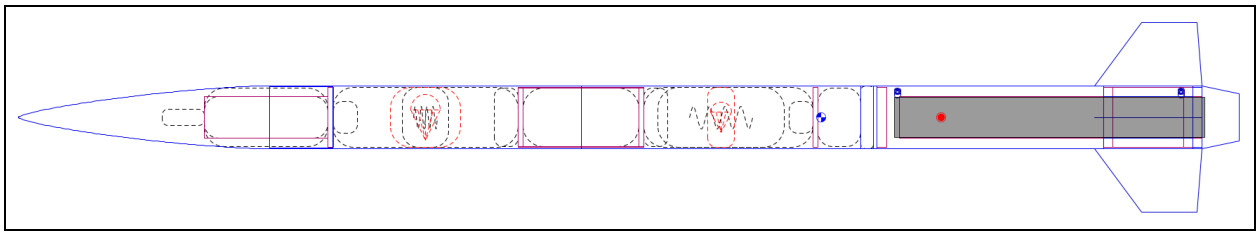


Figure 6. Preliminary OpenRocket layout of the full-scale vehicle with an AeroTech M2500T [9].

4.6 Trade Decision Records

Table 31 records each decision in one place: what was chosen, what it was chosen over, the evidence, how far the evidence can be trusted, and what would reopen the decision. Confidence is Low where the lead is inside scoring noise or the matrix is being rebuilt, Medium where the lead holds when any one weight moves by one point, and High where the choice follows from a requirement.

Table 31. Trade decision register.

ID	Decision	Alternatives	Evidence	Confidence	Reopen if
DR-01	Fly the baseline architecture (A0); carry airbrakes and the SDL experiment as gated options	A0 to A3 (Section 4.2)	Draft judgment matrix: 134 against 128, 123, 122	Low	Any gate in Table 17 fails or passes; updated apogee prediction
DR-02	Airframe: in-house prepreg carbon fiber	7 materials and methods (Table 43)	121 against 113 (+8); holds under ±1 weight changes	Medium	Layup trials fail, or ERC schedule slips; SRAD bonus weight falls to 0
DR-03	Nose cone: Haack series	6 shapes (Table 50)	92 against 87 (+5); drag	Medium	OpenRocket drag per shape disagrees
DR-04	Fins: clipped delta planform	3 planforms (Table 51)	97 against 87 (+10); flutter margin	Medium	Flutter calculation (AERO-003) fails
DR-05	Fin cross-section: double wedge	3 sections (Table 52)	80 against 78 (+2); tie-breaker maximum velocity	Low	Flutter or manufacturing trial result
DR-06	Motor: AeroTech M2500T (provisional)	5 COTS motors (Table 26)	Survey use (26 of 56 entries) and OpenRocket; matrix being rebuilt	Low	Rebuilt matrix; gate G2 (apogee decision)
DR-07	Recovery architecture: dual deployment	None; follows from the recovery requirements (REC-001 to REC-005)	DTEG §6	High	Rule change
DR-08	Parachutes: IRIS Ultralight (26 in drogue, 96 in main)	3 types (Table 45)	92 against 84 (+8)	Medium	Mass assumption changes the drogue

ID	Decision	Alternatives	Evidence	Confidence	Reopen if
					rate below 20 m/s (Table 61)
DR-09	Avionics layout: hybrid panel assembly	3 layouts (Table 46)	98 against 97 (+1); tie-breaker integration fit	Low	Trial print or bay length fails
DR-10	Airbrake mechanism: linear track	4 concepts, including none (Table 44)	99 against 89; ties with no airbrakes at weight 3 of 5 on apogee control	Low	Any gate in Table 17 fails
DR-11	Payload experiment: thermodynamic cycle	5 experiments (Table 47)	134 against 132 (+2); tie-breaker scientific value and hypothesis	Low	Mass estimate or SDL rubric changes

Table 32 lists decisions that appear in the design but were made without a trade study. Each needs a short record: the alternatives that were considered and the reason for the choice. The owners fill in the last column; none of the reasons are assumed here.

Table 32. Decisions made without a trade study.

Decision	Current choice	Where it appears	What the record needs
Flight computers	TeleMetrum v4.0 and RRC3+	Sections 5.5.4, 5.3.3	Alternatives and reason; fix the primary and backup labels
Ejection charge type	Black powder (FFFFg)	Appendix C	Alternatives and DTEG compliance
Nose cone material	Fiberglass, for GPS signal	Section 5.5.1	Alternatives and the signal evidence
GPS location	Separate tracker in the nose cone	Section 5.5.4	Alternatives and a signal test
Single-stage vehicle	Single stage (SYS-003)	Table 9	Reason; IREC permits multistage
Mandrel and fixturing	6061 aluminum mandrel; soft jaws	Appendix D	CTE argument and the airframe tolerance (note in D.4)
Frame material for airbrakes and payload	6061 aluminum	Sections 5.5.5, 5.5.6	Alternatives and a mass check
Airbrake control law	PI controller with integral clamping and a servo slew-rate limit	Section 5.3.1	Alternatives and a simulation result
Motor retention	Retainer sized by the structures team	Section 5.5.2	Design and alternatives; not yet designed

Decision	Current choice	Where it appears	What the record needs
Structural factor of safety	1.5, 3, or 1.4 to 2.0 in different places	Section 5.5.1, Table 10	One value for STR-001 and its reason

4.7 Tradeoffs and Assumptions

Each selected choice gains something and gives something up. Table 33 states both and the assumption each one depends on. Table 34 lists the design assumptions A1 to A5. Each assumption can be tested, and the last column says what happens if it is wrong. Assumptions about the competition and the launch site are in Table 3. The experiment's own assumptions will be added when it is confirmed.

Table 33. Key tradeoffs.

Choice	Gains	Gives up	Relies on
Baseline-first architecture	A complete, flyable vehicle at every stage; the fewest failure modes; options stay open behind gates.	Apogee error of the baseline (+1,484 ft, preliminary) until the airbrakes pass their gates; a possible late change.	A1, A2; gates G1 to G5 (Section 3.4)
SRAD airframe	Control of mass, CG and geometry; lower weight and high structural performance.	Build and test time; cost and schedule risk against budget.	A3
SRAD airbrakes	Control of apogee toward 10,000 ft after motor selection.	More failure modes, 1.7 kg of added mass, a stability shift.	A1, A2
Dual-deployment recovery	Two events add redundancy and a more controlled landing.	Two deployment events to verify and test; more mass.	A5
Non-deployable payload	No added mechanism, fewer failure modes, simpler design.	Payload limited to what fits in the vehicle.	A4
AeroTech M2500T motor	Most used in past entries; meets the rail-exit and impulse requirements.	Predicted apogee 1,484 ft above nominal; limited supply.	C4 (Table 3)

Table 34. Design assumptions.

ID	Assumption	If the assumption is wrong
A1	Apogee control works as modelled (motor, brake response, sensor reading).	The apogee benefit shrinks and the airbrakes become dead mass.
A2	Airbrakes fail to a safe state without altering stability.	Stability rules out flight.
A3	SRAD airframes stay within budget and allocated build time.	SRAD airframes become unachievable.

ID	Assumption	If the assumption is wrong
A4	The payload stays non-deployable, and its mass and size fit IREC requirements.	The mass and stability budgets shift, and the vehicle may fly without payload.
A5	Both recovery trigger events are reliably achievable.	Dual-deployment recovery becomes dangerous and unreliable.

5. Preliminary Design

5.1 System Layout

Figure 7 shows the architecture. The vehicle has four baseline subsystems inside the airframe envelope, arranged from nose to tail: the payload bay behind the nose cone (with the GPS module in the nose), the recovery section, the avionics bay, and the propulsion section with the 98 mm motor, four fins and a boattail. The airbrake section, drawn with a dashed outline, is the capability we added by design decision and sits between the avionics bay and the propulsion section. Each subsystem is allocated one or more functions from Section 3 (function numbers from Figure 5). The avionics bay is the hub: it reads the flight state, sends altitude and velocity to the airbrakes, triggers the two recovery channels, and downlinks position to the ground station. The airframe carries every mechanical load path, including the rail buttons that hold the vehicle on the launch rail. Safety-critical functions sit in the avionics, recovery, airbrake and airframe subsystems, so the interfaces between the first three get the earliest verification. The figure shows order and connections only. Lengths, diameters and masses are not set here and come from the detailed design.

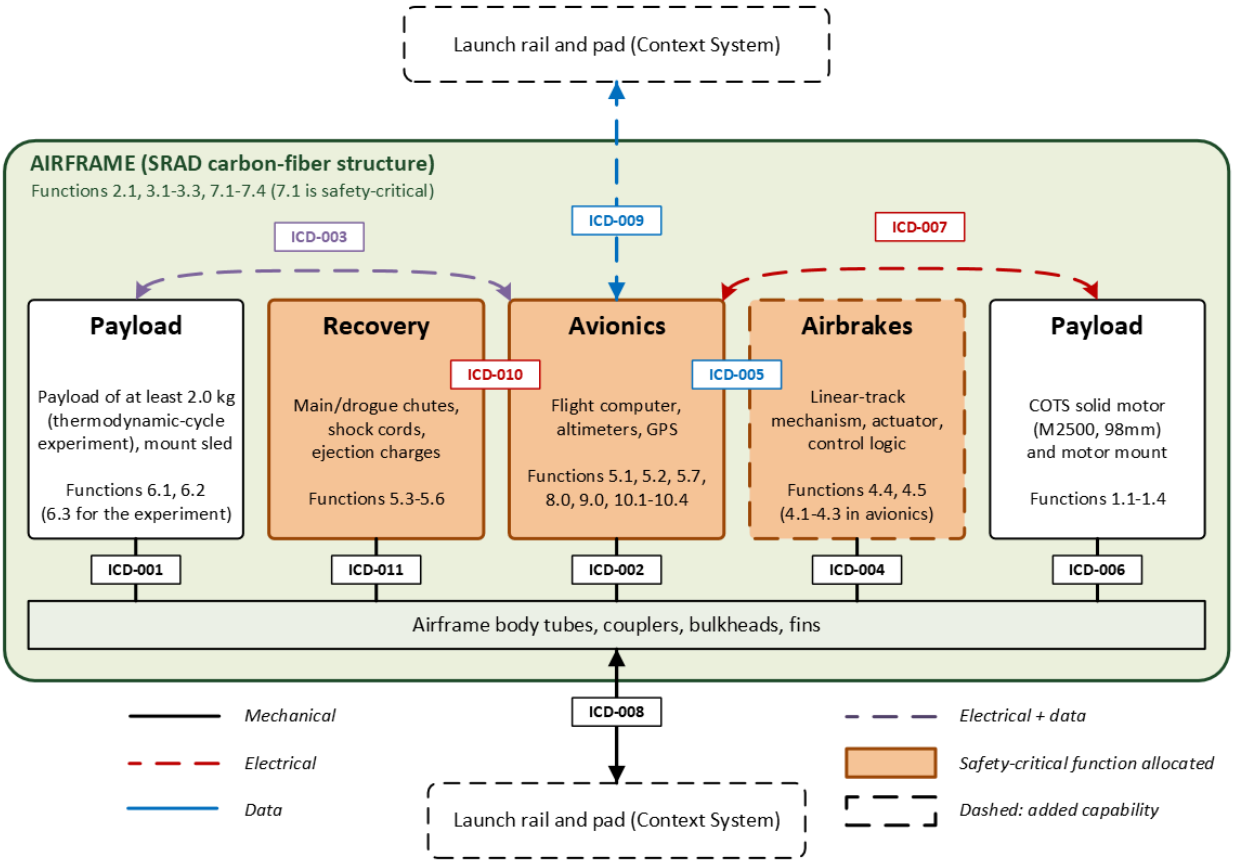


Figure 7. System architecture with function allocation and interfaces (ICD numbers match Table 36). ICD-010 and ICD-011 are proposed.

Table 35 lists the allocation of Level 2 functions to subsystems that Figure 7 shows. Thirty-nine of the 41 functions are allocated. Functions 2.2 and 10.5 are properties of the whole vehicle: no single subsystem owns them, so SYS-005, SYS-006 and SYS-009 carry them at system level.

Table 35. Allocation of functions to subsystems.

Subsystem	Functions allocated	Safety-critical functions	Requirement prefixes
Airframe	2.1, 3.1, 3.2, 3.3, 7.1, 7.2, 7.3, 7.4	7.1	STR, AERO, INT
Propulsion	1.1, 1.2, 1.3, 1.4	None	PT
Recovery	5.3, 5.4, 5.5, 5.6	5.3	REC
Avionics	5.1, 5.2, 5.7, 8.1–8.4, 9.1–9.4, 10.1–10.4; and 4.1–4.3 if airbrakes are flown	5.1, 5.2, 10.2, 10.3	AV, TST
Payload	6.1, 6.2; 6.3 if the experiment is flown	None	PAY
Airbrakes (option)	4.4, 4.5	4.5	AB
Whole vehicle (not allocated to one subsystem)	2.2 Achieve Stable Departure; 10.5 Protect Personnel from Hazards	None	SYS

5.2 Subsystem Interfaces

Table 36 lists each interface, what crosses it, and the requirements it supports. All nine interfaces in the Master SRR are at “Assumed” status: none is designed or verified, and each still has TBD parameters (bolt patterns, connector types, data rates, cutout sizes). We add two interfaces that the SRR either lumps into other interfaces or leaves out: the electrical initiation path from the avionics to the recovery charges (ICD-010), and the structural load path from the airframe to the recovery anchors and separation joints (ICD-011). Each interface moves from Assumed (an expectation only) to Designed (drawn and dimensioned) to Verified (confirmed by a fit check or test), as INT-006 requires.

Table 36. Subsystem interfaces.

ICD	Between	Type	What crosses the interface	Traces to	Status
ICD-001	Payload and airframe	Mechanical	Payload sled bolts to a structural bulkhead. Bolt pattern and torque TBD.	PAY-001, PAY-003, INT-001	Assumed
ICD-002	Airframe and avionics/recovery bay	Mechanical	Avionics sled mounts inside the coupler. Rail and sled dimensions TBD.	INT-001, AV-010	Assumed
ICD-003	Payload and avionics	Electrical and data	Power and data harness, if the payload needs it. Connector, voltage and data rate TBD.	PAY-005, INT-002	Assumed

ICD	Between	Type	What crosses the interface	Traces to	Status
ICD-004	Airframe and airbrakes	Mechanical	Mechanism actuates through an airframe cutout. Cutout size and deployed profile TBD.	AB-006, AERO-005	Assumed
ICD-005	Avionics and airbrakes	Data	Real-time altitude and velocity feed to the control logic. Rate and latency TBD.	AB-001, AB-002, AB-004	Assumed
ICD-006	Airframe and propulsion	Mechanical	Motor mount retains the motor against the aft structure. Ring dimensions and retention TBD.	PT-004, PT-005	Assumed
ICD-007	Avionics/ground equipment and propulsion	Electrical	Ignition signal to the motor igniter. Voltage and continuity check TBD.	PT-006, TST-003	Assumed
ICD-008	Airframe and launch rail	Mechanical	Rail buttons constrain the vehicle through boost. Button spacing and rail size TBD (rail at least 17 ft [2]).	SYS-006, SYS-011, PT-006	Assumed
ICD-009	Avionics and ground station	Data	GPS and telemetry downlink; post-flight data pull. Frequency and rate TBD.	AV-002, AV-004	Assumed
ICD-010	Avionics and recovery	Electrical	Initiation signals to the drogue and main charges over two independent channels, each with its own power.	REC-005–008	Proposed
ICD-011	Airframe and recovery	Mechanical	Anchors, shock cords and separation joints (shear pins) carrying deployment loads.	REC-009, REC-010, STR-004	Proposed

5.3 How the Architecture Meets Key Requirements

This section checks the architecture against the key requirements, using the OpenRocket model of the full-scale vehicle at Midland conditions (C3 in Table 3) and hand calculations. Table 37 lists each requirement with the value required, the current predicted value, its source, the margin and the status. Items marked TBD or Open have not been calculated yet. The subsections explain the four requirement areas. The OpenRocket values in this section are preliminary and unverified: the model has not been checked against the final airbrake mass, airframe diameter or payload mass (Section 4.5), and every value from it should be read as an estimate. Of the 22 rows in Table 37, 14 meet the requirement, 4 are open, 1 is not met without airbrakes, and 3 await a model update, a flight test or a confirmation.

Table 37. Key requirements against the current design (OpenRocket, Midland conditions).

Requirement	Value required	Current predicted value	Source	Margin	Status
Apogee and propulsion					

Requirement	Value required	Current predicted value	Source	Margin	Status
SYS-001	7,000 to 13,000 ft AGL, nominal 10,000 ft [1]	11,484 ft AGL	OpenRocket, Midland	1,516 ft under the ceiling; 1,484 ft over nominal	Meets
M1 (apogee error)	Within ±500 ft of 10,000 ft	+1,484 ft	OpenRocket, Midland	984 ft outside	Not met without airbrakes
AB-001	Airbrakes remove the baseline overshoot plus dispersion (at least 1,484 ft nominal)	TBD	RocketPy Monte Carlo	TBD	Open
PT-008	≤ 40,960 N·s total impulse [3]	9,671 N·s (M2500T)	Motor data [10]	31,289 N·s	Meets
SYS-005	Liftoff thrust-to-weight ≥ 5:1	10.6:1 (average thrust)	OpenRocket	+5.6	Meets
SYS-006	Rail exit ≥ 25 m/s at the effective rail length (D&T-5.3.1) [2]	31.9 m/s (104.6 ft/s) at 4.36 m	OpenRocket, Midland	+6.9 m/s	Meets
Stability					
SYS-007	≥ 7.5% of length subsonic (max Mach 0.96), launch through motor burnout (D&T-10.2.1) [2]	11.3% (2.19 cal) at rail exit	OpenRocket, no wind	+3.8 points (+2.8 over the M2 target)	Meets
SYS-008	≤ 18% at launch; ≤ 25% in flight (D&T-10.3.1) [2]	11.3% at launch; 17.9% (3.46 cal) peak	OpenRocket, no wind	6.7 and 7.1 points	Meets
AERO-002	Fin semi-span ≥ 0.8 airframe diameters	1.0 diameters (6.0 in)	OpenRocket geometry	+0.2 diameters	Meets
AERO-003	Flutter velocity ≥ 150% of maximum velocity (≥ 1,634 ft/s); team-derived, D&T-8.2.2 requires the calculation be reported	TBD	Hand calculation	TBD	Open
Rail buttons (D&T-8.6.2)	At least two rail buttons, both shown in the OpenRocket file	One modeled	OpenRocket file	n/a	Model to update
AB-005, AERO-005	SYS-007 floor held in every airbrake state	TBD	OpenRocket, CFD	TBD	Open

Requirement	Value required	Current predicted value	Source	Margin	Status
Recovery					
REC-001	Drogue event within 100 ft of apogee	Barometric apogee detection on both altimeters	Design	n/a	Verify by test (SS1)
REC-002	Drogue descent 20 to 40 m/s at main deployment	22.3 m/s (73 ft/s), 26 in drogue	Hand calculation	+2.3 m/s over the floor	Meets, near floor
REC-003	Main deployment ≤ 1,500 ft AGL	1,000 ft AGL	Appendix C	500 ft	Meets
REC-004	Landing descent < 11 m/s (36 ft/s)	6.0 m/s (19.5 ft/s), 96 in main	Hand calculation	5.0 m/s	Meets
REC-005	Two independent deployment channels	TeleMetrum and RRC3+, separate batteries and switches	Section 5.5.4	n/a	Meets by design
STR-004	Recovery attachments hold 1,684 N main and 1,595 N drogue (Table 49)	Shock cord rated 16.0 kN (3,600 lbf)	Hand calculation	9.5× on the main	Meets; pull test pending
M6, drift	Land inside the range recovery area	820 m (2,700 ft) at 10 mph; 1,640 m (5,400 ft) at 20 mph	Hand calculation	TBD	Open (SYS-012)
Payload and mass					
PAY-006	Payload ≥ 2.0 kg (4.4 lb)	2.3 to 2.7 kg (Section 5.5.6)	Section 5.5.6	+0.3 to +0.7 kg	Meets
Boilerplate (R&R-3.4.5.4) [1]	≤ 0.52 kg (1.15 lb) non-functional mass in the payload	0 kg in the payload; 0.57 kg nose ballast kept separate	OpenRocket mass model	See 5.3.4	Confirm
SYS-010	Liftoff mass ≤ cap, preliminary 28 kg (TBR, M3)	24.12 kg (53.2 lb)	OpenRocket mass model	3.88 kg (14%)	Meets

5.3.1 Apogee

OpenRocket predicts an apogee of 11,484 ft AGL at Midland (Figure 8a). The M2500T delivers 9,671 N·s over a 3.9 s burn from the RMS-98/10240 case [10], for an average liftoff thrust-to-weight ratio of 10.6:1 (SYS-005) and a peak acceleration of 17 g. The vehicle leaves the rail at 31.9 m/s (SYS-006), measured at the effective rail length of 4.36 m (17 ft of rail less the 0.83 m from the aft end to the rail button in the model), and reaches 1,089 ft/s (Mach 0.96) at burnout. The prediction is inside the SYS-001 window, with 1,516 ft to spare under the 13,000 ft ceiling, but 1,484 ft above the nominal target.

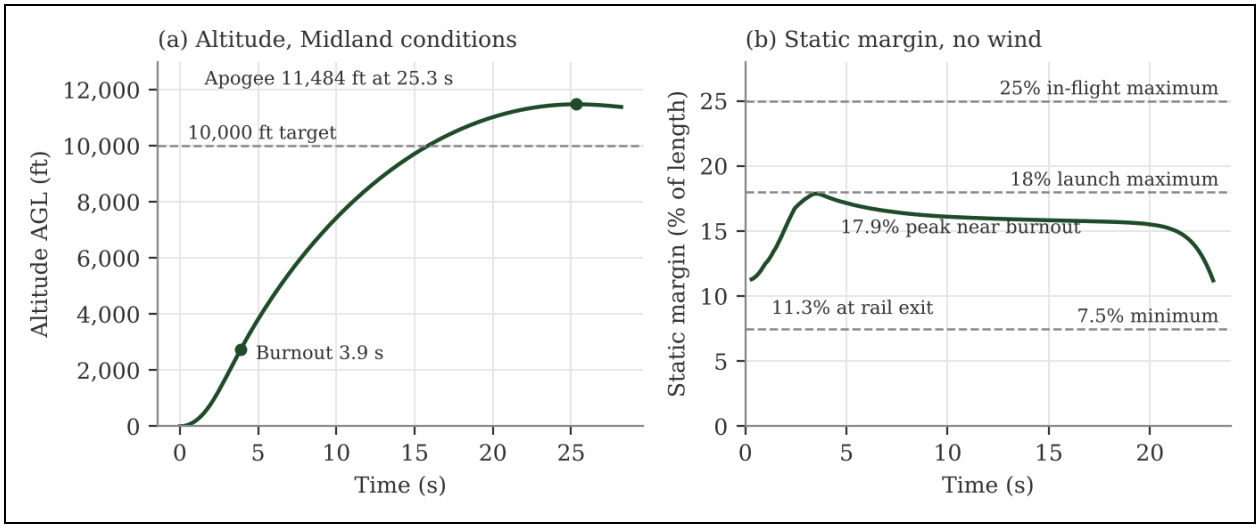


Figure 8. OpenRocket results for the full-scale vehicle: (a) altitude at Midland conditions, (b) static margin with no wind.

Launch conditions matter here. The same model at sea level in a standard atmosphere with no wind reaches 10,907 ft, 577 ft lower, mostly because the air at Midland in June is thinner through the whole climb. Every competition prediction in this report therefore uses the Midland conditions in C3 of Table 3.

Airbrakes can only remove altitude, so the baseline has to overshoot in every expected condition (Table 5, conflict 4). It does, but more than the airbrakes may be able to correct. The linear track assembly (Section 5.5.5) deploys after motor burnout under a PI controller with integral clamping and a servo slew-rate limit. AB-001 now has a number to meet: full deployment has to remove at least 1,484 ft at nominal conditions, plus the high side of the dispersion in motor impulse, mass, drag and wind. That authority comes from the RocketPy Monte Carlo before CDR. If the airbrakes hold the error to ±100 ft, the score rises from 176.9 to at least 338.3 points. The score uses the barometric apogee from the first COTS altimeter the team names (R&R-3.6.2, 3.6.6), the TeleMetrum, so the airbrake controller and the official logger should agree on apogee before FS2. Two changes before CDR will also move the baseline. Section 5.5.5 gives a 1.7 kg airbrake assembly where the model carries 0.97 kg for the mechanism and its electronics, and the payload in the model sits at the top of its range (Section 5.3.4). Both added masses lower the apogee. If the updated baseline still overshoots by more than the airbrake authority, the options are nose or payload ballast, or an alternate motor (Section 4.5).

5.3.2 Stability

SYS-007 and SYS-008 set the range as a percentage of vehicle length [2]: at least 7.5% subsonic (10% if supersonic at any point) from launch through motor burnout (D&T-10.2.1), and no more than 18% at launch and 25% through the flight (D&T-10.3.1). The 2027 DTEG ends the minimum-margin window at burnout; the 2026 edition ran it to drogue deployment, so the SYS-007 wording in Table 8 should be updated. One caliber on this vehicle (6.0 in on 116.3 in) equals 5.16% of length, so the limits are 1.45, 3.49 and 4.84 calibers. Measure M2 adds a 1-point buffer inside each limit: at least 8.5%, and at most 17% at launch and 24% in flight.

The inputs are the OpenRocket mass model with the full payload installed, as R&R-3.4.1.3 allows [1]. Liftoff mass is 24.12 kg, with the CG at 74.6 in and the CP at 87.8 in from the nose tip at rail exit. The

margin starts at its minimum, 2.19 calibers (11.3%), as the vehicle leaves the rail. It rises as propellant burns from the aft end and the CG moves forward, and peaks at 3.46 calibers (17.9%) near burnout, when the vehicle has lost 4.71 kg and weighs 19.41 kg (Figure 8b). Peak Mach is 0.96, so the 7.5% floor applies; the 11.3% minimum would also clear the 10% supersonic floor if the real flight crosses Mach 1. Both the minimum and the peak sit inside the M2 band. Near apogee, below Mach 0.05, OpenRocket's margin drops under 1 caliber. That is a low-speed artifact of the method and falls outside the D&T-10.2.1 window, which ends at burnout, but it still shows in the full-flight plot that D&T-10.2.3 asks for.

The airbrakes sit aft of the CG, so deploying them moves the CP aft and raises the margin. Their mass also sits aft of the CG, which moves the CG aft and lowers the margin, so the net effect must be recomputed with the 1.7 kg assembly (Section 5.5.5).

5.3.3 Recovery

Dual deployment follows from the recovery requirements, so the checks here are deployment altitudes, descent rates and drift. The drogue fires at apogee on both altimeters, the TeleMetrum as primary and the RRC3+ as backup, each on its own battery and switch (REC-001, REC-005). REC-001 allows 100 ft of error, which is verified with the SS1 flight data. The main fires at 1,000 ft AGL (Appendix C), 500 ft under the REC-003 limit.

Descent rates use the drag equation from Section 5.5.3, solved for velocity, with the 26 in IRIS Ultralight drogue and the 96 in IRIS Ultralight main. Hot, thin air gives the faster and more conservative descent. As a check, the same method applied to the parachutes in the OpenRocket file matches its simulated descent rates within 1%. Under the drogue the vehicle descends at 22.3 m/s when the main deploys, inside the 20 to 40 m/s band of REC-002 but only 2.3 m/s above the floor. A drogue larger than 26 in, or a drag coefficient higher than assumed, would push the rate under 20 m/s, so the drogue should not grow past 26 in. Under the main, the landing rate is 6.0 m/s against the 11 m/s limit of REC-004 and measure M6.

Drift is the descent time multiplied by the wind speed, with no credit for the vehicle weathercocking into the wind on ascent. The descent takes 184 s: 133 s under the drogue and 51 s under the main. That gives 410 m (1,350 ft) of drift at 5 mph, 820 m (2,700 ft) at 10 mph and 1,640 m (5,400 ft) at 20 mph. The comparison against the range recovery area stays open with SYS-012 until ESRA publishes the range layout. Per D&T-4.7.3, the team also needs a written recovery plan with range and direction estimates, which these drift numbers feed.

5.3.4 Payload and Mass

The thermodynamic-cycle experiment has an estimated total mass of 2.3 to 2.7 kg, including its structural frame. That satisfies the 2.0 kg minimum of PAY-006, but at the lower end the margin is only 0.3 kg, so reducing the condenser size or the refrigerant fill carries a compliance risk. Under the current rules, entries must reach at least 95% of the required mass at official check-in, and the supplemental non-functional (boilerplate) mass for functional assemblies is capped at 0.52 kg (1.15 lb) [1]. Official verification also requires an independent mass measurement of the payload module taken out of the airframe and recovery structures. The OpenRocket simulation uses an upper-bound 2.69 kg for the experiment plus 0.14 kg of integration hardware, so the apogee in Section 5.3.1 is conservative if the final structure is lighter.

The model also carries 0.57 kg (1.25 lb) of static nose cone ballast outside the main payload compartment. This is general airframe ballast, not payload boilerplate. Counting it as payload ballast would break the 0.52 kg limit on non-functional mass, so it must be kept physically separate and documented. D&T-8.3.4 prohibits dense or hazardous ballast materials, including lead and steel [2]. The predicted static margin exceeds the minimum by 3.8 percentage points, so relocating or trimming this nose mass is a main way to adjust the baseline apogee.

Total liftoff mass is 24.12 kg (53.2 lb), which leaves a 3.88 kg (14%) reserve below the preliminary 28 kg limit in measure M3 for SYS-010. The AeroTech M2500T accounts for 8.06 kg of gross mass, including 4.71 kg of usable solid propellant [10]. The major airframe contributions are the fin assembly (1.48 kg), the fin-can structural epoxy (1.59 kg), the machined aluminum boattail (0.80 kg) and the internal airbrake retention bulkhead (0.68 kg). Scaling the liftoff mass to the 28 kg ceiling with the same thrust curve gives a rail-exit velocity of about 29 m/s, still above the 25 m/s threshold. Rail-exit speed therefore does not limit the liftoff mass; apogee performance is the binding constraint for this motor.

5.4 Safety-Critical Features

The safety-critical functions in Section 3.1 (Figure 5) are Detect Recovery Events (5.1), Command Deployment (5.2), Separate Vehicle Sections (5.3), Carry Flight Loads (7.1), Inhibit Unintended Actuation (10.2) and Respond to Faults (10.3), and, with the airbrakes, Constrain Drag Control (4.5). Each subsection below gives design features only; handling procedures belong in the operations documents. Table 38 summarizes the hazards, mitigations and verification by subsystem, and each subsection names the matching hazard rows in Appendix F.

5.4.1 Detect Recovery Events (5.1)

Detecting the recovery events is safety critical because every recovery event is triggered from the altitude reading, so an inaccurate or lost reading can cause a late or early deployment. The critical issue is that one measurement sits upstream of several other critical functions, so a single sensor error would pass through all of them. The design therefore uses redundant altimeters with measurement checks, each on its own power, and keeps the flight data after landing on SD card storage. Altitude accuracy (AV-013, proposed), recording and retention (AV-003, AV-004) and survival of the flight environment (AV-006) are verified by test. Appendix F rows: HZ-01 and HZ-02.

5.4.2 Command Deployment and Separate Vehicle Sections (5.2, 5.3)

Recovery is safety critical because it is the only thing that brings the vehicle down at a survivable speed, so a missed, early or one-sided deployment can endanger the public and lose the vehicle. The critical issue is that dual deployment needs two events to happen at their scheduled flight times, with nothing firing before or after. The design uses two functionally independent deployment channels, each with its own power source and trigger event. The charges stay inert until the vehicle is armed and are sized with a factor of safety of 2 over the force needed to shear the pins (Appendix C). Shock cords, bulkheads and attachments are sized for the peak deployment load. REC-005 to REC-008, REC-010 and STR-004 are verified by ground test of each channel, vacuum-chamber testing and pull tests. Appendix F rows: HZ-03 to HZ-06, HZ-08 and HZ-13.

5.4.3 Inhibit Unintended Actuation and Respond to Faults (10.2, 10.3)

Maintaining a safe state is safety critical because a single fault, or a loss of data or power, must leave the vehicle in a known, recoverable state instead of an unpredictable one. The critical issue is that every other critical function can fail, and without a defined response a single failure can spread. The design gives each flight-critical system a defined response to its single-point failures, so the loss of one recovery channel leaves the other able to complete the recovery. It also inhibits hazardous functions until the vehicle is armed, and disarms them after landing. It is verified under ENG-008 by analysis and by fault ground tests that aim to find failure points and modes. Appendix F rows: HZ-02, HZ-05, HZ-A1 and HZ-A2.

5.4.4 Carry Flight Loads (7.1)

Carrying the flight loads is safety critical because the airframe carries the thrust and the deployment loads, and a structural failure in flight is a catastrophic outcome. The critical issue is that the airframe is the foundation the other critical functions depend on: the recovery attachments, the airbrake mount and the payload mount all load into it. The design is a carbon-fiber prepreg airframe modeled in Ansys and made by hand at the ERC. STR-001 to STR-004 and AERO-003 are verified by analysis, with compression and tension tests and laminate inspection to validate the model and catch voids. The article will be tested and verified with compression, tension and internal pressure loads. Appendix F rows: HZ-06 to HZ-09.

5.4.5 Constrain Drag Control (4.5) (added by design decision)

Constraining the drag control is safety critical because the airbrakes act on the vehicle in flight. Drag deployed during boost would add loads on the airframe, a jammed or unpowered mechanism would leave the apogee uncontrolled, and a deployed state with a low stability margin could make the vehicle unstable. The design inhibits the airbrakes until burnout is detected (AB-003, AV-014), defaults to neutral drag on a fault or a loss of data or power (AB-004), and caps the commanded authority at the stability floor (AB-005, AERO-005). These are verified by test and analysis: hardware release tests under load, wind-tunnel tests of the mechanism, and CFD and OpenRocket runs for stability in every airbrake state. Appendix F rows: HZ-A1 to HZ-A3.

Table 38. Safety-critical features.

Subsystem	Key hazard or failure	Mitigation	Test or verification method	Section
SRAD airframe	Structural failure, buckling, bridging and laminate voids.	Size the geometry with a set factor of safety, model it in Ansys ACP, and test the layup against predicted forces.	Structural analysis, compression and tension tests, laminate inspection.	5.4.4
SRAD airbrakes	Premature deployment, jammed hardware, malfunctioning electronics.	Practice hardware tests with the servo, verify the electronic power needs and test individual sensors.	Wind tunnel tests for structural integrity and hardware release tests under load.	5.4.5
Dual-deployment recovery	Missed or early deployment; hardware separation.	Verify the independent channels, pack the parachutes carefully and model the fall trajectory.	Ground test of each deployment channel. Test the sensors with changes in altitude and in a vacuum chamber.	5.4.1–5.4.3

Subsystem	Key hazard or failure	Mitigation	Test or verification method	Section
Non-deployable payload	Payload damage; vehicle interference.	Proper payload installation and mechanical retention in flight (no accidental release).	Inspect the mount and verify payload integrity. Document the final CG and mass of the system.	5.3.4, 5.5.6

5.5 Preliminary Component and Material Selections

Each subsection gives the selected components or materials for one subsystem, the reason, and the key numbers. The mass cap from Section 3.3, the $20,000 budget (Appendix E) and the motor supply (C4 in Table 3) bound the choices. The mass numbers used in 5.3 come from the OpenRocket model.

5.5.1 Airframe and Structures

The main airframe will be manufactured from unidirectional prepreg carbon fiber. The preliminary laminate design will follow the aerodynamic and inertial loads expected in flight, with particular attention to axial compression, bending and buckling, which could cause structural failure. Classical Laminate Theory (CLT) and ANSYS Composite PrepPost (ACP) will evaluate the laminate stiffness, the stresses and the possible failure modes. The laminate will be evaluated with a preliminary factor of safety of 1.5. Ply orientations and wall thickness will be adjusted from the structural analysis to give enough strength without adding unnecessary mass.

The nose cone is currently being considered in fiberglass, because fiberglass transmits GPS signals better than carbon fiber. The preliminary design also includes a Haack-series nose cone and clipped-delta fins with a double-wedge cross-section, as selected in the trade studies (Section 4.4.1). Before the airframe design is final, the team will verify the laminate through structural analysis, manufacturing trials and physical testing, to find problems such as buckling, inadequate laminate strength and manufacturing defects. The mandrel and fixturing choices for the layup are in Appendix D.

5.5.2 Propulsion

The motor is the AeroTech M2500T in the RMS-98 reload case. Its total impulse is 9,671 N·s, delivered over a 3.9 s burn, which gives an average thrust of about 2,480 N. The motor's gross mass is 8.06 kg, of which 4.71 kg is solid propellant [10]. In the OpenRocket model this gives a liftoff thrust-to-weight ratio of 10.6:1 and a rail-exit speed of 31.9 m/s (Section 5.3.1). The impulse is far below the FAA Class 2 limit of 40,960 N·s [3]. The motor mount holds the motor against the aft structure (ICD-006), with the retention sized by the structures team. The alternates are the AeroTech M1939W and M2400T and the Cesaroni M3400-P (Section 4.3.2). Using the motor's own total impulse and not the casing class (10,240 N·s) matters for every simulation and for PT-001.

5.5.3 Recovery

The recovery system will use a 26 in IRIS Ultralight drogue for the initial descent and a 96 in IRIS Ultralight main for the final descent. Shock cords cushion the deployment energy and keep the vehicle together as one system, and they attach to eye bolts for a strong and reliable connection. The whole recovery system, including the drogue and main connections, weighs about 0.5 kg, so it adds little mass to the vehicle.

Based on the competition requirements, the drogue must give an initial descent rate of 20 to 40 m/s and the main must give a landing velocity below 11 m/s. The descent velocity of each parachute is found by rearranging the drag-force equation, mg = ½ ρ Cd A v², to solve for velocity:

$$v = \sqrt{\frac{2mg}{\rho C_d A}}, A = \pi D^2$$

Here m is the vehicle mass without propellant, g is 9.81 m/s², ρ is the air density (1.089 kg/m³ at Midland, TX), Cd is the drag coefficient of the IRIS Ultralight parachutes (2.2) and D is the parachute diameter. The results are 22.3 m/s under the drogue and 6.0 m/s under the main (Section 5.3.3), inside the required ranges and the internal requirements.

5.5.4 Avionics

The avionics subsystem uses a TeleMetrum v4.0 and an RRC3+ as the primary and backup flight computers that control the drogue and main parachute deployment. The electronics, batteries, switches and wiring are secured to a custom hybrid mounting panel assembly. This layout makes efficient use of the available space and keeps the components accessible, based on the comparison in Section 4.4.4. A separate GPS tracker is planned for the nose cone, to reduce interference from the carbon-fiber airframe. Each flight computer has its own battery so the primary and backup systems work independently. The wiring is arranged neatly and secured to prevent loose connections in flight, and screw switches allow the electronics to be armed from outside the avionics bay (Figure 9).

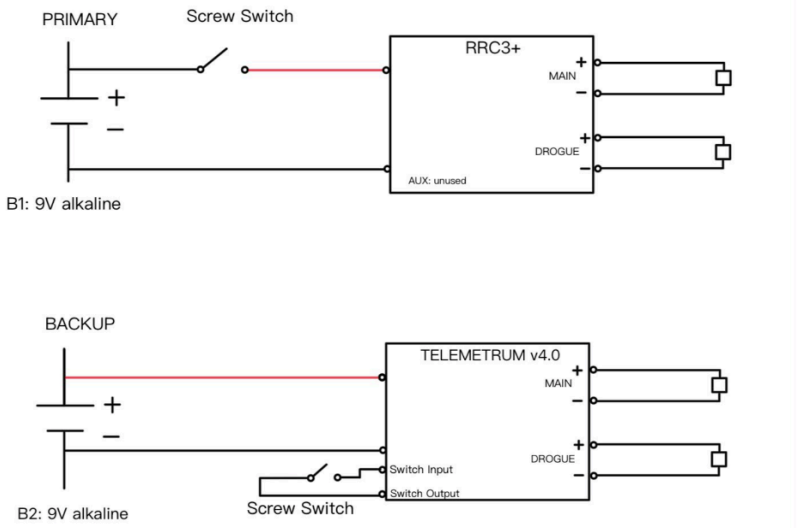


Figure 9. Avionics wiring layout: each flight computer has its own 9 V battery and screw switch and fires its own main and drogue charges.

5.5.5 Airbrakes (added by design decision)

The airbrake subsystem is a linear track assembly made entirely of 6061 aluminum alloy. This alloy was chosen because it is lightweight, strong, corrosion-resistant and relatively easy to machine, which suits the

moving parts of the mechanism and keeps the vehicle mass low. It is also widely available and affordable, which fits the $20,000 budget. The assembly has a total mass of 1,700 g (3.75 lb).

An actuator and a controller connect to the avionics through the data link in ICD-005. The actuator extends and retracts the airbrakes to control apogee, and the deployment time will be set by testing. Aerodynamic forces pass from the airbrakes through the linear track and the mounting brackets into the airframe. The design assumes the mechanism fits in the space available inside the airframe and withstands the aerodynamic forces in flight. The final dimensions, mounting strength and deployment performance still need to be verified.

5.5.6 Payload Experiment (added by design decision)

The thermodynamic cycle uses an off-the-shelf mini compressor, a custom condenser, an off-the-shelf expansion shell and a custom evaporator. The components are designed and selected to fit within the form factor. The payload frame is 6061 aluminum angle, which lets air flow around the payload so the condenser can work properly. The total payload mass is 2.3 to 2.7 kg, depending on the size of the condenser and the refrigerant. The design will use as much of the form factor as it can, and it includes pressure and temperature sensors and a computer. The mass meets the 2.0 kg minimum (PAY-006) with the margin shown in Section 5.3.4.

5.6 Engineering Budgets and Margins

Table 39 gathers every resource the design must stay within: the limit, the current estimate and the margin. Where the value is not yet known the cell says so, because an unknown margin is a risk to be managed and not a blank to be ignored. Appendix H gives the allocation behind the mass and cost rows and proposes a growth-allowance policy. Mass, stability and apogee values are from the preliminary OpenRocket model.

Table 39. Engineering budgets and margins.

Resource	Limit or requirement	Current estimate	Margin	Status and note
Liftoff mass	At most 28 kg (SYS-010, TBR)	24.12 kg	3.88 kg (14% of the cap)	6.64 kg of the model is not itemized; airbrake mass 1.7 kg against 0.97 kg in the model (Appendix H)
Static stability margin	7.5% to 18% at launch, at most 25% in flight; M2 buffer 8.5%	11.3% at rail exit; 17.9% peak	+3.8 points over the floor	No wind; the heavier airbrake mass is not yet included
Rail-exit speed	At least 25 m/s (SYS-006)	31.9 m/s	+6.9 m/s	About 29 m/s at 28 kg
Thrust-to-weight	At least 5:1 (SYS-005)	10.6:1	+5.6	Average thrust
Apogee	7,000 to 13,000 ft; M1 target ±500 ft of 10,000 ft	11,484 ft	1,516 ft under the ceiling; 984 ft outside M1	Preliminary; gate G1 for the airbrakes
Total impulse	At most 40,960 N·s (PT-008)	9,671 N·s	31,289 N·s	M2500T data sheet

Resource	Limit or requirement	Current estimate	Margin	Status and note
Payload mass	At least 2.0 kg (PAY-006); at most 0.52 kg of non-functional mass	2.3 to 2.7 kg	+0.3 to +0.7 kg	The low end leaves little margin
Drogue descent rate	20 to 40 m/s (REC-002)	22.3 m/s	+2.3 m/s over the floor	The thinnest recovery margin (Table 61)
Main landing rate	Below 11 m/s (REC-004)	6.0 m/s	5.0 m/s	Hand calculation
Cost	$20,000 (ENG-012)	$2,375.11 spent	$17,624.89 remaining	Secured funding $10,000 of $23,800 listed (Appendix E)
Electrical energy	Margin of at least [TBD]% (AV-012)	TBD	TBD	Avionics subteam to list loads (Appendix H)
Internal volume	Payload [TBD] in³ (PAY-002); recovery bay [TBD] (REC-012)	Bays 293.5 mm and 226 mm long, 99 mm inside diameter	TBD	Diameter conflict (Section 4.5)
Schedule	Design complete 2 weeks before CDR (M5)	See Section 5.8	TBD	Dates not final

5.7 Risk Management

Appendix F lists 25 hazards scored on impact and probability: 1 High, 19 Serious and 5 Medium. Table 40 lists the eight with the highest scores, with the next action that would lower each. The High risk, HZ-19, is the apogee window, and it is the reason the airbrake decision (Section 3.4) matters. Risk scoring follows Table 58; the scales still need confirmation (note after Table 59).

Table 40. Highest-scoring risks and next actions.

Hazard	Score	Why it ranks high	Next action (proposed)	Owner (proposed)
HZ-19 Apogee outside the window	12 High	Preliminary apogee is 1,484 ft above the target	Update the model; decide the baseline apogee (ballast, motor or airbrakes) through gates G1 and G2	Systems and propulsion
HZ-20 Verification not finished before flight	9 Serious	Test groups have TBR criteria; no test results yet	Fix criteria for each group in Table 11; schedule tests before SS1	Systems and Safety/Test
HZ-03 Drogue does not deploy	8 Serious	No deceleration if it fails	Ground ejection test after the charge basis is recalculated (Appendix C)	Recovery
HZ-05 Premature or uncommanded deployment	8 Serious	Hazard to people at the pad	Define the arming sequence; test that no output exists before arming (TST-001)	Recovery and avionics

Hazard	Score	Why it ranks high	Next action (proposed)	Owner (proposed)
HZ-07 Airframe structural failure	8 Serious	Factor of safety of 3 not yet verified by test	Apply factor of safety 3; coupon tests before layup	Structures
HZ-A3 Stability drops when airbrakes deploy	8 Serious	Added mass aft of the CG	Recompute the margin with 1.7 kg; CFD for gate G3	Aerodynamics
HZ-01 Altimeter reads wrong altitude	8 Serious	Every recovery event depends on it	Vacuum-chamber test; fix the accuracy for AV-013	Avionics
HZ-02 Avionics loses power or resets	8 Serious	No deployment command	Power budget (AV-012); vibration test	Avionics

5.8 Verification, Validation and Schedule

Verification shows that the vehicle meets its requirements; validation shows that it does the job the stakeholders need. Requirements are verified by the methods in Section 2.4 and the test groups in Table 11. Validation is by flight, in the sequence of Table 41. Every flight tests more of the system and sets the exit criteria for the next.

Table 41. Validation milestones.

Milestone	Date	What it validates	Exit criteria (proposed)
Design freeze	Before CDR (date TBD)	Requirements, interfaces and masses are baselined	TBD requirements closed; airbrake gates G1 to G5 decided; diameter settled
Subscale 1 (SS1)	November 28, 2026 (an outline note says December 28)	Recovery sequence, avionics, deployment timing and descent rates	Both events fire on time; flight data recovered
Full-scale 1 (FS1)	March 13, 2027	Full-scale stability, rail exit, apogee prediction; airbrakes if they have passed the gates	Stable flight; apogee within the window; mass and CG match the model
Full-scale 2 (FS2)	May 15, 2027	Competition configuration, payload, final apogee accuracy	Configuration frozen and documented (DOC-006)
Competition launch	June 2027 (team schedule June 20; ESRA June 14–19)	The mission	Flight-readiness review passed

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Appendix

A. Full Requirements List

The table lists all 105 requirements in the Master SRR [8] with the function each traces to (Section 3.2), followed by the seven proposed additions. A dagger (†) marks wording or source changed in this report. Requirements with a [TBD] value are open items tracked in the Master SRR Section 4.9.

Table 42. Master SRR requirements († changed in this report).

ID	Requirement	Source	Verif.	Function
SYS-001†	The vehicle shall achieve an apogee between 7,000 ft and 13,000 ft AGL, with a nominal target of 10,000 ft AGL	IREC R&R §2.8.1.4 (2026 ed.); competition scoring objective	Test	1.3, 4.3
SYS-002	The vehicle shall complete a controlled flight from launch through recovery without loss of the vehicle or required flight functions	Stakeholder safety need — umbrella requirement, satisfied by REC-008, STR-006, STR-007, AV-006	Test	System
SYS-003†	The vehicle shall operate as a single-stage launch vehicle	CSU design choice (IREC permits multistage, R&R §2.1)	Inspection	1.3
SYS-004†	The vehicle shall use a COTS-certified solid rocket motor permitted by the applicable IREC competition category	IREC DTEG §5.1.1.1 (2026 ed.)	Inspection	1.3
SYS-005	The vehicle shall achieve a liftoff thrust-to-weight ratio of at least 5:1	Team engineering requirement; launch safety	Analysis	1.3, 2.2
SYS-006	The vehicle shall achieve a minimum launch-rail exit velocity of 25 m/s	DTEG §5.3.1	Analysis	2.2
SYS-007†	The vehicle shall maintain a static stability margin of at least 7.5% of vehicle length while subsonic and at least 10% while supersonic, from launch through drogue deployment	DTEG §10.2.1 (2026 ed.)	Analysis	3.2
SYS-008†	The vehicle shall maintain a static stability margin no greater than 18% at launch and 25% during flight	DTEG §10.3.1 (2026 ed.)	Analysis	3.2
SYS-009	The vehicle shall remain within all applicable IREC, launch-range, and university safety constraints during launch and flight operations	IREC R&R; DTEG; range requirements — umbrella requirement, satisfied by STR-008, AV-011, DOC-001, PAY-004	Inspection	System, 10.5
SYS-010	The vehicle shall have a total liftoff mass not exceeding [TBD] kg	Team engineering constraint; informed by motor selection and structural margin	Test	System

ID	Requirement	Source	Verif.	Function
SYS-011	The vehicle shall have an overall length not exceeding 10 ft and a maximum body diameter not exceeding 6.0 in	Launch rail and transport interface constraint	Inspection	7.2
SYS-012	The vehicle shall be capable of safe launch operations in surface wind speeds up to [TBD] mph	IREC R&R / range launch commit criteria	Analysis	2.2
STR-001	The primary structure shall withstand all predicted flight loads with a structural factor of safety of 3	DTEG §8	Analysis	7.1
STR-002	The primary structure shall withstand the maximum predicted axial flight load without structural failure	Structural analysis	Analysis	7.1
STR-003	The primary structure shall withstand the maximum predicted lateral flight load without structural failure	Structural analysis	Analysis	7.1
STR-004	The recovery attachment structure shall withstand the maximum predicted recovery load with a factor of safety of 3	DTEG §8.4	Analysis	5.5
STR-005	The vehicle shall maintain structural interfaces between major vehicle sections throughout flight and recovery	System integration	Analysis	5.5
STR-006	The vehicle shall prevent unintended separation of structural sections during powered flight and coast	Flight safety	Analysis	7.1
STR-007	The vehicle shall prevent unintended separation of structural sections before commanded recovery events	Recovery safety	Analysis	5.3, 10.2
STR-008	The airframe shall not use PVC, PML Quantum Tube, or steel as primary structural materials	DTEG §8.3	Inspection	7.1
STR-009	The vehicle shall provide structural support for all flight-critical internal components such that component displacement under maximum predicted flight loads does not exceed [TBD] in (Section 4.9)	System integration	Inspection	7.3
STR-010	The vehicle shall maintain structural integrity under the thermal environment produced by the propulsion and recovery systems	Structural/thermal interface	Analysis	1.1, 7.4
AERO-001	The vehicle shall maintain the required static stability margin	DTEG §10; SYS-007/008	Analysis	3.2

ID	Requirement	Source	Verif.	Function
	throughout the defined flight envelope			
AERO-002	The vehicle shall have a fin semi-span of at least 0.8 times the largest airframe diameter	DTEG §10.2.4	Inspection	3.1
AERO-003	The vehicle shall have a predicted fin flutter velocity of at least 150% of the maximum expected vehicle velocity	DTEG §8.2.2	Analysis	7.1
AERO-004	The vehicle shall remain aerodynamically stable throughout powered flight and coast	Flight safety	Analysis	3.1
AERO-005	The vehicle shall maintain aerodynamic stability during all commanded airbrake operating states	Airbrake/vehicle interface	Analysis	3.2, 4.4
AERO-006	The vehicle shall remain within the applicable aerodynamic operating envelope throughout flight	Flight analysis	Analysis	3.3
AV-001	The avionics system shall acquire a valid GPS position fix no later than 10 minutes before launch	IREC R&R §2.4	Test	5.7
AV-002	The avionics system shall provide vehicle position information sufficient to support post-flight recovery	IREC R&R §2.4	Test	5.7, 9.3
AV-003	The avionics system shall record flight altitude data throughout the portion of flight required to determine apogee	IREC R&R §2.5	Test	9.2
AV-004	The avionics system shall retain recorded apogee data after landing until the data are retrieved	IREC R&R §2.5	Test	9.2
AV-005	The avionics system shall record timing information with a resolution of [TBD] ms or better to determine the occurrence of critical flight events (Section 4.9)	Verification requirement	Test	9.2
AV-006	The avionics system shall remain operational throughout the acceleration, vibration, temperature, and pressure profile defined in [TBD environmental test spec] (Section 4.9)	Avionics reliability requirement	Test	7.4
AV-007	The avionics system shall provide the information required by the recovery	Avionics/recovery interface	Test	5.1

ID	Requirement	Source	Verif.	Function
	system to initiate the required recovery events			
AV-008	The avionics system shall provide an unambiguous indication of flight-critical system readiness before launch	Flight Readiness Team stakeholder need	Demonstration	9.4, 10.1
AV-009	Flight-critical electrical connections shall maintain electrical continuity throughout the expected flight environment	DTEG §6.12	Test	8.3
AV-010	Flight-critical electrical connections shall include mechanical retention sufficient to prevent unintended disconnection during flight	DTEG §6.12	Test	8.3
AV-011	The vehicle shall use battery technology permitted by the applicable IREC requirements	DTEG §6.17.1.1	Inspection	8.1
AV-012	The avionics system shall provide electrical energy for all required flight functions for the complete mission duration with a design margin of at least [TBD]% beyond nominal mission duration (Section 4.9)	Avionics reliability	Analysis	8.1
REC-001†	The recovery system shall initiate the drogue recovery event at or within 100 ft of apogee	DTEG §6.3.1 (“at or near apogee”); ±100 ft tolerance is team-derived	Test	5.1, 5.2
REC-002	The recovery system shall produce a drogue descent rate between 20 m/s and 40 m/s at main-parachute deployment	DTEG §6.3.3	Test	5.4
REC-003	The recovery system shall initiate main-parachute deployment at or below 1,500 ft AGL	DTEG §6.4.1	Test	5.2
REC-004	The recovery system shall produce a main-parachute landing descent rate below 11 m/s (36 ft/s)	DTEG §6.4.3	Test	5.4, 5.6
REC-005	The recovery system shall provide two functionally independent deployment channels	DTEG §6.9	Inspection	5.2
REC-006	Each recovery deployment channel shall have an independent power source	DTEG §6.9	Inspection	5.2
REC-007	Each recovery deployment channel shall be capable of independently initiating its assigned recovery event	DTEG §6.9	Test	5.2

ID	Requirement	Source	Verif.	Function
REC-008	The recovery system shall prevent unintended deployment before the intended recovery event	Recovery safety	Test	10.2
REC-009	The recovery system shall maintain a continuous structural load path between the vehicle and recovery components during deployment and descent	Structural/recovery interface	Analysis	5.5
REC-010	The recovery system shall retain its parachutes and attachment hardware	DTEG §6.5; recovery safety	Test	5.5
REC-011	The recovery system shall protect required recovery components from damage caused by deployment-system thermal or exhaust products	DTEG §6.5	Inspection	7.4
REC-012	The recovery system shall provide physical volume for all required recovery components and their deployment paths within the finalized recovery-bay envelope (Section 4.9)	System integration	Inspection	7.3
REC-013	The recovery system shall provide a defined recovery state following loss of one recovery deployment channel	DTEG §6.9; redundancy	Analysis	10.3
REC-014	The main and drogue parachutes shall be visually distinguishable from the vehicle and surrounding environment after deployment	DTEG §6.7.1	Inspection	5.7
PT-001	The propulsion system shall provide sufficient total impulse to enable the vehicle to satisfy the required apogee performance	SYS-001; mission performance	Analysis	1.3
PT-002	The propulsion system shall provide sufficient thrust to satisfy the vehicle's minimum 5:1 liftoff thrust-to-weight ratio	SYS-005; launch safety	Analysis	1.3
PT-003	The propulsion system shall operate within the certified operating limits of the selected motor	IREC R&R §2.1; motor certification	Inspection	1.3
PT-004	The propulsion system shall remain mechanically secured to the vehicle throughout powered flight	SYS-009; structural/propulsion interface	Test	1.1
PT-005	The propulsion system shall withstand the thermal environment generated during motor operation without loss of required vehicle function	SYS-009; propulsion environment	Analysis	1.1

ID	Requirement	Source	Verif.	Function
PT-006	The propulsion system shall provide a mechanical interface compatible with the applicable launch system	Launch-system interface constraint	Inspection	2.1
PT-007	The propulsion system shall not cause the vehicle to exceed the applicable structural or aerodynamic flight envelope	SYS-009; vehicle safety	Analysis	1.4
AB-001†	The airbrake system shall control aerodynamic drag during the designated portion of flight to support the vehicle's apogee objective	SYS-013 (proposed); airbrake stakeholder need	Analysis	4.4
AB-002	The airbrake system shall respond to valid flight-state information within [TBD] ms to maintain the specified apogee-control performance (Section 4.9)	Airbrake control requirement	Analysis	4.4
AB-003†	The airbrake system shall prevent unintended aerodynamic deployment before the defined active-control flight state	DTEG §7.4.1 (2026 ed.)	Test	4.5, 10.2
AB-004†	The airbrake system shall transition to a defined safe state following loss of valid control data or control power	DTEG §7.3.1 (2026 ed.)	Test	4.5, 10.3
AB-005†	The airbrake system shall not reduce the vehicle's stability margin below the applicable requirement during operation	DTEG §7.2.1, §10.2.1 (2026 ed.)	Analysis	3.2
AB-006	The airbrake system shall remain mechanically attached to the vehicle throughout the expected flight environment	Structural interface	Test	4.4
AB-007	The airbrake system shall not interfere with required recovery-system operation	Recovery/airbrake interface	Analysis	4.5
PAY-001	The vehicle shall provide the physical interfaces required to accommodate the approved competition or sponsor payload	Sponsor/payload stakeholder need	Inspection	6.1
PAY-002	The vehicle shall provide payload volume of at least [TBD] in^3 and allowable mass capacity of at least [TBD] lb for the approved payload configuration (Section 4.9)	Payload integration	Inspection	6.1
PAY-003	The payload shall remain mechanically secured throughout the expected flight and recovery loads	Payload safety	Test	6.1

ID	Requirement	Source	Verif.	Function
PAY-004	Payload integration shall not reduce the vehicle's required stability margin below the applicable requirement	DTEG §10; vehicle integration	Analysis	3.2
PAY-005	Payload integration shall not prevent required propulsion, avionics, airbrake, structural, or recovery functions from operating	System integration	Analysis	6.1
INT-001	Each major subsystem shall maintain its defined mechanical interface with adjacent vehicle subsystems throughout the expected flight environment	System integration	Inspection	7.3
INT-002	Each flight-critical electrical interface shall maintain required electrical continuity throughout the expected flight environment	Avionics integration	Test	8.3
INT-003	Subsystem interfaces shall provide a minimum clearance of [TBD] in for installation, operation, and removal of required components (Section 4.9)	Maintainability/integration	Inspection	7.3
INT-004	Subsystem interfaces shall not introduce a failure mode that prevents an independent subsystem from performing its required function	System reliability	Analysis	10.3
INT-005	The final vehicle configuration shall maintain the required center-of-gravity and center-of-pressure relationship after integration of all subsystems	DTEG §10; vehicle stability	Analysis	3.2
INT-006	The vehicle shall provide defined interfaces for electrical power, data, structural attachment, and recovery connections where required by the applicable subsystem	System architecture	Inspection	System
ENG-001	Each flight-critical requirement shall have an assigned verification method before flight-readiness approval	Systems engineering process; INCOSE/NASA guidance	Inspection	Prog.
ENG-002	Each flight-critical subsystem shall have defined requirements, interfaces, and verification criteria before final design release	Systems engineering process	Inspection	Prog.
ENG-003	The vehicle mass properties shall be experimentally determined for the final flight configuration before launch	Flight-performance/stability analysis	Test	3.2

ID	Requirement	Source	Verif.	Function
ENG-004	The vehicle center of gravity shall be experimentally verified for the final flight configuration before launch	Stability requirement	Test	3.2
ENG-005	The predicted center of pressure and static stability margin shall be calculated for the final flight configuration	DTEG §10	Analysis	3.2
ENG-006	All flight-critical design changes shall be documented before the modified configuration is approved for flight	Configuration management	Inspection	Prog.
ENG-007	The verified flight configuration shall be traceable to the configuration presented for competition flight	Configuration control	Inspection	Prog.
ENG-008	Each flight-critical system shall have a defined failure response for identified single-point failures	Reliability/safety	Analysis	10.3
ENG-009	Each flight-critical system shall undergo the verification activities required to demonstrate compliance before flight	Verification planning	Inspection	Prog.
ENG-010	Requirements, design elements, interfaces, and verification evidence shall maintain traceability throughout the project	MBSE/INCOSE practice	Inspection	Prog.
DOC-001	The team shall submit all required IREC competition documentation by the applicable competition deadlines	IREC R&R §2.8	Inspection	Prog.
DOC-002	The team shall maintain the Master Requirements Traceability Matrix throughout the design process	Systems engineering process	Inspection	Prog.
DOC-003	The team shall provide the required preliminary design documentation before the applicable design review	IREC / senior design process	Inspection	Prog.
DOC-004	The team shall provide the required critical design documentation before the applicable design review	IREC / senior design process	Inspection	Prog.
DOC-005	The team shall maintain records demonstrating verification of all flight-critical system requirements	Verification/validation	Inspection	Prog.
DOC-006	The team shall document the final verified vehicle configuration before competition flight	Configuration management	Inspection	Prog.
DOC-007	The team shall provide the required post-flight data to competition	IREC competition requirement	Inspection	9.2

ID	Requirement	Source	Verif.	Function
	officials according to applicable IREC procedures			
DOC-008	The team shall maintain documentation sufficient to support transfer of the vehicle design and engineering knowledge to future project teams	Future-team stakeholder need	Inspection	Prog.
TST-001	The project shall provide a method to verify operation of each recovery deployment channel before flight	Recovery safety	Demonstration	5.2
TST-002	The project shall provide a method to verify avionics readiness before flight	Flight readiness	Demonstration	9.4
TST-003	The project shall provide a method to verify electrical continuity of flight-critical circuits before flight	DTEG §6.12	Test	9.4
TST-004	The project shall provide a method to verify the final vehicle mass before flight	Performance/stability	Test	3.2
TST-005	The project shall provide a method to verify the final vehicle center of gravity before flight	Stability	Test	3.2
TST-006	The project shall provide a method to verify recovery deployment functionality without requiring a competition flight	Verification/safety	Test	5.2
TST-007	The project shall retain objective evidence for each completed flight-critical verification activity	Verification management	Inspection	Prog.
TST-008	The project shall provide a method to determine whether critical flight events occurred within their specified requirements using recorded flight data	Post-flight verification	Test	9.2
Proposed additions				
PT-008	The propulsion system shall have a total impulse of no more than 40,960 N·s.	FAA JO 7400.2R §31-1-5 [3]	Inspection	1.3
PAY-006	The vehicle shall carry a payload with a mass of at least 2.0 kg (4.4 lb).	IREC R&R §2.3.1 [1]	Inspection	6.1
ENG-011	All fabrication shall follow the facility safety protocols and equipment procedures of the CSU ERC and EMEC.	CSU ERC/EMEC rules	Inspection	Program

ID	Requirement	Source	Verif.	Function
ENG-012	Total project expenditure shall not exceed the $20,000 baseline budget.	Team budget	Inspection	Program
AV-013	The avionics system shall measure altitude with an uncertainty of no more than ±50 ft up to 13,000 ft AGL (TBR).	Derived from SYS-001, REC-001	Test	4.1, 5.1
AV-014	The avionics system shall detect motor burnout and shall not enable airbrake deployment before burnout is detected.	Derived from AB-003; DTEG §7	Test	4.2
SYS-013	The vehicle shall provide a means of controlling apogee after motor burnout so that apogee error stays within ±500 ft of 10,000 ft AGL (TBR).	Derived from SYS-001 and measure M1; Section 3.4	Analysis	4.3, 4.4

B. Subsystem Decision Matrices

Each matrix uses the same layout and scale. Criteria are rows, concepts are columns, weights run from 1 to 5, scores run from 1 to 10 (higher is better), and the weighted total is the sum of weight times score. The highest total is shaded. Section 4.4 cites and discusses each matrix. Tables 43 to 47 cover the airframe material, airbrakes, recovery, avionics and payload; the nose cone and fin matrices are in Appendix D.

Table 43. Decision matrix: airframe material.

Criterion	Weight (1–5)	Carbon fiber, prepreg layup	Carbon fiber, wet layup	Fiberglass, prepreg layup	Fiberglass, wet layup	Off-the-shelf carbon fiber	Off-the-shelf fiberglass	6061 aluminum
Cost	1	2	4	4	6	8	8	6
Manufacturability	3	6	4	6	4	10	10	2
Performance	5	9	7	7	5	8	7	3
Bonus scoring (SRAD)	4	10	10	10	10	0	0	7
Customizability (ply orientations)	2	8	10	8	10	2	2	4
Total (weighted)		**121**	**111**	**113**	**103**	**82**	**77**	**63**

Table 44. Decision matrix: airbrake mechanism.

Criterion	Weight (1–5)	Linear track	Linkage	Rack and pinion	No airbrakes
Ease of integration	4	6	6	6	6
Manufacturability	5	4	5	2	6
Compactness	5	4	2	2	6
Apogee control	5	7	6	5	1
Total (weighted)		**99**	**89**	**69**	**89**

Table 45. Decision matrix: recovery parachute.

Criterion	Weight (1–5)	IRIS Ultralight	Printed nylon	Rip stop
Cost	1	9	7	9
Weight	2	8	8	8
Performance	4	10	9	8
Packability	3	9	8	9
Total (weighted)		**92**	**83**	**84**

Table 46. Decision matrix: avionics layout.

Criterion	Weight (1–5)	Vertical panel	Horizontal stack	Hybrid
Cost	1	9	7	10
Manufacturability	3	9	10	8
Ease of integration	4	8	7	9
Compactness	4	7	8	7
Total (weighted)		**96**	**97**	**98**

Table 47. Decision matrix: SDL payload experiment.

Criterion	Weight (1–5)	Thermodynamic cycle	RF attenuation	Internal flight environment mapping	Atmospheric particulate sampling	Soil radiation sampling
Scientific value	5	10	8	6	8	6
Clarity of hypothesis	4	9	6	6	8	8
Ground testability	2	9	10	10	7	10
Reliability	4	7	10	10	7	7
Package and mass difficulty	1	2	8	8	8	3
Total (weighted)		**134**	**132**	**122**	**122**	**113**

Table 48. Evidence needed to replace judgment scores.

Matrix	Criterion	Score basis today	Evidence that would replace it	Owner (proposed)
Airframe material	Performance	Judgment	Specific strength and stiffness from laminate analysis (CLT, Ansys ACP)	Structures
Airframe material	Bonus scoring (SRAD)	Sourced, citation missing	Cite the SRAD scoring rule in the R&R	Systems
Airframe material	Manufacturability	Judgment	Layup trial results and ERC schedule	Structures
Airbrake mechanism	Apogee control	Judgment	RocketPy authority for each mechanism (gate G1)	Airbrake lead
Recovery parachute	Performance	Calculated for descent rate; other parts judgment	Descent-rate calculation and deployment tests	Recovery

Matrix	Criterion	Score basis today	Evidence that would replace it	Owner (proposed)
Avionics layout	Manufacturability, compactness	Judgment	Trial print and bay-length check	Avionics
Payload experiment	Scientific value, clarity of hypothesis	Judgment	Mapping to the SDL rubric	Payload
Payload experiment	Reliability, package and mass difficulty	Judgment	Bench test and component mass list	Payload
Nose cone shape	Drag	Judgment	OpenRocket drag for each shape	Aerodynamics
Fin shapes	Flutter margin, stability	Judgment	Flutter calculation (AERO-003) and OpenRocket margin for each fin	Structures and aerodynamics

C. Black Powder Charge Calculations

Table 49 gives the charge sizing for the main and drogue bays. The charge must produce a pressure that separates the vehicle against the shear pins, friction and inertial forces. The design pressure is the pressure that gives the required separation force over the bay cross-section, times a factor of safety of 3. The charge mass follows from the ideal-gas relation $W = \Delta P \cdot V / (RT)$, where ΔP is the pressure rise the charge must produce, V is the free volume of the bay, R is the gas constant of the combustion products and T is the combustion temperature. The table uses RT = 219 kJ/kg, the standard value for FFFFg black powder. Each charge is fired through the electrical initiation path in ICD-010, from its own altimeter channel (Section 5.5.4).

Table 49. Black powder charge sizing for the main and drogue bays.

Parameter	Main parachute	Drogue parachute	Comments
Deployment altitude (AGL)	1,000 ft	10,000 ft (apogee)	Pueblo launch site (5,404 ft MSL)
Bay dimensions (L × ID)	293.5 mm × 152.4 mm	226 mm × 152.4 mm	Internal chamber dimensions
Free volume (V)	0.00535 m³	0.00412 m³	Net pressurized volume
Shear pins	4 × 2-56 nylon (M2)	4 × 2-56 nylon (M2)	Total holding force: 487.4 N
Required separation force	504.3 N	504 N	Includes friction and inertial forces
Design ejection pressure	82.9 kPa	82.9 kPa	Includes factor of safety of 3
Absolute bay pressure	Not used	Not used	Charge mass uses the pressure rise
Mass of black powder (FFFFg)	2.03 g	1.56 g	$W = \Delta P \cdot V/(RT)$, RT = 219 kJ/kg
Shock cord peak load × 3	TBD (CDR)	TBD (CDR)	Tether and bulkhead design load

D. Nose Cone, Fin and Motor Studies

D.1 Nose cone shape. Table 50 compares six shapes on drag, stability effect and manufacturability. The Haack series scored highest (92 against 87 for the power series).

Table 50. Decision matrix: nose cone shape.

Criterion	Weight (1–5)	Conical	Ogive	Ellipsoid	Power series	Parabolic series	Haack series
Drag	5	6	7	5	9	9	10
Stability effect	2	7	8	7	9	8	9
Manufacturability	3	8	8	7	8	7	8
Total (weighted)		**68**	**75**	**60**	**87**	**82**	**92**

D.2 Fin planform and cross-section. Table 51 and Table 52 compare the fin shapes. The clipped delta won the planform (97 against 87). The cross-section is a near tie between the double wedge (80) and the rounded section (78); the tie-breaker is in Section 4.4.1.

Table 51. Decision matrix: fin planform.

Criterion	Weight (1–5)	Clipped delta	Trapezoidal	Tapered swept
Apogee	5	8	7	8
Stability	3	7	7	9
Flutter margin	4	9	7	5
Total (weighted)		**97**	**84**	**87**

Table 52. Decision matrix: fin cross-section.

Criterion	Weight (1–5)	Rounded	Double wedge	Airfoil
Stability	3	6	8	8
Maximum velocity	4	6	7	9
Manufacturability	4	9	7	4
Total (weighted)		**78**	**80**	**76**

D.3 Motor. The motor survey is in Table 26 (Section 4.3.2). The decision matrix is being rebuilt so that each criterion has its own scores.

D.4 Manufacturing choices. The airframe layup needs a mandrel and a way to hold it for machining. Table 53 and Table 54 compare the choices the team considered. The team marks 6061 aluminum for the mandrel and soft jaws for the fixturing.

Table 53. Mandrel material choices.

Mandrel material	Pros	Cons
6061 aluminum (marked as selected)	Easy to manufacture; high coefficient of thermal expansion (CTE); lightweight; relatively cheap.	Tolerances may vary with temperature; easy to ding or scratch.
1018 steel	Less tolerance variation with temperature; less likely to be scratched.	Hard to manufacture; lower CTE; heavy; more expensive.

Table 54. Fixturing choices.

Fixturing type	Pros	Cons
Soft jaws (marked as selected)	Extremely flexible geometry; cheap and accessible.	Requires extra machining.
Vacuum plate	Extremely versatile; requires no extra machining.	Hard to fix complicated geometries; hard to gain access.

E. Budget and Timeline

Table 55 gives the budget by subsystem, and Table 56 gives the funding sources and whether each is secured or predicted. Of the $20,000 budget, $2,375.11 has been spent and $17,624.89 remains. Secured funding is $10,000, 42% of the $23,800 in listed sources: the Colorado Space Grant ($5,000) and Relativity ($5,000). The other $13,800 (58%) is predicted, so the budget carries funding risk until the fundraising lands. The listed sources exceed the $20,000 budget by $3,800, which is a margin only if the predicted sources arrive. In-kind support is not counted: donated composite materials from Ability Composites, OnlineMetals coupons, Prodigm machine access and a 20% BigRedBee discount. Table 57 lists the flights, and the first three are validation events.

Table 55. Budget by subsystem.

Subsystem	Predicted	Actual to date	Remaining
Structures	$6,000.00	$966.80	$5,033.20
Payload	$800.00	$0.00	$800.00
Avionics	$1,500.00	$1,231.31	$268.69
Recovery	$900.00	$0.00	$900.00
Propulsion	$3,000.00	$0.00	$3,000.00
Airbrakes	$800.00	$177.00	$623.00
Fees and miscellaneous	$7,000.00	$0.00	$7,000.00
Total	**$20,000.00**	**$2,375.11**	**$17,624.89**

Table 56. Funding sources.

Source	Amount	Status	Notes
Senior design (MECH 486)	$7,500	Predicted	Course allocation
Relativity	$5,000	Secured	Gift; confirm amount, terms and any spending deadline with Jay
Colorado NASA Space Grant Consortium	$5,000	Secured	Awarded; held in two university accounts, up to $800 of it for travel; must be spent by early May 2027
RamFunder	$4,000	Predicted	CSU crowdfunding campaign
Travel grant	$1,200	Predicted	Competition travel
RamRide	$500	Predicted	Volunteer fundraiser
Food fundraisers	$600	Predicted	

Source	Amount	Status	Notes
Total	**$23,800**		**Secured $10,000 (42%); predicted $13,800 (58%)**

Table 57. Flight timeline.

Event	Tentative date	Validation event
Subscale 1 (SS1) launch	November 28, 2026	Yes
Full-Scale 1 (FS1) launch	March 13, 2027	Yes
Full-Scale 2 (FS2) launch	May 15, 2027	Yes
Final competition launch	June 2027 (team schedule: June 20; ESRA: June 14–19)	Competition flight

F. Risk Matrix / FMEA

The risk matrix lists the hazards from the failure modes and effects analysis (FMEA). Impact is scored from 1 to 4 and probability from 1 to 4, and the risk score is their product. Table 58 defines the scales and the risk levels. Table 59 lists the baseline hazards first (HZ-01 to HZ-21), then the hazards that exist only because of the airbrakes (HZ-A1 to HZ-A3) and the payload experiment (HZ-P1). Each safety-critical feature in Section 5.4 names its rows.

Table 58. Impact and probability scales and risk levels.

Impact (I)	Probability (P)	Risk score (I × P) and level
4 Catastrophic	4 Frequent	10 to 16: High
3 Critical	3 Probable	6 to 9: Serious
2 Marginal	2 Remote	1 to 5: Medium
1 Negligible	1 Improbable	

Table 59. Hazard list and risk matrix (I = impact, P = probability).

ID	Failure mode	Effect	I	P	Score	Risk	Mitigation (design)	Linked requirement
Baseline hazards								
HZ-01	Altimeter reads wrong altitude	Late or early drogue/main deployment	4	2	8	Serious	Redundant altimeters with cross-check; vented ports; vacuum-chamber test	AV-003, AV-007, REC-001, REC-003
HZ-02	Avionics loses power or resets in flight	No deployment command; ballistic descent	4	2	8	Serious	Independent power per channel; connector retention; vibration test	AV-006, AV-009 to AV-012, REC-006
HZ-03	Drogue does not deploy	No deceleration; high-speed descent	4	2	8	Serious	Two independent channels; charge sized with FOS 3; ground ejection test	REC-001, REC-005, REC-007, REC-013
HZ-04	Main does not deploy	Landing at drogue speed (20–40 m/s)	3	2	6	Serious	Independent main channel; same checks as HZ-03	REC-003, REC-004, REC-007
HZ-05	Premature or uncommanded deployment	Separation at speed; hazard to people at the pad	4	2	8	Serious	Inert until armed; no output before the programmed event	REC-008, AV-008, TST-001
HZ-06	Unintended separation of airframe sections	Loss of vehicle	4	2	8	Serious	Joint sizing; shear-pin force checked against flight loads	STR-005, STR-006, STR-007

ID	Failure mode	Effect	I	P	Score	Risk	Mitigation (design)	Linked requirement
HZ-07	Airframe structural failure	Breakup in flight	4	2	8	Serious	Factor of safety 3; Ansys model; coupon and section tests; laminate inspection	STR-001 to STR-003, STR-010
HZ-08	Recovery attachment failure	Sections fall separately	4	2	8	Serious	Design for peak load with FOS of 3 (load calculated at CDR); pull test	STR-004, REC-009, REC-010
HZ-09	Fin flutter or fin joint failure	Loss of stability	4	2	8	Serious	Flutter margin at least 150%; fin joint test	AERO-002, AERO-003
HZ-10	Static margin outside limits	Unstable or over-stable flight	4	1	4	Medium	Measure mass and CG; recompute margin before flight	SYS-007, SYS-008, ENG-003 to ENG-005
HZ-11	Mass above cap, low T/W or slow rail exit	Unstable departure from the rail	3	1	3	Medium	Mass budget with reserve; weigh before flight	SYS-005, SYS-006, SYS-010
HZ-12	Motor retention or thermal failure	Motor moves or burns through structure	4	1	4	Medium	Retainer sized with FOS of 3; certified COTS motor; thermal protection	SYS-004, PT-003 to PT-005
HZ-13	Ejection gas damages recovery parts or structure	Burned cord or parachute	3	2	6	Serious	Protect recovery components from ejection products	REC-011, STR-010
HZ-14	Vehicle lands outside the area or drifts toward people	Public hazard; vehicle lost	3	2	6	Serious	Staged descent rates; GPS tracking	REC-002, REC-004, AV-002
HZ-15	GPS fix not acquired or lost	Vehicle not located	2	2	4	Medium	Fix confirmed before launch; position sent after landing	AV-001, AV-002
HZ-16	Black powder handling incident	Injury at the pad	4	1	4	Medium	Charge sized by calculation and no larger; inert until armed; handling per operations procedures	REC-008, SYS-009
HZ-17	Fabrication injury at the ERC or EMEC	Injury; loss of shop access	3	2	6	Serious	Facility protocols and PPE	None yet (ENG-011 proposed)
HZ-18	As-flown configuration differs from verified	Unverified vehicle flies	3	2	6	Serious	Document changes before approval; trace the flown configuration	ENG-006, ENG-007, DOC-006

ID	Failure mode	Effect	I	P	Score	Risk	Mitigation (design)	Linked requirement
HZ-19	Apogee outside the 7,000–13,000 ft window	Lost score	3	4	12	High	OpenRocket and motor selection; airbrakes if confirmed	SYS-001
HZ-20	Verification not finished before flight	Flight delayed or rushed	3	3	9	Serious	Finish design 2 weeks before CDR; rehearsals and scaled flights	ENG-009
HZ-21	Funding shortfall	Scope cut	3	2	6	Serious	Track spending against the $20,000 budget	Team and sponsor constraint
Exist only because of the airbrakes (added by design decision)								
HZ-A1	Airbrakes deploy prematurely (airbrake only)	Drag and loads during boost	4	2	8	Serious	Inhibit control during powered flight	AB-003
HZ-A2	Airbrakes jam or lose data or power (airbrake only)	Wrong drag; apogee error	3	2	6	Serious	Default to neutral drag on a fault	AB-004, AB-006
HZ-A3	Stability drops when deployed (airbrake only)	Unstable flight	4	2	8	Serious	Cap authority at the stability floor; CFD and wind-tunnel check	AB-005, AERO-005
Exists only because of the payload experiment (added by design decision)								
HZ-P1	Payload shifts or detaches (payload only)	CG shift or interference	3	2	6	Serious	Mechanical retention; document final CG and mass	PAY-003, PAY-004

G. Systems Engineering Calculations

G.1 Liftoff mass limit from the rail-exit requirement (SYS-006). A vehicle accelerating along the rail reaches $v^2 = 2aL$ at the end of its effective length L. Here v is the rail-exit speed (25 m/s), a is the acceleration along the rail, and L is the rail length the vehicle travels before it leaves. The acceleration is the thrust per unit mass minus the part of gravity acting along the rail: $a = T/m - g \cos \theta$, where T is the average motor thrust, m is the liftoff mass, g is 9.81 m/s², and θ is the rail angle from vertical. Solving for mass gives $m = T / (a + g \cos \theta)$, the heaviest vehicle that still reaches 25 m/s, with $a = v^2 / (2L)$. Table 60 gives the inputs and results.

Table 60. Mass limit from the rail-exit requirement.

Quantity	Value	Basis
Required rail-exit velocity v	25 m/s	SYS-006, DTEG §5.3.1 [2]
Effective rail length L	13 ft = 3.96 m	DTEG example; real value depends on rail-button placement
Required acceleration $a = v^2/(2L)$	78.9 m/s²	Calculated
Rail angle θ from vertical	6°	Assumed; 4–8° to be swept in OpenRocket
Average motor thrust T	about 2,500 N	M2500T class designation [10]; constant-thrust estimate
Maximum liftoff mass $m = T/(a + g \cos \theta)$	28.2 kg (62 lb)	Calculated
Thrust-to-weight at that mass	9.0	Greater than the 5:1 of SYS-005, so SYS-005 is not binding
Exit velocity if T/W were exactly 5:1	17.6 m/s	a = 4g; below 25 m/s
OpenRocket rail-exit speed at 24.12 kg	31.9 m/s	Full thrust curve; 4.36 m effective rail (Section 5.3.1)
OpenRocket rail-exit speed at 28 kg	about 29 m/s	Same thrust curve (Section 5.3.4)

The hand estimate assumes constant average thrust, constant mass, and no drag or rail friction. The OpenRocket result with the full thrust curve and the 4.36 m effective rail gives about 29 m/s at 28 kg, so the 28 kg cap is conservative. For this motor the apogee prediction, not the rail-exit speed, limits the liftoff mass (Section 5.3.4). The cap in SYS-010 stays TBR until the final rail geometry and the final airbrake mass are in the model.

G.2 Apogee score sensitivity (M1). The 2026 apogee score is $350 - (350 / (0.3 \times \text{target})) \times |\text{target} - \text{actual}|$ [1]. For a 10,000 ft target the slope is 0.117 points per foot, so each foot of error costs that much and a miss of 100 ft costs 11.7 points, 250 ft costs 29.2 points, 500 ft costs 58.3 points and 1,000 ft costs 116.7 points. The score reaches zero at 7,000 ft and 13,000 ft. At the current predicted error of +1,484 ft the score is 176.9 points; if the airbrakes hold the error to ±100 ft it is at least 338.3 points.

G.3 Descent-rate check. The descent speed of a parachute follows from the balance of weight and drag, $v = \sqrt{(2mg / (\rho C_d A))}$ (Section 5.5.3). Table 61 applies it to the 26 in drogue and the 96 in main with C_d = 2.2 and ρ = 1.089 kg/m³, for the three vehicle masses that appear in the team's documents. The speeds

depend on the mass assumed, and the drogue margin over the 20 m/s floor of REC-002 is the thinnest margin in the recovery system.

Table 61. Descent speed for three vehicle mass assumptions.

Vehicle mass (kg)	Source of the mass	Drogue (26 in), m/s	Margin over 20 m/s	Main (96 in), m/s	Margin under 11 m/s
18.15	Recovery draft (Section 5.5.3)	20.8	+0.8	5.64	+5.4
19.41	OpenRocket mass at burnout (Section 5.3.2)	21.5	+1.5	5.83	+5.2
20.80	Implied by Section 5.3.3 (22.3 and 6.0 m/s)	22.3	+2.3	6.04	+5.0

H. Resource Allocation

This appendix allocates the main resources among subsystems and shows how much of each is itemized, so that the margins in Section 5.6 have numbers behind them. Mass values come from the preliminary OpenRocket model and the subsystem sections; they are estimates until parts are weighed.

H.1 Growth allowance (proposed policy). Early mass estimates are low more often than high, so the team should hold a reserve and shrink it as design matures. The percentages below are proposed for the team to set; they are not taken from a standard. A 15% allowance at this stage on the 24.12 kg model is 3.6 kg, just under the 3.88 kg of reserve below the 28 kg cap.

Table 62. Proposed mass growth allowance by stage.

Stage	Mass growth allowance on itemized mass (proposed)	Rule
PDR (now)	15%	Reserve held against the cap
CDR	10%	All parts itemized with a source
Before FS1	5%	Parts weighed where built
Competition	0%	Measured mass only (ENG-003, TST-004)

H.2 Mass allocation.

Table 63. Mass allocation from the preliminary model.

Item	Mass (kg)	Source
AeroTech M2500T, gross (includes 4.71 kg propellant)	8.06	Motor data [10]
Payload experiment (2.69 kg) and integration hardware (0.14 kg)	2.83	OpenRocket model, Section 5.3.4
Nose cone ballast	0.57	OpenRocket model
Fin assembly	1.48	Section 5.3.4
Fin-can structural epoxy	1.59	Section 5.3.4
Machined aluminum boattail	0.80	Section 5.3.4
Airbrake retention bulkhead	0.68	Section 5.3.4
Airbrake mechanism and electronics (model value; 1.7 kg in Section 5.5.5)	0.97	OpenRocket model
Recovery system (parachutes, shock cords, hardware)	0.5	Section 5.5.3, estimate
Itemized subtotal	17.48	
Not itemized in this report (the owners should list airframe tubes, nose cone shell, avionics bay and other parts)	6.64	Remainder of the model
Liftoff mass in the model	24.12	OpenRocket

Item	Mass (kg)	Source
Preliminary cap (SYS-010, TBR)	28.00	Appendix G
Reserve under the cap	3.88	The airbrake mass difference (1.7 − 0.97 = 0.73 kg) would use 19% of it

H.3 Geometry and volume. The diameter is the largest open geometric conflict in the design (Table 64).

Table 64. Geometry and volume allocation.

Dimension	Value	Source	Status
Vehicle length	116.3 in	OpenRocket model	Preliminary
Outside diameter	6.0 in (152.4 mm)	OpenRocket model	Matches the 152.4 mm bay in Appendix C
Recovery bay length, main and drogue	293.5 mm and 226 mm	Appendix C	Based on a 152.4 mm inside diameter
Payload form factor	100 × 100 × 300 mm (141 mm diagonal)	Section 4.3.6	Fits a 152.4 mm bay
Maximum diameter and length	6.0 in; 10 ft (proposed)	SYS-011	Proposed

H.4 Electrical energy. Each flight computer has its own 9 V battery (Figure 9). The loads and the AV-012 margin have not been calculated. The avionics subteam fills Table 65, and the margin is set from it.

Table 65. Electrical energy budget (to be completed).

Load	Voltage	Current	Duty	Energy for the mission
Flight computer 1 (TeleMetrum)	TBD	TBD	TBD	TBD
Flight computer 2 (RRC3+)	TBD	TBD	TBD	TBD
GPS tracker	TBD	TBD	TBD	TBD
Airbrake actuator and controller (option)	TBD	TBD	TBD	TBD
Payload compressor and computer (option)	TBD	TBD	TBD	TBD

H.5 Cost shares. Table 66 shows each subsystem's share of the $20,000 budget.

Table 66. Budget shares by subsystem.

Subsystem	Budget	Share of $20,000	Spent to date
Structures	$6,000.00	30%	$966.80
Payload	$800.00	4%	$0.00
Avionics	$1,500.00	7.5%	$1,231.31
Recovery	$900.00	4.5%	$0.00
Propulsion	$3,000.00	15%	$0.00
Airbrakes	$800.00	4%	$177.00
Fees and miscellaneous	$7,000.00	35%	$0.00

I. Traceability Matrix

Table 67 traces every requirement from its need code through its function, subsystem, interface, hazard row and verification method to its test group. “—” means there is no entry. Need codes are shown only for the requirements listed in Table 8. Section 3.5 summarizes the coverage.

Table 67. Requirements traceability matrix. * proposed. Verif.: T test, A analysis, I inspection, D demonstration.

ID	Need	Function	Subsystem	ICD	Hazard	Verif.	Test group
SYS-001	N2, N9	1.3, 4.3	System	—	HZ-19	T	Flight*
SYS-002	N11	System	System	—	—	T	Flight*
SYS-003	—	1.3	System	—	—	I	—
SYS-004	N6, N8	1.3	System	—	HZ-12	I	—
SYS-005	N2	1.3, 2.2	System	—	HZ-11	A	—
SYS-006	N7, N10	2.2	System	ICD-008	HZ-11	A	—
SYS-007	N6, N7, N8	3.2	System	—	HZ-10	A	—
SYS-008	N7	3.2	System	—	HZ-10	A	—
SYS-009	N4, N6, N11	System, 10.5	System	—	HZ-16	I	—
SYS-010	—	System	System	—	HZ-11	T	Propulsion and mass properties
SYS-011	—	7.2	System	ICD-008	—	I	—
SYS-012	—	2.2	System	—	—	A	—
STR-001	N2, N7	7.1	Airframe	—	HZ-07	A	Structural
STR-002	—	7.1	Airframe	—	HZ-07	A	Structural
STR-003	—	7.1	Airframe	—	HZ-07	A	Structural
STR-004	—	5.5	Airframe	ICD-011	HZ-08	A	Structural
STR-005	—	5.5	Airframe	—	HZ-06	A	—
STR-006	—	7.1	Airframe	—	HZ-06	A	Structural
STR-007	—	5.3, 10.2	Airframe	—	HZ-06	A	Structural
STR-008	—	7.1	Airframe	—	—	I	—
STR-009	—	7.3	Airframe	—	—	I	—
STR-010	—	1.1, 7.4	Airframe	—	HZ-07, HZ-13	A	Structural

ID	Need	Function	Subsystem	ICD	Hazard	Verif.	Test group
AERO-001	—	3.2	Airframe	—	—	A	—
AERO-002	N7	3.1	Airframe	—	HZ-09	I	—
AERO-003	N7	7.1	Airframe	—	HZ-09	A	—
AERO-004	—	3.1	Airframe	—	—	A	—
AERO-005	—	3.2, 4.4	Airframe	ICD-004	HZ-A3	A	—
AERO-006	—	3.3	Airframe	—	—	A	—
AV-001	—	5.7	Avionics	—	HZ-15	T	Avionics
AV-002	—	5.7, 9.3	Avionics	ICD-009	HZ-14, HZ-15	T	Avionics
AV-003	—	9.2	Avionics	—	HZ-01	T	Avionics
AV-004	—	9.2	Avionics	ICD-009	—	T	Avionics
AV-005	—	9.2	Avionics	—	—	T	Avionics
AV-006	—	7.4	Avionics	—	HZ-02	T	Avionics
AV-007	—	5.1	Avionics	—	HZ-01	T	Avionics
AV-008	N7, N8	9.4, 10.1	Avionics	—	HZ-05	D	—
AV-009	—	8.3	Avionics	—	HZ-02	T	Avionics
AV-010	—	8.3	Avionics	ICD-002	HZ-02	T	Avionics
AV-011	—	8.1	Avionics	—	HZ-02	I	—
AV-012	—	8.1	Avionics	—	HZ-02	A	—
REC-001	N7, N11	5.1, 5.2	Recovery	—	HZ-01, HZ-03	T	Recovery
REC-002	N7, N11	5.4	Recovery	—	HZ-14	T	Recovery
REC-003	N7, N11	5.2	Recovery	—	HZ-01, HZ-04	T	Recovery
REC-004	N2, N11	5.4, 5.6	Recovery	—	HZ-04, HZ-14	T	Recovery
REC-005	N7, N11	5.2	Recovery	ICD-010	HZ-03	I	—
REC-006	—	5.2	Recovery	ICD-010	HZ-02	I	—
REC-007	—	5.2	Recovery	ICD-010	HZ-03, HZ-04	T	Recovery
REC-008	N11	10.2	Recovery	ICD-010	HZ-05, HZ-16	T	Recovery

ID	Need	Function	Subsystem	ICD	Hazard	Verif.	Test group
REC-009	—	5.5	Recovery	ICD-011	HZ-08	A	—
REC-010	—	5.5	Recovery	ICD-011	HZ-08	T	Recovery
REC-011	—	7.4	Recovery	—	HZ-13	I	—
REC-012	—	7.3	Recovery	—	—	I	—
REC-013	—	10.3	Recovery	—	HZ-03	A	—
REC-014	—	5.7	Recovery	—	—	I	—
PT-001	N2	1.3	Propulsion	—	—	A	—
PT-002	—	1.3	Propulsion	—	—	A	—
PT-003	—	1.3	Propulsion	—	HZ-12	I	—
PT-004	—	1.1	Propulsion	ICD-006	HZ-12	T	Propulsion and mass properties
PT-005	—	1.1	Propulsion	ICD-006	HZ-12	A	—
PT-006	—	2.1	Propulsion	ICD-007, ICD-008	—	I	—
PT-007	—	1.4	Propulsion	—	—	A	—
AB-001	N2	4.4	Airbrakes	ICD-005	—	A	—
AB-002	—	4.4	Airbrakes	ICD-005	—	A	—
AB-003	N7, N11	4.5, 10.2	Airbrakes	—	HZ-A1	T	Airbrakes*
AB-004	N7, N11	4.5, 10.3	Airbrakes	ICD-005	HZ-A2	T	Airbrakes*
AB-005	N6, N7	3.2	Airbrakes	—	HZ-A3	A	—
AB-006	—	4.4	Airbrakes	ICD-004	HZ-A2	T	Airbrakes*
AB-007	—	4.5	Airbrakes	—	—	A	—
PAY-001	—	6.1	Payload	ICD-001	—	I	—
PAY-002	—	6.1	Payload	—	—	I	—
PAY-003	—	6.1	Payload	ICD-001	HZ-P1	T	Payload*
PAY-004	—	3.2	Payload	—	HZ-P1	A	—
PAY-005	—	6.1	Payload	ICD-003	—	A	—
INT-001	—	7.3	All (interfaces)	ICD-001, ICD-002	—	I	—

ID	Need	Function	Subsystem	ICD	Hazard	Verif.	Test group
INT-002	—	8.3	All (interfaces)	ICD-003	—	T	Avionics
INT-003	—	7.3	All (interfaces)	—	—	I	—
INT-004	—	10.3	All (interfaces)	—	—	A	—
INT-005	—	3.2	All (interfaces)	—	—	A	—
INT-006	—	System	All (interfaces)	—	—	I	—
ENG-001	—	Prog.	Programme	—	—	I	—
ENG-002	—	Prog.	Programme	—	—	I	—
ENG-003	—	3.2	Programme	—	HZ-10	T	Propulsion and mass properties
ENG-004	—	3.2	Programme	—	HZ-10	T	Propulsion and mass properties
ENG-005	—	3.2	Programme	—	HZ-10	A	—
ENG-006	—	Prog.	Programme	—	HZ-18	I	—
ENG-007	—	Prog.	Programme	—	HZ-18	I	—
ENG-008	—	10.3	Programme	—	—	A	—
ENG-009	—	Prog.	Programme	—	HZ-20	I	—
ENG-010	N1	Prog.	Programme	—	—	I	—
DOC-001	N1, N6, N8, N9	Prog.	Programme	—	—	I	—
DOC-002	—	Prog.	Programme	—	—	I	—
DOC-003	—	Prog.	Programme	—	—	I	—
DOC-004	—	Prog.	Programme	—	—	I	—
DOC-005	—	Prog.	Programme	—	—	I	—
DOC-006	—	Prog.	Programme	—	HZ-18	I	—
DOC-007	—	9.2	Programme	—	—	I	—
DOC-008	N3	Prog.	Programme	—	—	I	—
TST-001	—	5.2	Programme	—	HZ-05	D	Recovery

ID	Need	Function	Subsystem	ICD	Hazard	Verif.	Test group
TST-002	—	9.4	Programme	—	—	D	—
TST-003	—	9.4	Programme	ICD-007	—	T	Avionics
TST-004	—	3.2	Programme	—	—	T	Propulsion and mass properties
TST-005	—	3.2	Programme	—	—	T	Propulsion and mass properties
TST-006	—	5.2	Programme	—	—	T	Recovery
TST-007	—	Prog.	Programme	—	—	I	—
TST-008	—	9.2	Programme	—	—	T	Avionics
PT-008*	N6, N7	1.3	Propulsion	—	—	I	—
PAY-006*	N6, N9	6.1	Payload	—	—	I	—
ENG-011*	N4, N5	Prog.	Programme	—	HZ-17	I	—
ENG-012*	N12	Prog.	Programme	—	—	I	—
AV-013*	—	4.1, 5.1	Avionics	—	—	T	Avionics
AV-014*	—	4.2	Avionics	—	—	T	Avionics
SYS-013*	N2	4.3, 4.4	System	—	—	A	—

J. PDR Content Map

A generic aerospace PDR covers a common set of content. Table 68 maps that content to the sections of this report, so a reviewer used to that structure can find each item, and states honestly what is complete and what is still partial. The categories follow NASA systems engineering practice [7], [11], applied by analogy to a student rocket. The team should confirm with its mentor which standard governs the review.

Table 68. Mapping of generic PDR content to this report.

Generic PDR content	Where in this report	Status	What is missing
System overview, mission, ConOps and requirements traceability	Sections 1 to 3; Appendix A and Appendix I	Mostly complete	11 requirements with TBD values; 7 functions without requirements
Subsystem technical designs	Section 5.5; Appendices C and D	Partial	Preliminary only; motor matrix being rebuilt; mounts, retention and interface values TBD
Engineering budgets and margins	Section 5.6; Appendices G and H	Partial	6.64 kg of mass not itemized; energy and volume budgets TBD
Specialty engineering and mission assurance (safety, reliability, standards)	Sections 5.4, 2.3, 5.7; Appendix F	Partial	FMEA cause column and scales to confirm; 6 requirements with no hazard link
Verification, validation and test planning	Sections 2.4, 5.8; Appendix I	Partial	Facilities and criteria are TBR; no results yet
Programmatics: schedule, budget and risk	Sections 5.7, 5.8; Appendix E	Partial	Dates not reconciled; funding changes to confirm